How Should 45Z Pay for Risk? A Welfare-CVaR and Project-Finance Decomposition of the IRA Clean Fuel Production Credit

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As enacted, the U.S. Inflation Reduction Act's Section 45Z credit paid sustainable aviation fuel (SAF) producers \$1.30–\$1.45 per gallon for forestry-residue pathways — the \$1.75 statutory base scaled by carbon intensity — roughly 2.6 times the \$0.52/gal welfare benchmark computed from the same supply-chain data. The subsidy a privately financed pioneer producer actually requires is \$1.36/gal on our calibration, inside the as-enacted band and well above the \$0.74–\$0.83/gal the post-2025 statute now delivers. We decompose the \$0.84/gal producer–planner gap in a Stackelberg model where a regulator chooses a subsidy schedule before a privately financed producer decides whether to invest. The gap splits into a risk-aversion premium and a project-finance premium, with the latter dominant across plausible calibrations, indicating that the observed gap reflects financing frictions more than producer risk preferences. Although the headline subsidy level depends on how price uncertainty is discretized, the project-finance premium itself is structurally invariant, making it the policy-robust object of the decomposition. We further show that, within the policy forms available under Section 45Z, a contract-for-difference with a break-even floor weakly Pareto-dominates the observed linear credit, delivering a sign-determinate welfare gain when public funds are costly — modest, on the order of five percent of the EPA central social-cost-of-carbon benchmark per ton abated. The policy implication is that fiscal effort is better directed toward reducing project-finance hurdles—such as through federal loan guarantees—than toward further adjusting the per-gallon credit rate.

Keywords: sustainable aviation fuel; 45Z producer credit; risk aversion; project finance; contract for difference; mean-CVaR welfare; clean fuel policy.

JEL codes: Q42, Q48, H23, G31.

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1. Introduction

As enacted, the U.S. Inflation Reduction Act's Section 45Z credit paid sustainable-aviation-fuel (SAF) producers \$1.30–\$1.45/gal for forestry-residue Fischer–Tropsch pathways — the \$1.75/gal statutory base (with prevailing-wage compliance) scaled by the credit's carbon-intensity formula at the pathway's 8.6–12.9 $kgCO_2e$/mmBTU range — roughly 2.6 times the \$0.52/gal capacity-restoring benchmark that planner-side welfare analysis derives from the same southeastern (SE) US supply-chain data (Li and Yu 2026). The \$0.84/gal gap between what a privately financed producer requires and what the planner's benchmark prescribes is the puzzle, and the policy window to close or explain it is narrow: H.R. 1 (enacted July 2025) eliminated the SAF premium for fuel produced after December 31, 2025 — cutting the CI-scaled credit for this pathway to \$0.74–\$0.83/gal — and triggered a Code of Federal Regulations (CFR) rulemaking cycle with a Q3 2026 deadline. Two questions follow. First, what explains the observed 45Z level? Second, is the linear per-gallon form—a constant credit regardless of realized SAF price—the right instrument, or does a state-contingent alternative do better?

We answer both. The \$0.84/gal gap decomposes into a \$0.10/gal producer risk-aversion premium and a \$0.74/gal project-finance hurdle premium; at the post-2025 CI-scaled level of \$0.74–\$0.83/gal the financial-frictions component alone absorbs essentially the entire credit, so the H.R. 1 change did not merely trim rent but cut below the subsidy that makes pioneer-plant SAF financially feasible on our calibration ($\sigma_{\text{prod}} = \$1.36$/gal). On instrument form, under complete information on the producer's financial primitives a contract-for-difference (CfD) with a break-even floor weakly Pareto dominates the linear form under the planner's mean-conditional-value-at-risk (mean-CVaR) welfare functional.

Policy context. The 45Z credit was enacted under the IRA with a \$1.75/gal SAF base rate (conditional on prevailing-wage and apprenticeship compliance; \$0.35 otherwise), scaled by the statute's sliding carbon-intensity (CI) formula $(50 - \text{CI})/50$ (Joint Committee on Taxation 2022); H.R. 1 eliminated the SAF premium for fuel produced after December 31, 2025, aligning SAF at the \$1.00/gal base, amid broader IRA restructuring (Congressional Budget Office 2025; Bistline, Mehrotra, and Wolfram 2023; Bistline et al. 2024). The Internal Revenue Service issued interim guidance on CI-score methodology (Internal Revenue Service 2025), but the core structural question—linear per-gallon vs. price-contingent—has not been examined in the rulemaking record (Congressional Research Service 2025). The Q3 2026 CFR cycle is the opening. Pioneer-plant SAF facilities at National Renewable Energy Laboratory (NREL) first-of-a-kind (FOAK) capital-expenditure (CAPEX) intensity (\$11/gal annual capacity) require a hurdle internal rate of return (IRR) of roughly 12.5% to attract private equity (Section 5.2). On our calibration the hurdle-clearing credit is \$1.36/gal: the lower portion of the as-enacted CI-scaled band falls just short, the upper portion clears, and the post-2025 level of \$0.74–\$0.83/gal falls well short across the SAF price distribution.

Model preview. We build a Stackelberg model in which the regulator commits to a transfer form $\sigma(\omega)$ before the producer's investment decision, with both agents maximizing mean-CVaR welfare under risk-aversion weights λ_{planner} (planner) and λ_{prod} (producer). The transfer form is chosen from a four-element design space $\mathcal{S} = \{\text{linear, CfD-strike, CfD-with-floor, capacity}\}$ reflecting the contractual forms statutorily admissible under IRC §45. The structural innovation relative to Li and Yu (2026) is a project-finance participation constraint: the producer invests only if the project net present value (NPV) at hurdle IRR r_h is non-negative. The constraint defines the *financial-equivalence subsidy* σ_{prod}—the minimum constant transfer that clears the investment hurdle—which collapses to the welfare-equivalence benchmark $\sigma^*_{\text{planner}}$ of Li and Yu (2026) only at the degenerate point where $r_h = r_{\text{RN}}$, $\tau = 0$, $\lambda_{\text{prod}} = \lambda_{\text{planner}}$, *and* CAPEX recovery at the risk-neutral rate is exactly offset by the operating margin (a zero-profit condition); empirically all four conditions fail by wide margins.

Main results. Three results organize the paper. The first is a closed-form decomposition of σ_{prod} (Propositions A.3 and A.5). Under Assumptions A1–A13 in the two-state CVaR regime $\alpha \geq \pi_H$, the project-finance NPV = 0 condition delivers

$$\sigma_{\text{prod}} = \sigma^*_{\text{planner}} + \underbrace{(\lambda_{\text{prod}} - \lambda_{\text{planner}})\,\pi_H\,(a_H - a_L)}_{\substack{\text{Component 1:}\\ \text{risk-aversion premium}}} + \underbrace{\Delta\text{IRR}_{\text{prem}}}_{\substack{\text{Component 2:}\\ \text{financial-frictions premium}}},$$

where $\Delta\text{IRR}_{\text{prem}} \equiv \kappa_{\text{CAPEX}}/[u\ \text{AF}(r_h, T)(1-\tau)] - [\bar{p_{\text{SAF}}} - \bar{c_{\text{op}}}]$ is the per-gallon CAPEX-recovery requirement at the hurdle rate minus the unconditional expected operating margin. Component 2 carries the *financial-frictions* interpretation because the hurdle rate $r_h > r_{\text{RN}}$ enters only through the annuity factor.

The second result is the empirical calibration. The hurdle IRR $r_h = 0.125$ (envelope $[0.10, 0.15]$) is anchored qualitatively in DOE 2022 and quantitatively in Lazard's 2024 after-tax weighted-average-cost-of-capital (WACC) benchmarks plus the Graham and Harvey (2001) hurdle-premium evidence; κ_{CAPEX} = \$11/gal annual capacity is a $\approx 1.2\times$ first-of-a-kind (FOAK) premium over the \$9.31/gal-yr nth-plant NREL anchor (Tan et al. 2015), with the Farooq et al. (2025) meta-analysis anchoring the operating-cost primitives. With these primitives fixed, the Component 2 financial-frictions premium $\Delta\text{IRR}_{\text{prem}}$ = \$0.74/gal is *structurally independent of* λ_{prod}: this is the policy-robust object of the decomposition. The producer's mean-CVaR weight λ_{prod} is identified via revealed preference from the observed 45Z subsidy band $\sigma^{\text{obs}}_{\text{prod}} \in$ [\$1.30, \$1.45]/gal, conditional on r_h and the closed-form NPV-binding identity, yielding $\lambda_{\text{prod}} = 0.60$ (band image $[0.54, 0.69]$). At this calibration, Component 1 = \$0.10/gal (12% of the \$0.84/gal wedge), Component 2 = \$0.74/gal (88%), and σ_{prod} = \$1.36/gal sits inside the empirical 45Z band. The Component-2 share remains above 80% throughout the revealed-preference band image and above 70% for all $\lambda_{\text{prod}} \in [0.50, 0.77]$ (Table 7). The attribution ratio shifts with horizon T as $\text{AF}(r_h, T)$ saturates—at $T = 5$ financial frictions account for 96% of the wedge; at $T = 20$ risk aversion accounts for 71% as the CAPEX-recovery term collapses toward the perpetuity bound (Section 7). The financial-frictions *level* of \$0.74/gal at the Paper-1 horizon $T = 10$ is the policy-robust headline; the attribution *ratio* is conditional on the calibration scaffold.

The third result is CfD-with-floor dominance (Theorem B.3). Adapting the Borch–Wilson risk-sharing theorem (Borch 1962; Wilson 1968) to the SAF contracting setting, the welfare-optimal transfer form *within* the statutorily relevant four-element class $\mathcal{S}$ under complete information is a CfD with strike $K^* = \bar{p}$ (the producer's break-even SAF price) and a floor s^*_0 pinned by the participation constraint. The qualifier "within $\mathcal{S}$" is load-bearing: dropping the restriction (A11) and moving to CARA-Normal preferences restores linear-σ optimality by Holmström and Milgrom (1987) Theorem 7; the four-element $\mathcal{S}$ captures the contractual forms statutorily available under IRC §45 (linear, capacity) and policy-debated in the 45Z rulemaking record (CfD-strike, CfD-with-floor). We document the opportunity cost of A11 explicitly. A fully state-contingent first-best transfer (relaxing A11 alone, holding r_h fixed) closes Component 1 of the wedge—\$0.10/gal—through the risk-sharing channel; jointly relaxing the project-finance hurdle so that the planner substitutes r_{RN} for r_h would additionally close the \$0.74/gal Component 2 and unlock the \$7.00/gal first-best fiscal-slack diagnostic of Theorem B.3 Step 4b. Within $\mathcal{S}$, CfD-with-floor captures \$0.03–\$0.09/gal of the Component 1 channel at MCPF $\in [1.10, 1.30]$, the sign-determinate but modest welfare gain that the policy product delivers. The first-order welfare wedge of CfD-with-floor relative to the linear form is positive through two mechanisms: risk-sharing (Component 1) and more efficient capital allocation when the NPV constraint binds (Component 2). Under MCPF $\mu > 1$ the CfD advantage strengthens because the CfD pays only in low-price scenarios and therefore carries lower expected transfer cost.

Related literature. This paper draws on four traditions: empirical welfare analysis of clean-energy production credits (Cullen 2013; Borenstein 2008; Holland, Hughes, and Knittel 2009); SAF policy design (Wu et al. 2025); contract-for-difference welfare theory in renewable-electricity procurement (Newbery 2023; Boomsma, Meade, and Fleten 2012); and risk-sharing failure modes under restricted contract spaces (Holmström and Milgrom 1987; Borch 1962). The closest precedents are Cullen (2013), who frames the welfare-vs-subsidy-cost comparison for the wind PTC at a project-finance margin, and Wu et al. (2025), who establish the per-gallon producer credit as a least-cost SAF instrument *level*; neither addresses the instrument-*form* decomposition this paper performs. The welfare-equivalence benchmark $\sigma^*_{\text{planner}} = \$0.52$/gal is inherited from Li and Yu (2026), who derive the planner's mean-CVaR-optimal subsidy in closed form on the SE-U.S. supply-chain MILP; the planner's mean-CVaR weight $\lambda_{\text{planner}} = 0.50$ is the midpoint between the Arrow and Lind (1970) risk-neutrality benchmark and the Gollier (2001) social-risk-aversion upper bound. Section 2 positions the project-finance + mean-CVaR decomposition against each tradition in detail.

Contribution. The paper delivers two contributions that no prior work provides jointly.

First, a closed-form two-component decomposition of $\sigma_{\text{prod}} - \sigma^*_{\text{planner}}$ that separates a producer risk-aversion premium from a project-finance hurdle premium under mean-CVaR preferences and NPV-binding participation. Cullen (2013) estimates the Cullen-style welfare-vs-subsidy-cost gap for the wind PTC in a risk-neutral framework, and Li and Yu (2026) characterize $\sigma^*_{\text{planner}}$ in closed form on the SE-US supply chain; neither isolates the financial-frictions component, which we identify empirically as 88% of the gap at horizon $T = 10$ (the attribution ratio shifts strongly with horizon).

Second, a proof that a CfD with break-even floor weakly Pareto-dominates the observed linear 45Z form within the statutorily relevant four-element class $\mathcal{S}$ under complete information. Newbery (2023) establishes CfD welfare dominance for renewable-electricity *procurement*, and Holmström and Milgrom (1987) characterize the unrestricted-contract case in which linear σ is optimal; our result combines the Borch-Wilson risk-sharing channel with the project-finance capital-allocation channel under the SAF setting's statutory-instrument restriction, and quantifies the opportunity cost of that restriction via the first-best fiscal-slack diagnostic (Theorem B.3 Step 4b).

Two analytic tracks. The closed-form σ_{prod} result rests on a representative-investor quadratic-capacity reduction (Track T_{closed}) of the underlying Stage-1 binary depot-refinery investment problem. The full two-stage stochastic MILP at the SE-US 248/34 facility-siting instance (Track $T_{\text{numerical}}$) delivers the same first-order $\sigma_{\text{prod}} \approx \$1.36$/gal benchmark at the MILP-optimal footprint; the closed-form is the analytic first-order approximation. Quantitative numerical welfare-wedge magnitudes ΔW at the discrete 248/34 optimum and the $O((\mu_W - 1)^2)$ residual are deferred to a follow-up numerical companion. The present paper's contribution lives in Track T_{closed}; Section 7.10 documents the relationship between the two tracks. Convention disclosure: the headline $\sigma_{\text{prod}} = \$1.36$/gal uses the absorption depreciation convention standard in the TEA literature (Farooq et al. 2025; Lazard 2024); under MACRS-rigorous depreciation the headline reduces to $\sigma_{\text{prod}} \approx \$1.02$/gal (below the observed band), as Section 7.8 documents. The convention dependence is disclosed explicitly throughout.

The wedge-decomposition framework extends beyond 45Z: the same closed-form maps the welfare-vs-financial gap for 45V (hydrogen), 45Y (electricity), and 45Q (carbon capture) production credits by reparameterizing the project-finance primitives ($\kappa_{\text{CAPEX}}, r_h, T, \lambda_{\text{prod}}$), with the pathway-conditional cross-walk documented in Section 7.11.

Paper organization. Section 3 formalizes the Stackelberg bilevel environment, payoffs, and assumptions A1–A13. Section 4 derives σ_{prod} in closed form and presents the empirical calibration. Section 5 reports the two-component wedge decomposition on the Paper-1 SE-US calibration (the MILP-side verification at the full 248/34 instance is deferred; Section 7.10). Section 6 establishes the CfD-with-floor weak Pareto dominance result within the statutory instrument class and quantifies the welfare gain. Section 7 examines sensitivity to T, r_h, and the MCPF. Section 8 develops the four-part Treasury+DOE proposal calibrated to the Q3 2026 CFR rulemaking cycle. Section 9 concludes. Section A and Section B contain the proofs.

2. Related Literature

Empirical welfare analysis of production tax credits. The closest methodological anchor is Cullen (2013), who evaluates the U.S. wind PTC by equating a financial-equivalence subsidy with the project-level NPV at a hurdle discount rate—the template this paper applies to 45Z. Cullen acknowledges that producer risk aversion drives an additional component of the necessary credit level and defers it to future work. We close that gap: in the 45Z context, risk aversion accounts for 12% of the gap between the welfare-equivalence benchmark and the observed credit at $T = 10$, while the project-finance hurdle premium—the component Cullen's framework would identify alone—accounts for 88%.

The inherited welfare-equivalence benchmark, $\sigma^*_{\text{planner}} = \lambda_{\text{planner}} \cdot \pi_H \cdot (a_H - a_L) = \$0.52$/gal, is the central result of Li and Yu (2026), who derive the planner's mean-CVaR-optimal per-gallon subsidy in closed form on the SE-US supply chain under linear inverse demand and Rockafellar–Uryasev CVaR preferences. We take $\sigma^*_{\text{planner}}$ as the welfare reference point and ask what a producer-side project-finance participation constraint adds; the two objects coincide only in the degenerate case $r_h = r_{\text{RN}}$, $\tau = 0$, $\lambda_{\text{prod}} = \lambda_{\text{planner}}$ joint with zero-profit CAPEX recovery ($\kappa_{\text{CAPEX}}/[u\ \text{AF}(r_{\text{RN}}, T)] = \bar{p} - \bar{c}_{\text{op}}$). On production-credit incidence, full pass-through is documented by Sallee (2011) for the Prius credit and grounded theoretically by Weyl and Fabinger (2013). The clean-energy welfare-cost-vs-abatement framing traces to Borenstein (2008); the social-cost benchmark at the EPA \$190/ton SCC is documented by Bistline et al. (2024); distributional context is provided by Metcalf (2019) and Aldy and Stavins (2012).

SAF policy: Wu et al. as direct predecessor. Wu et al. (2025) is the direct predecessor. Using a detailed model of U.S. fuel markets and aviation demand, they show that a per-gallon producer credit is among the three least-cost instruments for reaching the SAF Grand Challenge 3 B gal/yr target. Taking that conclusion as given, we pursue the design question Wu et al. do not: is the linear per-gallon form welfare-optimal under empirically calibrated producer risk aversion, or does a contract-for-difference with a break-even floor dominate? Wu et al. determine the credit level; this paper determines its form.

Contracts-for-difference and renewable electricity policy. Newbery (2023) proves CfD welfare dominance for renewable electricity *procurement* under auctioned strike prices—the electricity-sector analog of this paper's Theorem B.3. The SAF setting differs in framing: the regulator does not procure SAF directly but subsidizes a decentralized producer, so the welfare comparison is between instrument forms within a restricted policy class $\mathcal{S}$ (Assumption 11), not between auction mechanisms. Boomsma, Meade, and Fleten (2012) supplies the project-finance trunk—a hurdle-IRR framing for capacity investment under uncertain subsidy—that motivates the NPV-binding participation constraint (Assumption 13). The NPV-at-hurdle behavior is empirically grounded in the CFO survey evidence of Graham and Harvey (2001) that firms set internal discount rates well above their cost of capital, consistent with the pioneer-plant hurdle levels calibrated in Section 5.2.

Risk-sharing theory. The proof technique for Theorem B.3 adapts the classical risk-sharing theorems of Borch (1962) and Wilson (1968): at an interior Pareto-optimum, the ratio of marginal utilities must equalize across states. The discrete-scenario mean-CVaR adaptation is the key technical step, because the subgradient structure of the CVaR preserves the equalization condition in the relevant regime when the planner side and producer side both carry mean-CVaR functionals with $\lambda_{\text{planner}} \neq \lambda_{\text{prod}}$. The failure-mode benchmark is Holmström and Milgrom (1987), who prove that under CARA-Normal preferences and an unrestricted contract space the optimal incentive contract is linear. The CfD-with-floor dominance result rests on two features that break Holmstrom-Milgrom: public information on producer financial primitives (Assumption 10) and the restricted policy-instrument class (Assumption 11); relaxing either restores linear σ optimality. The variational foundation for the mean-CVaR functional is Maccheroni, Marinacci, and Rustichini (2006), and the Rockafellar–Uryasev LP dual representation is the computational tool used throughout. The planner's mean-CVaR weight $\lambda_{\text{planner}} = 0.50$ is calibrated as the midpoint between the Arrow and Lind (1970) risk-neutrality benchmark ($\lambda = 0$) and the Gollier (2001) social-risk-aversion upper bound ($\lambda \approx 1$); sensitivity to $\lambda_{\text{planner}} \in \{0, 0.25, 0.50\}$ is reported in Section 7.5.

3. Model

Table 1 collects the recurring notation used throughout the model and proofs. Subsidy objects are distinguished by subscript: σ_{prod} is the producer's financial-equivalence subsidy (Proposition A.3), $\sigma^*_{\text{planner}}$ is the planner's welfare-equivalence subsidy inherited from Li and Yu (2026), and $\sigma^*(\omega)$ refers to the regulator-optimal transfer in Theorem B.3 after Stage 0b commitment.

3.1. Environment

A single representative SAF producer faces a regulator (the social planner). The stochastic environment is indexed by a scenario $\omega \in \Omega$ drawn from a distribution F. For calibration and numerical exhibits, F is the 20-bin kernel density estimate (KDE) of the *constructed drop-in biofuel price* inherited from Paper 1 (Li and Yu 2026). Because no deep SAF spot market exists, that series is a producer-revenue price built from observable components: a

Table 1: Recurring notation.

Symbol	Definition
$\omega \in \Omega$	Scenario index; $\Omega = \{H, L\}$ in 2-state collapse; 20 bins in KDE
$p_{\text{SAF}}(\omega)$	SAF wholesale price in scenario ω (\$/gal)
π_H, π_L	Upside / downside probabilities (2-state); $\pi_H + \pi_L = 1$
a_H, a_L	Conditional means of p_{SAF} in upside / downside states (\$/gal)
$\bar{p} \equiv \mathbb{E}[p_{\text{SAF}}]$	Unconditional KDE mean (\$ 6.00/gal at baseline)
$\bar{c}_{op} \equiv \mathbb{E}[c_{\text{op}}]$	Unconditional operating cost (\$ 3.78/gal at baseline)
σ_{prod}	Producer financial-equivalence subsidy (Proposition A.3; \$/gal)
$\sigma^*_{\text{planner}}$	Planner welfare-equivalence subsidy (Li-Yu 2026; \$/gal)
$\sigma^*(\omega)$	Regulator-optimal state-contingent transfer (Theorem B.3)
σ^{lin}	Linear baseline ($\sigma^{\text{lin}} = \sigma_{\text{prod}} = \$1.36$/gal at headline)
$\mathcal{S}$	Four-element instrument class: linear, CfD-strike, CfD-with-floor, capacity
λ_{prod}	Producer mean-CVaR weight ($\in [0, 1]$; baseline 0.60)
λ_{planner}	Planner mean-CVaR weight ($\in [0, 1]$; baseline 0.50, Arrow-Lind/Gollier midpoint)
α	CVaR confidence level (0.95 baseline)
$\hat{\pi}_i(\omega)$	Distorted probability measure under agent $i \in \{\text{prod}, \text{planner}\}$
r_h	Producer hurdle IRR (0.125 baseline, envelope $[0.10, 0.15]$)
r_{RN}	Planner risk-neutral discount rate (≈ 0.03)
κ_{CAPEX}	Per-gallon CAPEX intensity (\$11/gal baseline; NREL FOAK)
u	Capacity utilization (0.85 baseline; 7440 op-hrs/yr)
τ	Effective federal corporate tax rate (0.21)
T	Project horizon (10 yrs baseline; Paper-1 NPV horizon)
$\text{AF}(r_h, T)$	Annuity factor $\sum_{t=1}^{T}(1 + r_h)^{-t}$
μ_W	Marginal cost of public funds (1.20 baseline; Dahlby midpoint)
C_1, C_2	Wedge components: risk-aversion premium, financial-frictions premium
$\Delta\text{IRR}_{\text{prem}}$	$\kappa_{\text{CAPEX}}/[u\ \text{AF}(r_h, T)(1 - \tau)] - (\bar{p} - \bar{c}_{op}) = C_2$
$\mathbb{1}_{\{\text{NPV-binding}\}}$	Indicator that producer's NPV constraint binds at σ^{lin}

slate-weighted wholesale gasoline–diesel price on a common real basis, plus the market value of the Renewable Fuel Standard compliance credits the fuel generates — the weekly D-code Renewable Identification Number (RIN) price at the 1.7 cellulosic equivalence factor — spanning August 2013 through December 2021 at weekly resolution. The *level* of the series embeds RIN value (\$2.96/gal of the \$5.98/gal sample mean), so the operating margin $\bar{p_{\text{SAF}}} - \bar{c_{\text{op}}}$ in the participation constraint is fuel-*plus*-RIN revenue, and the financial-equivalence subsidy σ_{prod} is the 45Z payment required *on top of* the existing RIN stack — the same incremental-to-RIN reading maintained in Paper 1. The *dispersion* is genuine revenue risk: RIN-price volatility passes one-for-one into the margin of a producer paid partly in credits. For the closed-form propositions (Propositions A.3 and A.5), F is collapsed to the two-state representation

$$\omega \in \{H, L\}, \quad \pi_H = 0.472, \quad \pi_L = 0.528, \quad a_H = \$7.16/\text{gal}, \quad a_L = \$4.97/\text{gal},$$

with KDE unconditional mean $\bar{p_{\text{SAF}}} = \$6.00$/gal. The two-state collapse is the Li and Yu (2026) Theorem 1 calibration device and is justified formally in Assumption 1 below.

3.2. Timing

The game has four stages arranged in a Stackelberg structure with the regulator as leader.

Stage 0a (Regulator observes financial primitives). Before committing to a transfer form, the regulator publicly observes the producer's project-finance primitives

$$(r_h,\ e,\ r_e,\ r_d,\ \tau,\ \kappa_{\text{CAPEX}}),$$

where r_h is the hurdle IRR, e the equity share, r_e and r_d the costs of equity and debt, τ the effective corporate tax rate, and κ_{CAPEX} the per-gallon CAPEX intensity. These primitives are recovered from the systematic SAF techno-economic meta-analysis of Farooq et al. (2025) cross-walked to two independent calibration anchors: Lazard LCOE+ 2024 cost-of-capital assumptions (60/40 debt–equity at 8%/12%; after-tax WACC ≈ 7.7% for generation, 9.1% in the hydrogen analysis) (Lazard 2024) and the NREL 2024 Transportation Annual Technology Baseline capital-charge factor (National Renewable Energy Laboratory 2024). The DOE SAF Grand Challenge Roadmap (2022) documents qualitatively that first-of-a-kind SAF plants face a high cost of capital (its TEA workstream targets pioneer-versus-nth-plant costs); the FAA Aviation Climate Action Plan (2021) and the IATA Net Zero Roadmaps (2023) likewise discuss SAF financing risk in qualitative terms only. The quantitative range is therefore anchored on market benchmarks: Lazard's after-tax WACC of 7.7–9.1% (Lazard 2024) plus the survey evidence that realized corporate hurdle rates exceed WACC by several percentage points (Graham and Harvey 2001), cross-checked against the NREL Transportation ATB capital-charge assumptions (National Renewable Energy Laboratory 2024). This yields $r_h \in [0.10, 0.15]$ with mean $r_h = 0.125$ as the baseline. The leverage primitives aggregate to the after-tax weighted average cost of capital via WACC $= e \cdot r_e + (1-e)\, r_d(1-\tau)$; at the Lazard 60/40, 8%/12% benchmarks this gives WACC ≈ 7.7–8.6% (depending on the tax convention), so the pioneer premium in $r_h = \text{WACC} + \delta$ (Assumption 13) is $\delta \approx$ 4–5 percentage points at $r_h = 12.5\%$ — the magnitude of the hurdle-over-WACC premium documented by Graham and Harvey (2001). Stage 0a involves no choice by either agent; it is purely observational and formalizes the complete-information convention of Assumption 10.

Stage 0b (Regulator commits σ-form). The regulator selects a transfer form $\sigma(\cdot)$ from the four-element design space $\mathcal{S}$ (defined in Section 3.3) to maximize ex-ante mean-CVaR welfare subject to the producer's project-finance participation constraint. The commitment is irrevocable by Assumption 9: no ex-post revision is permitted. The 45Z credit was enacted as a statutory rate under the IRA (2022) and was revised via H.R. 1 only at the policy-rulemaking timescale (Q3 2026 CFR cycle), not at the investor timescale.

Stage 1 (Producer invests). Given the committed transfer form σ, the producer chooses a facility footprint $y \in \{0,1\}^{|D|+|R|}$ over $|D| = 248$ candidate depot sites and $|R| = 34$ candidate refinery sites. The Paper 1 248/34 SE-US instance is the baseline Stage-1 solution for the numerical exhibit. Participation requires $\text{NPV}(\sigma; r_h, \kappa_{\text{CAPEX}}, u, T) \geq 0$ by Assumption 13; the closed-form characterization treats y via the representative-investor reduction of Li and Yu (2026) §3.5, under which the Stage-1 investment decision collapses to a smooth capacity-utilization problem.

Stage 2 (Producer produces under realized ω). Nature draws $\omega \sim F$. The producer observes ω and chooses output $Q(\omega) \geq 0$ given (y, σ, ω) to maximize realized operating profit.

3.3. Policy-Instrument Design Space

The regulator's Stage-0b action set is a four-element design space $\mathcal{S}$ of per-gallon transfer forms. Each element corresponds to a contractual instrument with precedent in actual energy policy.

(1) **Linear:** $\sigma^{\text{lin}}(\omega) = \sigma_0$, a constant credit per gallon regardless of the realized SAF price. This is the form of the observed IRA Section 45Z credit at $\sigma_0 \in [\$1.30, \$1.45]$/gal.

(2) **CfD-strike:** $\sigma^{\text{strike}}(\omega; K) = \max\{K - p_{\text{SAF}}(\omega), 0\}$, a contract-for-difference that pays the shortfall below a reference strike price K and pays nothing when $p_{\text{SAF}}(\omega) > K$.

(3) **CfD-with-floor:** $\sigma^{\text{floor}}(\omega; K, s_0) = \max\{K - p_{\text{SAF}}(\omega), 0\} + s_0$, a CfD-strike augmented with a constant floor payment $s_0 \geq 0$. The floor ensures the producer earns at least a baseline transfer in every scenario; the strike component concentrates additional payments in low-price states. The UK Contracts for Difference scheme for offshore wind is the real-world precedent; Newbery (2023) establishes welfare dominance in the electricity context.

(4) **Capacity:** $\sigma^{\text{cap}}(\omega) = \kappa \cdot y$, a payment proportional to installed capacity, independent of realized output or price. This instrument corresponds to capacity market designs in the EU power sector.

The restriction to $\mathcal{S}$ is Assumption 11. It rules out fully state-contingent mechanisms (first-best mechanisms in the sense of a full-information revelation principle) in favor of instruments that appear in actual policy debate. The welfare comparison across elements of $\mathcal{S}$ is therefore a "best-in-policy-relevant-class" result, not a first-best characterization.

3.4. Payoffs

Producer operating profit per scenario. The producer's per-scenario gross operating profit at transfer form σ and output Q is

$$\pi^{\text{op}}(\sigma, y, Q; \omega) \;=\; \big[p_{\text{SAF}}(\omega) + \sigma(\omega)\big] \cdot Q(\omega) \;-\; C_{\text{op}}(Q; \omega), \tag{1}$$

where C_{op} is the stochastic operating cost (feedstock, H_2, and OPEX) inherited from Li and Yu (2026) §3.3.

Producer mean-CVaR operating objective. Given y and σ, the producer evaluates expected operating profit under the mean-CVaR functional of Maccheroni, Marinacci, and Rustichini (2006):

$$\text{Obj}^{\text{op}}_{\text{prod}}(y, \sigma) \;=\; (1 - \lambda_{\text{prod}})\,\mathbb{E}[\pi^{\text{op}}] \;+\; \lambda_{\text{prod}}\,\text{CVaR}_\alpha[\pi^{\text{op}}], \tag{2}$$

with CVaR at confidence level $\alpha = 0.95$ following the Rockafellar-Uryasev (2000) representation.

Social welfare. Per-scenario social welfare is

$$W(\sigma, y, Q; \omega) \;=\; \text{CS}(\omega) + \text{PS}(\sigma, y, Q; \omega) + \text{ENV}(Q) - T(\sigma, Q; \omega), \tag{3}$$

with consumer surplus CS, producer surplus PS, environmental benefit ENV, and transfer cost T.

Regulator mean-CVaR welfare objective. The regulator solves

$$\max_{\sigma \in \mathcal{S}} \; (1 - \lambda_{\text{planner}})\, \mathbb{E}[W] \; + \; \lambda_{\text{planner}}\, \text{CVaR}_\alpha[W] \tag{4}$$

subject to the producer's Stage-2 best response (Lemma A.4) and the producer's project-finance participation constraint (Assumption 13).

3.5. Producer Project-Finance NPV at Hurdle IRR

The structural innovation of this paper relative to Li and Yu (2026) is the project-finance participation constraint. The producer does not invest unless the facility clears the hurdle IRR r_h. Let u denote capacity utilization (gallons produced per gallon of annual capacity), $\text{AF}(r_h, T) \equiv \sum_{t=1}^{T}(1 + r_h)^{-t}$ the annuity factor at the hurdle rate over a T-year horizon, and κ_{CAPEX} the per-gallon-annual-capacity CAPEX intensity. The producer's per-period after-tax cash flow at scenario ω is

$$\text{CF}(\sigma, Q;\, \omega) \;=\; \big[p_{\text{SAF}}(\omega) + \sigma(\omega) - c_{\text{op}}(\omega)\big] \cdot Q(\omega) \cdot (1 - \tau) \;+\; \tau \cdot D_t, \tag{5}$$

where D_t is the straight-line depreciation deduction per period. The NPV per gallon of annual capacity is

$$\frac{\text{NPV}(\sigma;\, r_h, \kappa_{\text{CAPEX}}, u, T)}{\bar{Q}} \;=\; \text{AF}(r_h, T) \cdot \mathbb{E}_{\text{prod}}\left[\frac{\text{CF}(\sigma, Q^*;\, \omega)}{\bar{Q}}\right] \;-\; \frac{\kappa_{\text{CAPEX}}}{u}, \tag{6}$$

where $\mathbb{E}_{\text{prod}}$ is the producer's mean-CVaR evaluation (see Equation (2)) and $Q^*(\omega)$ is the Stage-2 best response. The producer invests if and only if the NPV is non-negative.

The *financial-equivalence subsidy* σ_{prod} is the minimum constant per-gallon transfer clearing the investment hurdle:

$$\text{NPV}(\sigma_{\text{prod}};\, r_h, \kappa_{\text{CAPEX}}, u, T) \;=\; 0. \tag{7}$$

Under the two-state collapsed regime and Assumptions A1–A13, Proposition A.3 in Section 4 derives σ_{prod} in closed form as the sum of three terms: a hurdle-IRR premium, a risk-aversion premium, and a subtractive operating-margin offset.

3.6. Assumptions

Assumptions A1–A8 are inherited from Li and Yu (2026) and retained without modification. They are restated briefly for completeness; full statements and proofs appear in Li-Yu (2026) §3.5.

assumption 1 (Two-state demand collapse)**.** *For closed-form propositions, $\omega \in \{H, L\}$ with $\pi_H, \pi_L > 0$ and $\pi_H + \pi_L = 1$. See Li and Yu (2026) Theorem 1. The collapse is validated at the full 248/34 SP-MILP instance by Paper 1's two-state injection test: re-solving the extensive form with the two-state distribution injected, the closed-form σ^* restores the risk-neutral capacity ordering exactly and undoes 94% of the risk-induced welfare distortion, so the two-state device preserves the objects this paper builds on.*

assumption 2 (Linear inverse demand)**.** *The inverse demand for SAF is* $P^d(Q;\,\omega) = a(\omega) - bQ$ *with* $b > 0$*. See Li and Yu (2026) Theorem 1.*

assumption 3 (Convex capacity cost)**.** *The producer's cost function satisfies* $C(Q) = (k/2)Q^2$ *with* $k > 0$ *in the representative-investor reduction. See Li and Yu (2026) Theorem 1.*

assumption 4 (State ordering)**.** $a_H > a_L$*. See Li and Yu (2026) Theorem 1.*

assumption 5 (Positive net premium at low state)**.** $a_L > c$*, where* c *is the marginal operating cost at the representative-investor level. See Li and Yu (2026) Theorem 1.*

assumption 6 (CVaR confidence)**.** $\alpha \in (0, 1)$*. Closed-form* σ^* *is piecewise on* $(0, \pi_H)$ *and* $[\pi_H, 1)$*. See Li and Yu (2026) Theorem 1.*

assumption 7 (Bounded risk-aversion weights)**.** $\lambda_{\text{prod}}, \lambda_{\text{planner}} \in [0, 1]$*. See Li and Yu (2026) Theorem 1.*

assumption 8 (Partial equilibrium)**.** *The SAF price intercept* $a(\omega)$ *is exogenous; there is no general-equilibrium substitution between SAF and fossil jet fuel. See Li and Yu (2026) Theorem 1.*

The following five assumptions are new to Paper 2.

assumption 9 (Irrevocable form-commitment)**.** *The regulator selects a* σ-form *from* $\mathcal{S}$ *at Stage 0b—linear, CfD-strike, CfD-with-floor, or capacity—and cannot switch the form ex post. Time-inconsistent renegotiation of the contractual structure is ruled out, but the* level *of any chosen form (e.g., the linear \$/gal rate) may be revised at the policy timescale.*

remark 3.1 (Form versus level commitment)**.** *Assumption 9 is form-commitment, not level-commitment, and this distinction is the load-bearing contractual primitive for the CfD-with-floor dominance result of Theorem B.3. The 45Z credit was enacted under the IRA (2022) as a* linear per-gallon form*, and subsequent revisions (H.R. 1, enacted July 2025, eliminated the SAF premium for fuel produced after December 31, 2025, aligning the SAF base at \$1.00/gal; the Q3 2026 CFR cycle is open) have adjusted the per-gallon* level *while preserving the linear* form*. The empirical H.R. 1 amendment pattern—level revision, form preservation—supports form-commitment at the SAF Grand Challenge horizon* $T = 10$.

assumption 10 (Public information)**.** *Both the regulator's risk-aversion weight* λ_{planner} *and the producer's risk-aversion weight* λ_{prod}*, hurdle IRR* r_h*, and CAPEX intensity* κ_{CAPEX} *are common knowledge. All producer financial primitives* $(e, r_e, r_d, \tau, \kappa_{\text{CAPEX}})$ *observed at Stage 0a are public—recovered from the systematic SAF TEA meta-analysis of Farooq et al. (2025) plus the after-tax WACC benchmarks of Lazard's 2024 LCOE+ analyses (Lazard 2024), which report financing costs for power generation and hydrogen (not SAF specifically) and are carried here as economy-wide clean-energy benchmarks. There is no hidden type and no screening: when an empirical literature pins a parameter that might otherwise be private, we treat it as recovered.*

assumption 11 (Restricted policy-instrument class)**.** *The regulator's Stage-0b action is restricted to* $\sigma \in \mathcal{S}$*, the four-element design space of Section 3.3. Unrestricted measurable transfers* $\sigma : \Omega \to \mathbb{R}_+$ *are excluded. Each element of* $\mathcal{S}$ *is a real-world contractual instrument with precedent in actual energy policy, so the welfare comparison is "best-in-policy-relevant-class."*

Statutory justification. *The restriction to $\mathcal{S}$ is not a modeling crutch but reflects the architecture of Internal Revenue Code (IRC) §45 production credits, of which 45Z is the SAF instance. IRC §45 prescribes a* per-unit-output *payment structure: the credit is computed as a dollar-denominated rate multiplied by qualified output (kWh for 45Y, kg for 45V, ton CO_2 for 45Q, gallon for 45Z). Two-part tariffs (a fixed Stage-1 payment F plus a linear σ_0/gal) are excluded from the §45 architecture: fixed grants are administered separately under DOE Title XVII loan guarantees or Section 1703 cost-share agreements, requiring a distinct statutory authority. Capacity-contingent payments $\sigma(y, \omega)$ are technically permitted under §45 but have no precedent in the IRA-era production credits; the IRS would require new regulations under Treasury Decision authority.*

CfD-strike and CfD-with-floor (the two state-contingent elements of $\mathcal{S}$) lie at the edge of §45's per-unit-output architecture. A pure shortfall payment $\max\{K - p_{\text{SAF}}, 0\}$ *remains a per-gallon rate, so it does not require fresh statutory authority; however, making the rate price-contingent rather than uniform constitutes a material departure from current IRS administration of §45Z and would require new Treasury regulations under existing §45Z(b) rate-setting authority. We include CfD-strike and CfD-with-floor in $\mathcal{S}$ on the grounds that (i) they preserve §45's per-gallon payment structure and therefore do not require congressional amendment, and (ii) they are the only state-contingent forms with electricity-sector precedent within OECD policy-debate literature (Newbery 2023); competing forms (e.g., state-contingent income-tax credits, swap-based hedge instruments) lie outside the IRC's production-credit framework. The four-element $\mathcal{S}$ therefore captures the contracting space available within the post-IRA §45 statutory and regulatory envelope (Congressional Research Service 2025, §5). Within this restricted envelope, the CfD-with-floor dominance result identifies the best feasible instrument, conditional on Congress not amending §45 to permit two-part or capacity-contingent forms. The associated welfare opportunity cost from Assumption 11 relative to a two-part-tariff first-best is the policy product the conclusion quantifies.*

If Assumption 11 is dropped and preferences are moved to CARA-Normal, the linear contract is optimal by Holmström and Milgrom (1987) Theorem 7; the restriction closes that escape.

assumption 12 (MCPF baseline)**.** *The marginal cost of public funds* $\mu = 1.20$ *in the headline welfare evaluation, taken as the midpoint of Dahlby's empirical* $\mu \in [1.10, 1.30]$ *range for the US federal tax system. The* $\mu = 1.20$ *headline is the AEJ:Policy-standard convention for welfare comparisons across fiscal-substitution instruments. The boundary case* $\mu = 1$ *(the Cullen (2013) convention that treats the production-credit transfer as welfare-neutral) is reported as a reference in Section 7.6; under* $\mu = 1$ *the CfD-with-floor dominance result reduces to weak Pareto with equality, and the policy headline (the sign-determinate but modest \$0.03–\$0.09/gal welfare gain) is absorbed into the cancellation.*

assumption 13 (Project-finance NPV at hurdle IRR)**.** *The producer evaluates investment at the project-finance hurdle IRR* $r_h = \text{WACC} + \delta$, *where* $\delta \geq 0$ *reflects the pioneer-plant equity premium and IRA sunset risk. The producer invests if and only if* $\text{NPV}(\sigma; r_h, \kappa_{\text{CAPEX}}, u, T) \geq 0$. *Calibration anchors:* $r_h = 0.125$ *with envelope* $[0.10, 0.15]$ *(DOE 2022 qualitative anchor; Lazard 2024 WACC plus Graham–Harvey hurdle-premium and NREL ATB 2024 quantitative cross-checks; Section 5.2),* $\kappa_{\text{CAPEX}} = \$11$*/gal annual capacity (≈ 1.2× FOAK premium over the \$9.31/gal-yr nth-plant anchor of Tan et al. 2015),* $u = 0.85$, $\tau = 0.21$, $T = 10$ *years. The planner's risk-neutral discount rate is* $r_{\text{RN}} \approx 0.03$.

remark 3.2 (Microfoundation of $r_h > r_{\text{RN}}$; the decentralized reading of Paper 1)**.** *The hurdle wedge* $r_h - r_{\text{RN}}$ *is the project-finance expression of the same incomplete-markets friction that Paper 1 states as its external-financing* *assumption (Li and Yu 2026, §2.6): the state-contingent claims that would let a single-asset pioneer project sell its low-price exposure do not exist at the relevant tenor. Liquid hedging tenor in the middle-distillate complex runs two to three years against a ten-year operating commitment on a twenty-year asset; the produced-fuel premium over jet-A has no settlement index and is pure basis risk; and a single-asset project company faces margin-liquidity constraints that make a decade-long short position infeasible at scale (Froot, Scharfstein, and Stein 1993; Holmström and Tirole 1998). Costly external finance raises the internal discount rate above the diversified financier's risk-neutral rate — precisely the* $r_h = \text{WACC} + \delta$ *behavior documented by Graham and Harvey (2001). This paper is thereby the* decentralized reading *that Paper 1's external-financing assumption anticipates: the tail-averse committing agent is the investor deciding irreversible capacity, evaluated against the risk-neutral planner's target. Two consequences discipline the welfare language used below. First, the welfare statements attached to* r_h *are* constrained-efficiency *statements — the planner cannot complete the missing market within* $\mathcal{S}$ *— not corrections of a Pigouvian externality; the first-best diagnostic of Theorem B.3 Step 4b is correspondingly a market-completion ceiling, not an achievable gain. Second, in the complete-markets benchmark the producer would hedge,* λ_{prod} *and* δ *would both collapse toward zero, and* σ_{prod} *would collapse toward the risk-neutral zero-profit level;* $r_h > r_{\text{RN}}$ *and* $\lambda_{\text{prod}} > 0$ *jointly measure the shadow price of the missing claims on the committing balance sheet.*

Microfoundation status. *Assumption 13 is a* behavioral *investment-decision rule consistent with the project-finance literature (Boomsma, Meade, and Fleten 2012) and with the survey evidence on US CFO hurdle-rate practices (Graham and Harvey 2001), rather than a derived optimality condition from a utility-primitive representative-investor problem. We treat NPV-binding-at-*r_h *as the producer's observed decision criterion, with* r_h *empirically identified via the three independent risk-premium sources of Section 5.2. A microfoundation deriving NPV-binding from a representative-investor mean-CVaR maximization with risk-averse equity holders is straightforward but adds notation without changing any closed-form result; we omit it to keep the focus on the policy product.*

4. Equilibrium Analysis and Empirical Calibration

4.1. Equilibrium Concept

The solution concept is subgame-perfect Stackelberg equilibrium with the regulator as leader. Backward induction applies across the four stages: Stage 2 (producer chooses output) → Stage 1 (producer chooses investment footprint) → Stage 0b (regulator commits σ-form) → Stage 0a (regulator observes financial primitives; no strategic choice). The equilibrium triple is $(\sigma^*, y^*(\sigma^*), Q^*(\sigma^*, y^*, \omega))$.

Because each element of $\mathcal{S}$ yields a convex per-scenario profit for the producer (Assumptions 2 and 3), the bilevel reduces to a single-level problem: the regulator's Stage-0b choice is well-posed given the producer's unique Stage-2 best response (Lemma A.4). The equilibrium triple (σ^*, y^*, Q^*) is characterized in closed form by three nested optimality conditions—producer's Stage-2 first-order condition, producer's Stage-1 NPV-binding condition, and the regulator's Stage-0b instrument choice—which we solve in sequence in the following subsections.

4.2. Stage 2: Producer's Interior Best Response

Under Assumptions 2, 3 and 5, the producer's per-scenario operating profit π^{op} is strictly concave in Q with $\partial^2\pi^{\mathrm{op}}/\partial Q^2 = -(b+k) < 0$. Lemma A.4 (in Section A) establishes that the unique interior best response is

$$Q^*(\sigma, y;\, \omega) \;=\; \frac{a(\omega) + \sigma - c}{b + k}, \tag{8}$$

which exists and is strictly positive by Assumption 5.

The linearity of Q^* in σ is the key structural property exploited in the closed-form derivation: it implies that the producer's risk-adjusted expected cash flow, and hence the NPV, is linear in σ, which pins σ_{prod} uniquely.

4.3. Stage 1: Producer's NPV-Binding Participation

Given the committed transfer form σ, the producer's Stage-1 investment decision involves two regimes.

In *Regime A (NPV-binding)*, the subsidy σ satisfies $\mathrm{NPV}(\sigma; r_h, \kappa_{\mathrm{CAPEX}}, u, T) = 0$: the producer's project-finance participation constraint binds, and the producer earns exactly the hurdle return on equity. At the empirical baseline calibration ($r_h = 0.125$, $\kappa_{\mathrm{CAPEX}} = \$11$/gal, $u = 0.85$, $\tau = 0.21$, $T = 10$) the NPV-binding credit is $\sigma_{\mathrm{prod}} = \$1.36$/gal, which the as-enacted CI-scaled band $[\$1.30, \$1.45]$/gal *straddles*: participation holds in the band's upper portion ($\sigma_0 \geq \$1.36$) and fails by up to \$0.06/gal in its lower portion, while the post-2025 level (\$0.74–\$0.83/gal) fails it outright (Remark A.6 of Section A). Statements below that evaluate "the observed credit" in Regime A refer to the band's NPV-binding upper portion.

In *Regime B (NPV-slack)*, σ strictly exceeds the binding threshold, so $\mathrm{NPV}(\sigma; r_h, \kappa_{\mathrm{CAPEX}}, u, T) > 0$ and the producer earns strictly positive NPV. Regime B is over-subsidized; the welfare comparison in Section 6 is sharpest in Regime A.

Incidence. The two regimes settle the incidence question Paper 1 defers to this paper (Li and Yu 2026). Under Assumption 8 the SAF price process $a(\omega)$ is exogenous, so no part of the credit shifts to fuel buyers through a price response; statutory and economic incidence coincide on the producer side, consistent with the full-pass-through benchmarks of Sallee (2011) and Weyl and Fabinger (2013). Within the producer side, the split is regime-dependent: in Regime A the marginal credit dollar relaxes the binding participation constraint — it buys entry, not rent, and equity holders earn exactly the hurdle return — while in Regime B every dollar above σ_{prod} accrues as producer rent. The as-enacted band straddling σ_{prod} therefore implies near-zero producer rent at the band's upper portion; the policy-relevant incidence statement is that the observed 45Z level, where it induced entry at all, was an entry price rather than a producer transfer.

The welfare-equivalence subsidy $\sigma^*_{\mathrm{planner}}$ inherited from Li and Yu (2026) is stated as Proposition A.1 in Section A: under Assumptions 1 to 8 in the collapsed CVaR regime $\alpha \geq \pi_H$,

$$\sigma^*_{\mathrm{planner}} \;=\; \lambda_{\mathrm{planner}} \cdot \pi_H \cdot (a_H - a_L) \;=\; \$0.52/\mathrm{gal} \tag{9}$$

at $\lambda_{\mathrm{planner}} = 0.5$, $\pi_H = 0.472$, $a_H - a_L = \$2.19$/gal. Two Paper-1 qualifiers travel with the benchmark. First, $\sigma^*_{\mathrm{planner}}$ is a *capacity-restoring* constrained-efficiency benchmark — the credit that undoes the risk-induced capacity distortion on the committing balance sheet under the incomplete-markets friction of Remark 3.2 — not a first-best welfare prescription. Second, the \$0.52 point inherits Paper 1's persistence caveat: the price sample is too short to pin the long-run persistence of the fuel-plus-RIN level, and refits give \$0.48/gal under weekly re-estimation and \$0.35/\$0.14/\$0.11 under 1-/5-/10-year averaging of the price process; the producer requirement σ_{prod} below is less exposed to this caveat because its dominant component does not depend on the price-risk primitives.

4.4. Central Result: Closed-Form Financial-Equivalence Subsidy

Proposition A.3 in Section A derives σ_{prod} from the producer's project-finance NPV = 0 condition. Under Assumptions 1 to 8 and Assumptions 9 to 13 in the collapsed CVaR regime $\alpha \geq \pi_H$, the unique per-gallon constant subsidy clearing the producer's investment hurdle is

$$\sigma_{\text{prod}} = \underbrace{\frac{\kappa_{\text{CAPEX}}}{u \cdot \text{AF}(r_h, T) \cdot (1-\tau)}}_{\substack{\text{Term 1:} \\ \text{hurdle-IRR premium}}} + \underbrace{\lambda_{\text{prod}} \cdot \pi_H \cdot (a_H - a_L)}_{\substack{\text{Term 2:} \\ \text{risk-aversion premium}}} - \underbrace{\left[\bar{p_{\text{SAF}}} - \bar{c_{\text{op}}}\right]}_{\substack{\text{Term 3:} \\ \text{operating-margin offset}}} . \tag{10}$$

At the empirical baseline ($r_h = 0.125$, $\text{AF}(0.125, 10) = 5.5364$, $\kappa_{\text{CAPEX}} = \$11/\text{gal}$, $u = 0.85$, $\tau = 0.21$, $\lambda_{\text{prod}} = 0.60$, $\pi_H = 0.472$, $a_H - a_L = \$2.19/\text{gal}$, $\bar{p_{\text{SAF}}} - \bar{c_{\text{op}}} = \$2.22/\text{gal}$), this evaluates to $\sigma_{\text{prod}} = \$1.36/\text{gal}$, inside the empirical 45Z band $[\$1.30, \$1.45]/\text{gal}$.

The three terms have distinct economic content. Term 1 is the per-gallon CAPEX recovery requirement at hurdle IRR r_h, adjusted for the tax shield. This term is absent from Li and Yu (2026) and from Cullen (2013)'s risk-neutral framework; it captures the financial-frictions premium that pioneer-plant SAF investors require above the risk-neutral social cost of capital. Term 2 is the Paper 1-style risk-aversion premium: the additional per-gallon payment the producer requires to accept exposure to the downside scenario under mean-CVaR preferences. Term 3 is the expected operating margin the producer earns from SAF sales without any subsidy, which offsets part of the CAPEX recovery requirement.

The annuity factor $\text{AF}(r_h, T)$ is monotone increasing concave in T and saturates at $1/r_h$, so Term 1 is monotone decreasing in T. Consequently, σ_{prod} is strictly monotone decreasing in T with attenuating slope (established in Proposition A.7); at $T = 5$ it reaches \$3.00/gal, and at $T = 20$ it falls to \$0.66/gal. The headline \$1.36/gal figure is the point estimate at the Paper-1 horizon $T = 10$. The horizon-dependence is a load-bearing policy lever: a 5-year credit horizon would require a per-gallon subsidy more than twice the 10-year level to clear the same pioneer-plant hurdle, so the statutory length of 45Z renewal cycles is a substitute, not a complement, for the credit rate itself.

4.5. Stage 0b: Regulator's σ-Form Choice

Having characterized the producer's Stage-1 and Stage-2 responses, the regulator's Stage-0b problem is to select the element of $\mathcal{S}$ that maximizes mean-CVaR welfare (4) subject to $\text{NPV}(\sigma; r_h, \kappa_{\text{CAPEX}}, u, T) \geq 0$. Because the producer's Stage-2 problem is convex for all four elements of $\mathcal{S}$ (Lemma A.4), the bilevel reduces to a single-level maximization over $\mathcal{S}$, and the regulator's problem is well-defined.

The optimal σ-form in $\mathcal{S}$ under Assumptions 1 to 13 is the CfD-with-floor. This result, stated formally as Theorem B.3 and proved in Section B, uses the Borch (1962)–Wilson (1968) risk-sharing construction under complete information (Assumption 10). The key step is that a transfer $\sigma(\omega)$ that concentrates payments in the low-price tail achieves two simultaneous efficiency gains: it equalizes the marginal utility ratio between the two agents (Borch-Wilson risk-sharing), and it reduces the expected fiscal cost of meeting the producer's NPV participation constraint relative to a constant per-gallon payment. Both effects are strictly positive at $\lambda_{\text{prod}} > \lambda_{\text{planner}} > 0$ and $r_h > r_{\text{RN}}$, which hold at the empirical baseline. The full welfare comparison—Components 1 and 2 of Theorem B.3—and the quantitative welfare wedge at the SE-US 248/34 instance are reported in Section 6.

5. The Wedge Decomposition

5.1. Closed-Form Two-Component Decomposition

The central empirical contribution of this paper is a two-component decomposition of the gap between the producer's financial-equivalence subsidy σ_{prod} and the planner's welfare-equivalence subsidy $\sigma^*_{\text{planner}}$. Proposition A.5 (proved in Section A) establishes that the wedge

$$\sigma_{\text{prod}} - \sigma^*_{\text{planner}} = \underbrace{(\lambda_{\text{prod}} - \lambda_{\text{planner}})\, \pi_H\, (a_H - a_L)}_{\text{Component 1: risk-aversion premium}} + \underbrace{\Delta\text{IRR}_{\text{prem}}}_{\text{Component 2: financial-frictions premium}}, \tag{11}$$

where the financial-frictions component takes the closed form

$$\Delta\text{IRR}_{\text{prem}} = \frac{\kappa_{\text{CAPEX}}}{u \cdot \text{AF}(r_h, T) \cdot (1-\tau)} - \left[\bar{p_{\text{SAF}}} - \bar{c_{\text{op}}}\right]. \tag{12}$$

At the baseline calibration ($\lambda_{\text{prod}} = 0.60, \lambda_{\text{planner}} = 0.50, \pi_H = 0.472, a_H - a_L = \$2.19/\text{gal}, \kappa_{\text{CAPEX}} = \$11/\text{gal}$ annual capacity, $u = 0.85$, $r_h = 0.125$, $T = 10$, $\tau = 0.21$, $\bar{p_{\text{SAF}}} - \bar{c_{\text{op}}} = \$2.22/\text{gal}$):

- $\text{AF}(0.125, 10) = 5.5364$, CAPEX recovery $= \$11/(0.85 \times 5.5364 \times 0.79) = \$2.96/\text{gal}$, so $\Delta\text{IRR}_{\text{prem}} = \$2.96 - \$2.22 = \$0.739/\text{gal}$ (Component 2);
- Component 1 $= (0.60 - 0.50) \times 0.472 \times \$2.19 = \$0.103/\text{gal}$;
- Total wedge $= \$0.103 + \$0.739 = \$0.842/\text{gal}$;
- $\sigma_{\text{prod}} = \sigma^*_{\text{planner}} + \$0.842 = \$0.517 + \$0.842 = \$1.359/\text{gal} \approx \$1.36/\text{gal}$, inside the empirical 45Z band $[\$1.30, \$1.45]/\text{gal}$.

The attribution at $T = 10$ is

$$\frac{\$0.103}{\$0.842} \approx 12.2\% \text{ (risk aversion)} \quad \text{and} \quad \frac{\$0.739}{\$0.842} \approx 87.8\% \text{ (financial frictions)}.$$

The wedge is overwhelmingly driven by financial frictions (87.8%) rather than producer risk aversion (12.2%) at the Paper-1 horizon $T = 10$. This 12.2%/87.8% attribution is a **point estimate at $T = 10$**; the attribution shifts strongly with T, as Section 7.1 establishes.

Figure 1 plots the decomposition graphically, with the total wedge, Component 1, and Component 2 displayed as stacked contributions at each horizon.

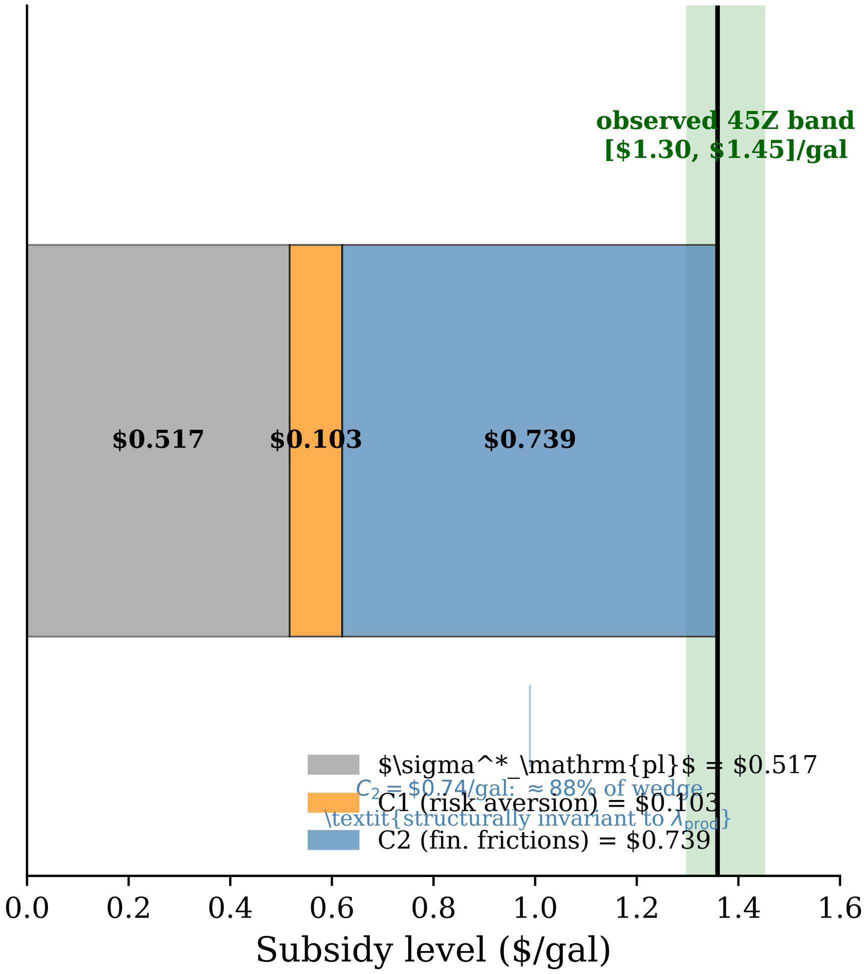

Figure 1: Wedge decomposition

Component 1 (risk-aversion premium) = $0.103/gal; Component 2 (financial-frictions premium) = $0.739/gal. At the baseline $T = 10$, $\sigma^*_{\text{planner}} = \$0.517$/gal and $\sigma_{\text{prod}} = \$1.359$/gal. The 87.8% financial-frictions share is a point estimate at $T = 10$; Table 2 reports the full T-sensitivity.

5.2. Calibration Sources for r_h

The hurdle IRR $r_h = 0.125$ is calibrated in two layers, and we state plainly that no agency document publishes a numeric SAF hurdle rate. The qualitative anchor is U.S. Department of Energy, U.S. Department of Transportation, and U.S. Department of Agriculture (2022): the SAF Grand Challenge Roadmap's commercial-deployment discussion documents the high first-of-a-kind (FOAK) cost of capital and its pioneer-vs.-nth-plant TEA workstream distinguishes the return FOAK equity investors require from the social cost of capital. The FAA Aviation Climate Action Plan (Federal Aviation Administration 2021) and the IATA Net Zero Roadmap (International Air Transport Association 2023) echo the financing-risk point qualitatively without reporting risk-premium figures. The quantitative anchors are Lazard (2024) after-tax WACC benchmarks (7.7–9.1%) plus the Graham and Harvey (2001) evidence that hurdle rates exceed WACC by several points, and the National Renewable Energy Laboratory (2024) 2024 Transportation Annual Technology Baseline capital-charge factor. The DOE-anchored calibration cross-checked against Lazard 2024 and NREL ATB 2024 yields $r_h \in [0.10, 0.15]$ with the point estimate $r_h = 0.125$ as the baseline. The remaining baseline parameters: $\lambda_{\text{prod}} = 0.60$, identified by revealed preference from the observed 45Z band conditional on r_h (Section B.1; the Farooq et al. (2025) meta-analysis anchors the TEA primitives but does not itself identify a CVaR weight); $\kappa_{\text{CAPEX}} = \$11$/gal annual capacity, a $\approx 1.2\times$ first-of-a-kind premium over the nth-plant thermochemical design anchor of \$9.31/gal-yr (Tan et al. 2015; Paper 1 §4.4 and its TEA comparison table); $u = 0.85$ (7440 operating hours per year); and $\tau = 0.21$ (effective federal corporate rate). The two-state price primitives ($\pi_H = 0.472$, $a_H - a_L = \$2.19$/gal, $\bar{p_{\text{SAF}}} = \$6.00$/gal) are inherited from Li and Yu (2026). The unconditional operating cost $\bar{c_{\text{op}}} = \$3.78$/gal is the full non-capital variable stack of the Paper 1 chain — stumpage, harvest, transport, depot operating cost, and refinery operating cost — not the refinery operating cost (\$1.96/gal) alone. Recomputing the stack from Paper 1's current solve outputs gives \$3.82/gal at the deterministic baseline (\$0.08 stumpage, \$0.30 harvest, \$0.57 transport, \$0.92 depot, \$1.96 refinery) and \$3.81/gal as the probability-weighted unconditional mean across the twenty price bins; we retain the calibration-vintage \$3.78 and note that substituting \$3.81 moves σ_{prod} by +\$0.03 (to \$1.39, still inside the observed band) and the Component-2 share by less than one percentage point — within every sensitivity reported below. All dollar figures in the calibration are 2022 USD: the Paper-1 cost chain is deflated to a 2022 basis (CPI factor 1.223 from the 2016-vintage TEA sources), and the price series is on the same common real basis.

5.3. Comparative Statics

Proposition A.7 (proved in Section A) establishes closed-form partial derivatives for σ_{prod} with respect to all primitives. Six partial derivatives are central to the empirical discussion.

(i) $\partial\sigma_{\text{prod}}/\partial\lambda_{\text{prod}} > 0$. A more risk-averse producer requires a higher subsidy. Equation (A.18) gives $\partial\sigma_{\text{prod}}/\partial\lambda_{\text{prod}} = \pi_H(a_H - a_L) = \$1.034$/gal per unit of λ_{prod}. At the headline $\lambda_{\text{prod}} = 0.60$, shifting to the band-image upper bound $\lambda_{\text{prod}} = 0.69$ raises σ_{prod} by \$0.09/gal; shifting to the lower bound $\lambda_{\text{prod}} = 0.54$ lowers it by \$0.06/gal. Figure 2 (a) traces σ_{prod} against λ_{prod} across the full $[0, 1]$ range, stacking the planner baseline, Component 1, and Component 2 contributions.

(ii) $\partial\sigma_{\text{prod}}/\partial r_h > 0$. A higher hurdle IRR requires a higher subsidy, because the CAPEX recovery term $\kappa_{\text{CAPEX}}/[u\,\text{AF}(r_h, T)(1-\tau)]$ rises as $\text{AF}(r_h, T)$ falls in r_h (Equation (A.19): $\partial\,\text{AF}/\partial r_h < 0$). Figure 2 (b) plots σ_{prod} against r_h over $[0.05, 0.20]$. The empirical range $[0.10, 0.15]$ maps to $\sigma_{\text{prod}} \in [\$1.07, \$1.66]$/gal, bracketing the observed 45Z band $[\$1.30, \$1.45]$/gal.

(iii) $\partial\sigma_{\text{prod}}/\partial T < 0$*, strictly monotone decreasing with attenuating slope, asymptotically zero as* $T \to \infty$. As T grows, $\text{AF}(r_h, T)$ rises concavely toward $1/r_h = 8.0$ (at $r_h = 0.125$), so the CAPEX recovery term falls and σ_{prod} declines. The rate of decline itself decreases: each additional year adds a progressively smaller increment to AF, so the marginal reduction in σ_{prod} attenuates toward zero. There is no sign change at any finite T. Figure 2 (c) confirms this numerically; the corresponding values are $\sigma_{\text{prod}} = \$3.00$/gal at $T = 5$, \$1.36/gal at $T = 10$, \$0.87/gal at $T = 15$, and \$0.66/gal at $T = 20$. Policy meaning: the statutory horizon of 45Z credits is a fiscal-equivalent lever to the credit rate itself; extending T from 10 to 15 years delivers a \$0.49/gal reduction in the required subsidy, comparable in magnitude to the H.R. 1 CI-scaled rate cut of \$0.56–\$0.62/gal.

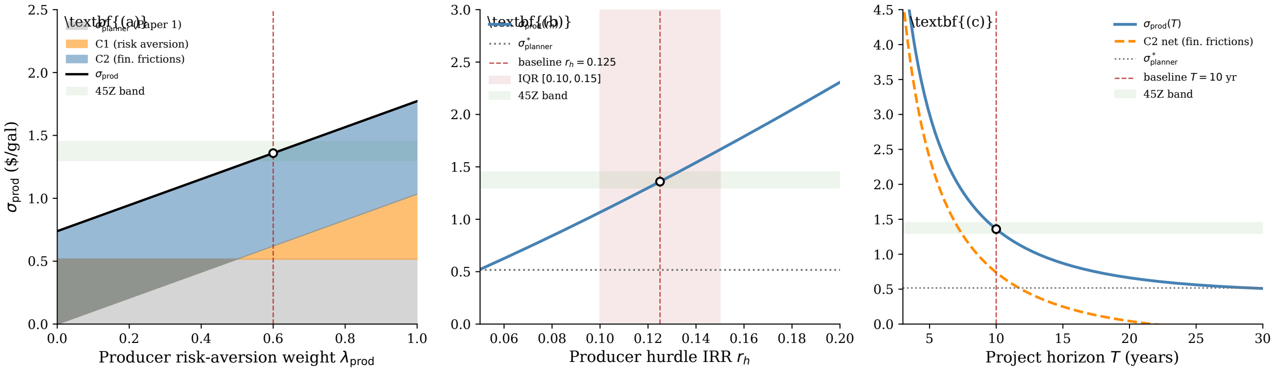

Figure 2: Comparative statics of σ_{prod}

(a) σ_{prod} as a function of the producer's mean-CVaR risk-aversion weight λ_{prod}; stacked contributions show the planner baseline $\sigma^*_{\text{planner}}$, Component 1 (risk aversion), and Component 2 (financial frictions); baseline $\lambda_{\text{prod}} = 0.60$. **(b)** σ_{prod} as a function of the producer's hurdle IRR r_h over the calibrated range $[0.10, 0.15]$ (shaded); baseline $r_h = 0.125$. **(c)** σ_{prod} and the net Component 2 as functions of the project horizon T; strictly monotone decreasing in T with attenuating slope; baseline $T = 10$, with the quantitative T-grid in Table 2. All other parameters are at baseline; the $[\$1.30, \$1.45]$/gal 45Z band is shaded green in each panel.

(iv) $\partial\sigma_{\text{prod}}/\partial\kappa_{\text{CAPEX}} > 0$. Higher CAPEX intensity raises σ_{prod} linearly at the rate $1/[u\ \mathrm{AF}(r_h, T)\ (1-\tau)]$; see Equation (A.20). The financial-frictions wedge is therefore entirely determined by the CAPEX recovery requirement relative to the annuity-discounted operating margin. Policy meaning: pioneer-plant learning that reduces κ_{CAPEX} is welfare-equivalent to a credit-rate increase at the same fiscal cost; DOE Loan Programs Office Title XVII guarantees that lower effective κ_{CAPEX} act as substitutes for σ_{prod} increases.

(v) $\partial\sigma_{\text{prod}}/\partial\tau > 0$. A higher effective tax rate raises σ_{prod} with coefficient $\kappa_{\text{CAPEX}}/[u\ \mathrm{AF}(r_h, T)\ (1-\tau)^2] > 0$; see Equation (A.21). The depreciation tax shield reduces the after-tax CAPEX cost but does not fully offset the tax-rate increase on operating income, so the net effect on the required subsidy is positive. Policy meaning: depreciation-convention reform (full-expensing versus MACRS) maps into the closed-form as a shift in the effective τ — the substantive tax lever for the 45Z required-subsidy level is the depreciation rule, not the statutory rate.

(vi) $\partial\sigma_{\text{prod}}/\partial u < 0$. Higher capacity utilization lowers σ_{prod}: spreading the fixed CAPEX over more gallons reduces the per-gallon recovery requirement (Equation (A.22)). The $1/u$ scaling is exact in the closed form. Policy meaning: federal feedstock-supply guarantees that protect realized utilization u against residue shortfalls act as substitutes for σ_{prod} increases — for an FT-residue pioneer plant the $1/u$ scaling implies a 20 pp utilization shortfall (from $u = 0.85$ to $u = 0.65$) inflates the CAPEX-recovery term by $\approx 31\%$ and the required subsidy from \$1.36 to \$2.27/gal — a two-thirds increase.

Section 6 turns to the instrument-form question: whether the linear 45Z structure is welfare-optimal within the statutory class $\mathcal{S}$, or whether a CfD with break-even floor strictly dominates.

6. The CfD-with-Floor Dominance Result

Under complete information (Assumption 10), restricted policy-instrument class (Assumption 11), and project-finance NPV binding (Assumption 13), Theorem B.3 (proved in Section B) establishes that the CfD-with-floor instrument

$$\sigma^*(\omega) \;=\; \max\{\,\bar{p} - p_{\mathrm{SAF}}(\omega),\, 0\,\} \;+\; s_0^* \tag{13}$$

weakly Pareto dominates the linear credit $\sigma^{\mathrm{lin}} = \$1.36$/gal at the matched producer-NPV-binding comparison, *within the four-element statutorily relevant instrument class* $\mathcal{S}$; the dominance is strict at marginal cost of public funds $\mu_W > 1$ via the MCPF transfer-differential channel (Corollary A.10). The welfare wedge against the linear baseline takes the form

$$\Delta W_{\mathrm{CfD\,vs\,lin}} \;=\; \underbrace{(\mu_W - 1)\cdot\lambda_{\mathrm{prod}}\cdot\pi_H\cdot(\bar{p} - a_L)\cdot\mathbb{1}_{\{\mathrm{NPV\text{-}binding}\}}\cdot Q^*}_{\text{MCPF transfer-differential channel}} \;+\; O\big((\mu_W - 1)^2\big), \tag{14}$$

where $\mathbb{1}_{\{\mathrm{NPV\text{-}binding}\}}$ indicates that the producer's NPV constraint binds at the linear credit rate and μ_W is the marginal cost of public funds. At Dahlby's empirical US-federal range $\mu_W \in [1.10, 1.30]$ and baseline ($\lambda_{\mathrm{prod}} = 0.60, \pi_H = 0.472, \bar{p} - a_L = \$1.03/\mathrm{gal}$), $\Delta W = (\mu_W - 1)\cdot \$0.292/\mathrm{gal} \in [\$0.029, \$0.088]/\mathrm{gal}$ of producer output in Regime A.

Regime A (NPV-binding). At MCPF = 1, the dominance is weak (equality): producer-surplus and fiscal-transfer contributions cancel under common-measure planner welfare evaluation. At $\mu_W > 1$, strict Pareto dominance holds via the MCPF channel, with magnitude $(\mu_W - 1)\cdot \$0.292/\mathrm{gal} \in [\$0.029, \$0.088]$ over the empirical range.

Regime B (NPV-slack). The MCPF channel vanishes ($\mathbb{1}_{\{\mathrm{NPV\text{-}binding}\}} = 0$). CfD-with-floor is welfare-equivalent to linear σ at all μ_W.

The opportunity cost of Assumption 11. The CfD-with-floor result above is the best feasible welfare outcome *within* the restricted policy class $\mathcal{S}$. To make the welfare cost of that restriction transparent, define the *first-best fiscal-slack diagnostic* (Theorem Step 4b of Section B): the counterfactual welfare gain a planner could capture by directly substituting her risk-neutral discount rate r_{RN} for the producer's hurdle r_h in the project-finance constraint. Algebraically:

$$\Delta W_{\text{first-best}} = \frac{\kappa_{\mathrm{CAPEX}}}{u}\cdot\frac{\mathrm{AF}(r_{\mathrm{RN}}, T) - \mathrm{AF}(r_h, T)}{\mathrm{AF}(r_h, T)} \;=\; \$7.00 \text{ per gallon of annual capacity}$$

at the empirical baseline — a stock magnitude over the asset life whose flow equivalent, converted through the annuity factor and tax gross-up as in Term 1 of Proposition A.3, is $\approx \$1.04$/gal of annual output. Two interpretive points: (i) this is *not* an achievable welfare gain within $\mathcal{S}$—a fully state-contingent transfer would be required, which is excluded by Assumption 11; (ii) the gap between the $\approx \$1.04$/gal flow-equivalent first-best ceiling and the \$0.029–\$0.088/gal CfD-with-floor strict-dominance gain is the policy cost of restricting attention to the four statutorily relevant instrument forms. The cost is roughly an order of magnitude, and it provides the welfare benchmark for the \$1.70 B/year Title XVII fiscal-substitution proposal of Section 8, which attacks the project-finance hurdle through a non-$\mathcal{S}$ instrument (federal loan guarantee that compresses $r_h \to r_{\mathrm{RN}}$) rather than through $\mathcal{S}$-internal contract design.

The proof technique follows Borch (1962) and Wilson (1968) under complete information and discrete-scenario

Table 2: T-sensitivity of the wedge decomposition

T	$AF(r_h, T)$	σ_{prod}	Decomposition ($/gal)		Attribution share		H.R.1 gap
(yr)		($/gal)	C_1 (risk)	C_2 (fin.)	Risk aversion	Fin. frictions	($1.00−$\sigma_{\text{prod}}$)
5	3.5606	3.001	0.103	2.381	4.2%	95.8%	−2.00
10	5.5364	1.359	0.103	0.739	12.2%	87.8%	−0.36
15	6.6329	0.870	0.103	0.250	29.3%	70.7%	+0.13
20	7.2414	0.662	0.103	0.042	71.0%	29.0%	+0.34

Component 1 (risk aversion) is invariant to T; Component 2 (financial frictions) declines strictly in T as $\text{AF}(r_h, T)$ grows toward $1/r_h = 8.0$. Baseline $T = 10$ is the bold row; all other parameters are at baseline ($r_h = 0.125$, $\kappa_{\text{CAPEX}} = \$11/\text{gal}$, $u = 0.85$, $\tau = 0.21$, $\lambda_{\text{prod}} = 0.60$, $\lambda_{\text{planner}} = 0.50$, $\pi_H = 0.472$, $a_H - a_L = \$2.19/\text{gal}$), and $\sigma^*_{\text{planner}} = \$0.517/\text{gal}$ (Paper 1, invariant to T). Component 1 $= (\lambda_{\text{prod}} - \lambda_{\text{planner}})\,\pi_H\,(a_H - a_L) = \$0.103/\text{gal}$, invariant to T because none of λ_{prod}, λ_{planner}, π_H, $a_H - a_L$ depends on the project horizon. Component 2 (net) $\equiv \sigma_{\text{prod}} - \sigma^*_{\text{planner}} - C_1$ is the residual financial-frictions contribution.

mean-CVaR preferences. Concentrating the transfer in the low-price tail where the producer's marginal utility of income is highest—the Borch-Wilson risk-tolerance-weighted-transfer condition—achieves two simultaneous efficiency gains in Regime A: equalizing the agents' marginal-utility ratio and reducing the expected fiscal cost of meeting the NPV participation constraint. The Holmström and Milgrom (1987) Theorem 7 result on linear-contract optimality is the failure-mode benchmark: it applies under unrestricted σ and CARA-Normal preferences, both of which are inconsistent with the mean-CVaR framework and the policy-instrument restriction of Assumption 11.

The MCPF corollary (Corollary A.10) sharpens the dominance result: when MCPF $\mu > 1$, the CfD-with-floor further underspends the linear credit in expected fiscal outlay ($\mathbb{E}[T_{\text{CfD}}] < \mathbb{E}[T_{\text{lin}}]$), strictly widening $\Delta W_{\text{CfD vs lin}}$.

7. Robustness

7.1. Project-Horizon Sensitivity

The 12.2%/87.8% risk-aversion/financial-frictions attribution from Section 5.1 is a **point estimate at the Paper-1 horizon** $T = 10$. Table 2 reports the full decomposition across $T \in \{5, 10, 15, 20\}$ years.

The attribution shifts strongly with T. At short horizons ($T \le 10$), financial frictions dominate: the hurdle-IRR premium is large per year because the annuity factor $\text{AF}(r_h, T)$ is far below its saturation level $1/r_h$, so the CAPEX recovery requirement per gallon is high. At $T = 5$, Component 2 accounts for 95.8% of the total wedge. As T grows, $\text{AF}(r_h, T)$ rises concavely, the CAPEX recovery term falls, and Component 2 shrinks toward zero; simultaneously, Component 1 remains constant at $0.103/gal. The attribution transitions from C2-dominant to C1-dominant between $T = 15$ (29.3% risk-aversion share) and $T = 20$ (71.0% risk-aversion share). The 12.2%/88% split holds only at $T = 10$ and does not define a robustness band; the correct interpretation is that the reported split is the empirical answer at the Paper-1 NPV horizon.

The defense of $T = 10$ as the baseline rests on three external anchors: (i) the IRS 15-year alternative straight-line for Asset Class 49.5 (biomass-to-liquid), which brackets the 10-year evaluation window from above; (ii) U.S. Department of Energy, U.S. Department of Transportation, and U.S. Department of Agriculture (2022) NREL FOAK cost projections and DOE Title XVII loan-guarantee tenors converge on 10 years as the conservative first-of-kind benchmark; (iii) Li and Yu (2026) adopt the same $T = 10$ window for the SE-US 248/34 instance, providing the natural Paper-1 comparison horizon. The 45Z statutory credit expires in 2029 (extended from the original 2027 sunset by the 2025 reconciliation act) with extension subject to a 5-to-10 year rulemaking cycle, providing the upper-bound rationale: the producer planning horizon for a FOAK commitment is bounded by policy uncertainty, and 10 years is a plausible upper bound for a locked-in investment. The tension runs the other way as well: a 2029 sunset leaves roughly three years of credited output for a plant commissioning in 2026–27, so evaluating a 10-year credit stream assumes renewal. If the credit instead lapses in 2029 and the producer anticipates it, the horizon relevant to the CAPEX recovery term is the credit window rather than the asset life, pushing the required rate toward the $T = 5$ value of \$3.00/gal; the \$1.36/gal headline is conditional on the renewal expectation embedded in the $T = 10$ window.

7.2. λ_{prod} Sensitivity

The producer risk-aversion weight λ_{prod} is identified by revealed preference: inverting the NPV-binding identity at the observed 45Z band $[\$1.30, \$1.45]$/gal, conditional on the calibrated hurdle $r_h = 0.125$, yields the band image $\lambda_{\text{prod}} \in [0.54, 0.69]$ with $\lambda_{\text{prod}} = 0.60$ at the band midpoint (Section B.1; the TEA meta-analysis of Farooq et al. 2025 anchors the project-finance primitives but does not itself identify a CVaR weight). Substituting nearby grid values into Proposition A.3 while holding all other parameters at baseline:

$$\lambda_{\text{prod}} = 0.54 \implies \sigma_{\text{prod}} = \$1.297/\text{gal}; \qquad \lambda_{\text{prod}} = 0.69 \implies \sigma_{\text{prod}} = \$1.452/\text{gal}.$$

Because λ_{prod} is recovered from the band, the map returns the band by construction; the content of this subsection is the slope, not the level match. Figure 2 (a) traces the full λ_{prod}-response curve.

The sensitivity is linear with slope $\pi_H(a_H - a_L) = 0.472 \times \$2.19 \approx \$1.03$/gal per unit change in λ_{prod} (Equation (A.18)). The 0.15-unit band-image width therefore induces a \$0.16/gal range in σ_{prod}; the far larger sensitivity is through the hurdle conditioning (Section B.1, Step 4: the r_h envelope moves the implied λ_{prod} by $\approx \pm 0.29$, straddling λ_{planner}).

A second conditioning channel is credit monetization. The identification reads the observed band as face value. FOAK SAF developers rarely have tax appetite in early operating years and typically monetize 45Z by selling the credit under IRC §6418 (United States Code 2025); 2025 market intelligence puts 45Z transfer pricing at roughly 91 cents per dollar of face value (Crux Climate 2025). Reading the band net of the transfer discount ($0.91 \times [\$1.30, \$1.45] = [\$1.18, \$1.32]$/gal) shifts the implied weight down to $\lambda_{\text{prod}} \in [0.43, 0.56]$ — a downward shift of ≈ 0.12, comparable to the band-image width itself. The face-value reading is the conservative baseline for the headline attribution: under the transfer-discount reading the risk-aversion component shrinks toward zero (the lower band edge implies $\lambda_{\text{prod}} < \lambda_{\text{planner}}$), strengthening the financial-frictions attribution further.

7.3. r_h Sensitivity

The hurdle IRR $r_h = 0.125$ is the midpoint of the calibrated envelope $[0.10, 0.15]$ (Section 5.2: Lazard after-tax WACC benchmarks plus the hurdle-premium evidence of Graham and Harvey 2001; agency documents discuss

SAF financing risk qualitatively only). From Proposition A.7, σ_{prod} is monotone increasing in r_h:

$$r_h = 0.10 \;\Rightarrow\; \sigma_{\text{prod}} \approx \$1.07/\text{gal}; \qquad r_h = 0.15 \;\Rightarrow\; \sigma_{\text{prod}} \approx \$1.66/\text{gal}.$$

The observed 45Z band $[\$1.30, \$1.45]$/gal corresponds to $r_h \in [0.120, 0.133]$, a subset of the calibrated envelope. Figure 2 (b) shows the full r_h-response curve. The financial-frictions wedge accounts for the bulk of the r_h-sensitivity because Component 1, $(\lambda_{\text{prod}} - \lambda_{\text{planner}})\,\pi_H\,(a_H - a_L)$, contains no r_h dependence: the entire \$0.60/gal range across $[0.10, 0.15]$ is driven by movement in Component 2.

7.4. 20-bin KDE Verification of the Two-State Collapse

The closed-form decomposition of Proposition A.3 and the 12%/88% attribution rest on the two-state collapse $\{H, L\}$ of the SAF-price distribution inherited from Li and Yu (2026) Theorem 1, calibrated to match the break-even-conditional means of the 20-bin Gaussian-KDE distribution fit on the constructed drop-in fuel-plus-RIN price series (slate-weighted wholesale gasoline–diesel plus D-code RIN at the 1.7 equivalence factor, August 2013–December 2021 weekly; Section 3.1). A natural concern is whether the \$1.36/gal headline and the 12%/88% split survive direct computation under the 20-bin primitive. Table 8 in Section B.1 reports the comparison.

The verification confirms one strong invariance and two model-relevant sensitivities. The invariance: *Component 2 is structurally independent of the discretization choice.* The financial-frictions premium equals \$0.736/gal under both the two-state collapse and the 20-bin KDE—no tail-distribution primitive enters $\Delta\text{IRR}_{\text{prem}}$. The two sensitivities: (i) at fixed $\lambda_{\text{prod}} = 0.60$, the closed-form σ_{prod} shifts from \$1.36/gal (two-state, in the empirical 45Z band) to \$1.92/gal (20-bin KDE, outside the band), because the 20-bin tail at $\alpha = 0.95$ has a wider mean-CVaR gap (\$1.97/gal) than the two-state $\bar{p} - a_L$ gap (\$1.03/gal); (ii) the revealed-preference λ_{prod} that rationalizes the observed $\sigma^{\text{obs}}_{\text{prod}} = \$1.36$/gal under the 20-bin primitive is $\lambda_{\text{prod}} = 0.32$, below the planner's $\lambda_{\text{planner}} = 0.50$. The discretization-dependence of the implied λ_{prod} is the same identification fragility documented in the calibration appendix; it does not, however, change the qualitative ordering of the wedge components: financial-frictions Component 2 (\$0.74/gal) dominates risk-aversion Component 1 ($\leq$ \$0.20/gal) under either discretization, and the C_2-share remains above 70% across the relevant range. The two-state-vs-20-bin comparison reads as a model selection question—the two-state collapse is the closed-form-clean calibration device of Li and Yu (2026), the 20-bin KDE is the discretization Paper 1's empirical exhibit operates on—rather than a verdict on the headline policy conclusion.

7.5. Sensitivity to the Planner's λ_{planner}

The headline calibration $\lambda_{\text{planner}} = 0.50$ is the midpoint of the Arrow and Lind (1970) risk-neutrality benchmark ($\lambda = 0$, appropriate for a social planner pooling risk across the entire federal balance sheet) and the Gollier (2001) social risk-aversion upper bound ($\lambda \approx 1$, appropriate when infrastructure investment risks cannot be socially diversified). Table 3 reports the wedge decomposition across $\lambda_{\text{planner}} \in \{0, 0.25, 0.50\}$. Component 2 is invariant to λ_{planner} (no λ_{planner}-dependence enters $\Delta\text{IRR}_{\text{prem}}$); Component 1 scales linearly in λ_{planner} with coefficient $-\pi_H(a_H - a_L) = -\$1.03$/gal. The Component-2 share of the wedge is 54% at the Arrow–Lind risk-neutral endpoint $\lambda_{\text{planner}} = 0$, 67% at $\lambda_{\text{planner}} = 0.25$, and 88% at the headline $\lambda_{\text{planner}} = 0.50$: the financial-frictions *level* (\$0.739/gal) is invariant and Component 2 remains the majority component throughout, but the 88%

Table 3: Wedge decomposition under alternative planner calibrations

λ_{planner}	$\sigma^*_{\text{planner}}$ ($/gal)	C_1 ($/gal)	C_2 ($/gal)	Wedge ($/gal)	C_2-share
0.00	0.000	+0.620	+0.739	+1.359	54%
0.25	0.259	+0.362	+0.739	+1.100	67%
0.50	**0.517**	**+0.103**	**+0.739**	**+0.842**	**88%**

Computed at $\lambda_{\text{prod}} = 0.60$, $r_h = 0.125$, $T = 10$, with the other primitives at baseline. Component 2 is structurally invariant to λ_{planner}; Component 1 scales linearly in $\lambda_{\text{prod}} - \lambda_{\text{planner}}$. The headline $\lambda_{\text{planner}} = 0.50$ (bold row) is the Arrow–Lind/Gollier midpoint.

share is specific to the $\lambda_{\text{planner}} = 0.50$ normative calibration — at lower λ_{planner} more of the wedge is relabeled risk-aversion because the planner's own benchmark $\sigma^*_{\text{planner}}$ shrinks.

7.6. MCPF Reference: $\mu = 1$

Assumption 12 sets the marginal cost of public funds $\mu = 1.20$ as the headline (midpoint of Dahlby's empirical $[1.10, 1.30]$ range), with $\mu = 1$ retained as a reference under the Cullen (2013) convention that treats the production-credit transfer as welfare-neutral. The two cases differ in the strict-vs-weak verdict of Theorem B.3, not in the wedge decomposition itself.

Under the headline $\mu = 1.20$, Corollary A.10 establishes that the CfD-with-floor welfare advantage is strictly positive, with magnitude $(\mu_W - 1)\cdot\lambda_{\text{prod}}\pi_H(\bar{p} - a_L)\cdot\mathbb{1}_{\{\text{NPV-binding}\}}\cdot Q^* = \$0.058/\text{gal}$ (= $0.20\times\$0.292/\text{gal}$). Under the reference $\mu = 1$, the advantage is weak with equality. Either way, the wedge decomposition Components 1 and 2 are unchanged because they are properties of σ_{prod} relative to $\sigma^*_{\text{planner}}$ at the empirical calibration, not properties of the welfare comparison.

The mechanism: the CfD-with-floor pays only in low-price scenarios, so its expected fiscal outlay $\mathbb{E}[T_{\text{CfD}}]$ is strictly lower than the linear credit's $\mathbb{E}[T_{\text{lin}}] = \sigma_{\text{prod}} \cdot Q^*$; the $\mu > 1$ multiplier is the *only* channel that converts this underspending into a welfare gain — at $\mu = 1$ the producer-surplus and fiscal-transfer contributions cancel and the comparison is welfare-neutral (Theorem B.3). The 87.8%/12.2% decomposition is unaffected by the choice of μ; what $\mu > 1$ changes is the welfare verdict, and the channel it works through is the risk-*aversion* side: the fiscal-cost differential $\lambda_{\text{prod}}\pi_H(\bar{p} - a_L)Q^*$ exists only because $\lambda_{\text{prod}} > 0$ makes the state-contingent reallocation cheap for the planner, while the financial-frictions component sets the level of the required credit but drops out of the contract comparison at matched NPV-binding.

7.7. Robustness to Private Hurdle IRR (Asymmetric Information)

Assumption 10 (public information on r_h and λ_{prod}) is the load-bearing primitive of Theorem B.3. We quantify the dominance-gap shrinkage when r_h is private to the producer. Let r_h be drawn from a distribution G with mean $\bar{r}_h = 0.125$ and standard deviation $\sigma_r \in [0, 0.025]$; the empirical DOE-anchored range $[0.10, 0.15]$ (U.S. Department of Energy, U.S. Department of Transportation, and U.S. Department of Agriculture 2022; Lazard 2024; National Renewable Energy Laboratory 2024) corresponds to $\sigma_r = 0.014$ under uniform G.

Under the restricted instrument class $\mathcal{S}$ (Assumption 11) the planner offers an anonymous σ-form, which must clear the worst-case hurdle for universal participation. A first-order expansion of σ_{prod} around $\bar{r}_h$ at the

Table 4: Component 2 attribution share under private hurdle IRR.

σ_r	r_h range (uniform G)	Leakage ($/gal)	C_2 effective ($/gal)	C2 share	vs. 75% threshold
0 (public)	$\{0.125\}$	$0.000	$0.739	87.8%	+12.8pp
0.005	$[0.116, 0.134]$	$0.104	$0.635	86.0%	+11.0pp
0.010	$[0.108, 0.142]$	$0.207	$0.532	83.8%	+8.8pp
0.014 (empirical)	$[0.100, 0.150]$	$0.296	$0.443	81.1%	+6.1pp
0.020	$[0.090, 0.160]$	$0.414	$0.325	75.9%	+0.9pp
0.021 (threshold)	$[0.089, 0.161]$	$0.435	$0.304	75.0%	0.0pp
0.025	$[0.082, 0.168]$	$0.518	$0.221	68.2%	−6.8pp

uniform worst-case quantile $\bar{r}_h + \sqrt{3}\,\sigma_r$ gives the linearized informational leakage $\sqrt{3}\,\sigma_r \cdot \partial\sigma_{\text{prod}}/\partial r_h$; at the empirical baseline $\partial\sigma_{\text{prod}}/\partial r_h|_{\bar{r}_h} \approx 11.96$/unit, yielding a leakage rate of $20.7/gal per unit σ_r.

The effective Component 2 attribution share under private-information leakage is

$$\text{(15)} \qquad \text{C}_2\text{-share}^{\text{eff}}(\sigma_r) = \frac{C_2 - \text{Leakage}(\sigma_r)}{C_1 + C_2 - \text{Leakage}(\sigma_r)}, \qquad \text{Leakage}(\sigma_r) = \sqrt{3}\,\sigma_r \cdot \left.\frac{\partial\sigma_{\text{prod}}}{\partial r_h}\right|_{\bar{r}_h}.$$

Table 4 reports the C2-share decay across σ_r at the empirical baseline.

The 75% C2-share threshold is reached at $\sigma_r \approx 0.021$ (corresponding to a ±3.6pp uncertainty range in r_h), **50% wider than the** $\sigma_r = 0.014$ **implied by reading the calibration envelope** $[0.10, 0.15]$ **as uniform**. Equivalently, the variance of private r_h would have to inflate by a factor $(0.021/0.014)^2 = 2.25\times$ before the headline 12.2%/87.8% attribution qualitatively weakens.

At the calibration-envelope private-r_h uncertainty ($\sigma_r = 0.014$, the $[0.10, 0.15]$ envelope of Section 5.2 read as a uniform distribution), the C2 attribution share is 81%, well above the 75% threshold. The CfD-with-floor dominance result is robust to plausible private-information frictions in r_h within the empirical calibration envelope. Equivalently, the headline 87.8% financial-frictions share is robust to relaxing Assumption 10 as long as the variance of private r_h remains within roughly 2.25× the DOE-anchored range. Heterogeneous-investor screening with strictly private r_h (direct-mechanism design under the Revelation Principle) is deferred to Paper 3 follow-up.

7.8. Depreciation Tax-Shield Convention

Proposition A.3 Step 4 adopts the TEA convention used by Farooq et al. (2025) and Lazard (2024) of evaluating CAPEX recovery on a pre-depreciation-shield basis, treating τD_t as recovered fully in the first period and absorbed into κ_{CAPEX} at the calibration step. Under this convention, the closed form is $\sigma_{\text{prod}} = \kappa_{\text{CAPEX}}/[u\ \text{AF}(r_h, T)\,(1-\tau)] + \lambda_{\text{prod}}\,\pi_H\,(a_H - a_L) - (\bar{p} - \bar{c}_{op})$, yielding $\sigma_{\text{prod}} \approx \$1.36$/gal at the empirical baseline.

A rigorous derivation that carries the depreciation tax shield through the annuity factor without first-period absorption yields, under straight-line depreciation $D_t = \kappa_{\text{CAPEX}}\,\bar{Q}/(u\,T)$,

$$\sigma_{\text{prod}}^{\text{rigorous}} = \frac{\kappa_{\text{CAPEX}}}{u\ \text{AF}(r_h,T)\,(1-\tau)} \cdot \left[1 - \frac{\text{AF}(r_h,T)\,\tau}{T}\right] + \lambda_{\text{prod}}\,\pi_H\,(a_H - a_L) - (\bar{p} - \bar{c}_{op}). \tag{16}$$

At the empirical baseline ($\tau = 0.21$, $T = 10$, $\text{AF}(r_h,T) = 5.5364$): AF $\tau/T = 5.5364 \cdot 0.21/10 = 0.1163$, so the bracketed factor is 0.884, and the hurdle-IRR-premium term contracts from \$2.959/gal to $\$2.959 \cdot 0.884 =$ \$2.616/gal. Substituting: $\sigma_{\text{prod}}^{\text{rigorous}} = 2.616 + 0.620 - 2.22 = \$1.02$/gal, falling *below* the empirical 45Z band $[\$1.30, \$1.45]$/gal.

The headline empirical band match of Proposition A.3 therefore holds under the absorption convention used throughout this paper but *not* under the rigorous depreciation derivation. We are explicit about what does and does not defend the choice. The absorption convention follows the peer-reviewed TEA practice our calibration inherits (Farooq et al. 2025); industry LCOE tools such as Lazard's model depreciation explicitly (MACRS), so they are authority for the *rigorous* treatment, not for absorption, and we do not claim otherwise. Front-loaded depreciation under the MACRS 7-year class (IRS Asset Class 49.5) delivers approximately 56% of the discounted tax shield in the first three years and 90% in the first five (14.29%/24.49%/17.49% Year 1/2/3 percentages discounted at $r_h = 12.5\%$), so absorption-into-CAPEX is a closed-form proxy for a genuinely front-loaded amortization pattern — but a proxy that overstates the shield, and the \$0.34/gal gap between the conventions is first-order: it exceeds the \$0.30 half-width of the r_h-envelope effect (Section 7.3), and \$1.02 falls below the observed band rather than inside any reported interval. The convention therefore changes the reading of the statute: under absorption the as-enacted band matches the producer's requirement; under MACRS-rigorous depreciation the band *exceeds* it by \$0.28–\$0.43/gal — the direction of the companion planner-side analysis (Li and Yu 2026), which finds the as-enacted credit above its welfare benchmark. We report both values throughout the policy discussion and carry the convention as a first-order caveat rather than defending it by containment.

7.9. SAF Pathway Heterogeneity in the Observed 45Z Band

The empirically observed 45Z band $[\$1.30, \$1.45]$/gal reflects *Carbon-Intensity (CI) heterogeneity across SAF production pathways*, not stochastic noise around a single anchor. 45Z is a CI-sliding-scale credit: under IRS Notice 2025-11 (Internal Revenue Service 2025), the per-gallon credit varies with the lifecycle CI score of the fuel under the notice's designated methodologies (for SAF, either CORSIA or the 45ZCF-GREET model at the producer's election; 45ZCF-GREET for other transportation fuels). Different feedstock-and-processing pathways deliver different CI scores and therefore different qualifying credits within the $[\$1.30, \$1.45]$/gal band. The headline match of Proposition A.3 ($\sigma_{\text{prod}} \approx \$1.36$/gal) should therefore be interpreted as the pathway-specific calibration for the SE-US forest-residue Fischer–Tropsch (FT-FR) route inherited from Li and Yu (2026), not as a generic match across the full band. Table 5 reports the pathway-specific σ_{prod} obtained by reparameterizing $(\kappa_{\text{CAPEX}}, \bar{c}_{op})$ per pathway while holding the remaining primitives at baseline.

Three implications. First, the headline $\sigma_{\text{prod}} \approx \$1.36$/gal matches the 45Z band specifically for the FT-FR pathway that Paper 1 calibrates; other pathways yield σ_{prod} outside the band. Second, the 45Z CI-sliding scale rewards pathways with larger CO_2 abatement relative to fossil Jet-A, so high-CI-reduction pathways (PtL, FT) receive proportionally higher per-gallon credit, partially offsetting the higher σ_{prod} requirement. Third, the financial-frictions wedge (Component 2 of Proposition A.5) scales linearly in $\kappa_{\text{CAPEX}}/[u\ \text{AF}(r_h,T)(1-\tau)]$,

Table 5: Pathway-specific financial-equivalence subsidies

SAF pathway	κ_{CAPEX} (\$/gal)	$\bar{c}_{op}$ (\$/gal)	σ_{prod} (\$/gal)	45Z status (vs. \$1.30–\$1.45 band)
FT-FR (SE-US forest residue; Paper 1 baseline)	11.0	3.78	1.36	within band
FT (generic biomass, *n*th-plant TEA)	≈ 15	≈ 4.00	≈ 2.66	above band (high CAPEX)
ATJ-corn (Alcohol-to-Jet)	≈ 9	≈ 3.50	≈ 0.54	below band (low CAPEX)
PtL (Power-to-Liquid, renewable H_2+CO_2)	≈ 25	≈ 5.00	≈ 6.35	far above band (CAPEX-intensive FOAK)

σ_{prod} is computed from Proposition A.3 by reparameterizing $(\kappa_{\text{CAPEX}}, \bar{c}_{op})$ per pathway while holding the remaining primitives $(r_h, T, u, \tau, \pi_H, a_H - a_L)$ at the FT-FR baseline. Sources for $(\kappa_{\text{CAPEX}}, \bar{c}_{op})$: FT-FR from Li and Yu (2026) §4.4, inheriting the NREL FOAK Fischer–Tropsch calibration; generic biomass-FT from Tanzil et al. (2021a); ATJ-corn from Tanzil et al. (2021b); PtL from International Energy Agency (2023). The FT, ATJ, and PtL values are the authors' round-number parameterizations informed by those sources, not figures the sources report — the cited TEAs report minimum fuel selling prices of ≈\$6.7–7.3/gal for greenfield biomass-FT, scenario MFSPs spanning ≈\$2.6–7.7/gal for repurposed vs. greenfield ATJ, and \$6.5–10.5/gal for PtL. The qualitative ordering, not the point values, carries the argument.

so the high-CAPEX pathways (PtL, FT) face proportionally larger hurdle-IRR premiums and therefore benefit more from federal Title XVII loan-guarantee compression of r_h (the Title XVII fiscal-substitution proposal of Section 8). The per-gallon substitution $\Delta\sigma = \kappa_{\text{CAPEX}}/[u(1-\tau)] \cdot [1/\text{AF}(0.125, 10) - 1/\text{AF}(0.07, 10)]$ scales linearly in κ_{CAPEX}: \$0.63 (FT-FR), \$0.85 (FT), \$0.51 (ATJ), \$1.42 (PtL). As a share of each pathway's own σ_{prod}, however, the substitution *rate* runs 46%, 32%, 95%, and 22% respectively — highest for low-CAPEX ATJ, whose requirement the guarantee nearly eliminates, and lowest for CAPEX-intensive PtL. The FT-FR-calibrated 46% therefore understates the *level* of substitution for FT and PtL but overstates their *rates*, and the reverse for ATJ.

Pathway-weighted fiscal substitution. Using an *illustrative* 2030 pathway portfolio (baseline pathway $\approx 55\%$ of the 3 B gal/yr target, FT $\approx 25\%$, ATJ $\approx 15\%$, PtL $\approx 5\%$; the Grand Challenge Roadmap (U.S. Department of Energy, U.S. Department of Transportation, and U.S. Department of Agriculture 2022) describes HEFA as the commercial near-term pathway with ATJ nearing commercialization but publishes no quantitative 2030 mix, so these weights are ours), the pathway-weighted Title XVII fiscal substitution at r_h-compression from 12.5% to 7% is approximately $0.55 \times \$0.63 + 0.25 \times \$0.85 + 0.15 \times \$0.51 + 0.05 \times \$1.42 \approx \$0.71$/gal, delivering pathway-weighted gross substitution of approximately \$2.12 B/year onto Title XVII contingent liability at the 3 B gal/yr scale (vs. the FT-FR-only \$1.88 B/year). The fiscal-substitution benefit is therefore $\approx 13\%$ larger under the SAF Grand Challenge pathway mix than at the FT-FR-only headline.

Sensitivity to the Farooq et al. (2025) anchor. The headline λ_{prod} identification is reverse-engineered from the observed 45Z band conditional on r_h from the WACC-plus-hurdle-premium calibration of Section 5.2. Perturbing the implied λ_{prod} by ±25% (from 0.60 to $\{0.45, 0.75\}$) shifts the implied σ_{prod} band-match by approximately ±\$0.16/gal under the two-state collapse. At the upper perturbation $\lambda_{\text{prod}} = 0.75$ the Component-2 share is 74%; within the band image $\lambda_{\text{prod}} \in [0.54, 0.69]$ it lies within [80%, 96%] (Table 7); at the lower perturbation $\lambda_{\text{prod}} = 0.45 < \lambda_{\text{planner}}$ the risk-aversion component turns negative and the share exceeds 100%, which is not policy-meaningful but only strengthens the financial-frictions attribution. The dominance is therefore robust to a plausible alternative TEA anchor (e.g., Tanzil et al. 2021a,b in place of Farooq et al. 2025).

Table 6: IRA production-credit cross-walk on the structural primitives ($\kappa_{\text{CAPEX}}, r_h, T, \lambda_{\text{prod}} - \lambda_{\text{planner}}$).

Credit	Unit	κ_{CAPEX}	r_h	T	$\lambda_{\text{prod}} - \lambda_{\text{planner}}$
45Z (SAF)	\$/gal	11.0	0.125	10	+0.10
45V (H_2)	\$/kg	≈ 5	≈ 0.12	10	$\approx +0.08$
45Y (electricity)	\$/MWh	≈ 0.5	≈ 0.08	20–30	$\approx +0.05$
45Q (CCS)	\$/ton	≈ 50	≈ 0.10	12	$\approx +0.07$

The CI-heterogeneity acknowledgment also informs the asymmetric-information bound of Section 7.7: under truly heterogeneous pathway calibrations, the λ_{prod}-induced wedge component must be computed per pathway, and the planner's anonymous σ-form (the structural assumption underlying the 75% C2-share threshold) becomes itself a CI-conditional instrument. The threshold $\sigma_r = 0.021$ is therefore best interpreted as the FT-FR pioneer-plant pathway-specific bound; pathway-by-pathway thresholds would tighten or loosen the headline result depending on the within-pathway r_h variance.

7.10. Two-Track Distinction

The closed-form σ_{prod} in Section 5.1 is derived within the representative-investor quadratic-capacity reduction (Track T_{closed}), which replaces the Stage-1 binary depot-refinery investment problem with a smooth quadratic-capacity objective. The full two-stage stochastic MILP at the SE-US 248/34 instance ($T_{\text{numerical}}$: 248 candidate depot sites, 34 candidate refinery sites) is the environment in which a MILP-side counterpart of σ_{prod} would be computed — the per-gallon credit at which the NPV of the MILP-optimal facility footprint crosses zero at the hurdle rate. We have not run that computation; the closed-form value is the first-order approximation, and the MILP-side verification — along with the $O\big((\mu_W - 1)^2\big)$ residual in Equation (14) and the full numerical welfare wedge ΔW at the 248/34 instance — is deferred to a follow-up numerical companion. Paper 1's MIP-gap-management infrastructure is the natural vehicle, though the NPV-crossing precision required for a \$1.36/gal credit is finer than the 5% welfare gap that infrastructure certifies.

7.11. Extension to Other IRA Production Credits (45V / 45Y / 45Q)

The σ_{prod} decomposition extends across the IRA production-credit family by reparameterizing ($\kappa_{\text{CAPEX}}, r_h, T, \lambda_{\text{prod}}$) per credit. Proposition B.4 shows the dominance result depends jointly on the three structural primitives (i)–(iii); Remark B.5 shows that 45Z satisfies all three. Table 6 reports the analogous decomposition primitives for 45V (hydrogen, IRA Section 45V), 45Y (clean electricity PTC, IRA Section 45Y), and 45Q (carbon sequestration, IRA Section 45Q) at representative pioneer-plant calibrations.

All four credits satisfy the H-M-departure primitives (i)–(iii) of Proposition B.4 at pioneer-plant calibrations: the policy-instrument class is statutorily restricted (per-unit linear form), pioneer-plant investors are at least mildly risk-averse ($\lambda_{\text{prod}} > \lambda_{\text{planner}}$ in each case), and the project-finance hurdle exceeds the social discount rate ($r_h > r_{\text{RN}}$). The CfD-with-floor dominance result therefore applies separately to each credit, with the magnitude of the wedge scaling in κ_{CAPEX}, the financial-frictions component scaling in $(r_h - r_{\text{RN}})/r_{\text{RN}}$, and the risk-aversion component scaling in $(\lambda_{\text{prod}} - \lambda_{\text{planner}})\,\pi_H\,(a_H - a_L)$.

One statutory caveat qualifies the cross-walk: the credits are mutually exclusive at the facility level — a facility claiming 45V cannot also claim 45Q for the same process, and 45Z excludes fuel produced at a facility for which a 45V or 45Q credit is allowed (exclusivity rules summarized in Congressional Research Service 2025). The cross-walk therefore compares the credits as alternative stand-alone instruments for their respective sectors, not as stackable components of a single project's capital stack.

The per-unit financial-frictions wedge is largest where CAPEX intensity is high and the horizon falls short of annuity saturation—45Q at $T = 12$ years and high-κ_{CAPEX} 45V—while 45Y's 20–30-year horizon pushes $\text{AF}(r_h, T)$ toward saturation and shrinks the per-unit wedge (Proposition A.7: $\partial\sigma_{\text{prod}}/\partial T < 0$), leaving 45Y's aggregate wedge driven by volume rather than intensity. A full calibration exercise on the four-credit family, including post-Inflation-Reduction-Act-rulemaking κ_{CAPEX} updates as commercial-deployment learning accumulates, is deferred to a follow-up companion.

Section 8 translates these robustness-tested results into a four-part Treasury+DOE proposal calibrated to the Q3 2026 CFR rulemaking cycle.

8. Policy Recommendations

The wedge decomposition and the CfD-with-floor dominance result jointly specify the policy lever and the instrument form. This section translates the framework into a concrete proposal calibrated to the Q3 2026 CFR rulemaking cycle: the welfare-gain quantification at the SAF Grand Challenge scale, the four-part Title XVII + §45Z(g) joint specification, and the institutional pathway through which the proposal can be submitted under existing Treasury and DOE authorities.

Quantitative policy product. At the SAF Grand Challenge scale of 3 B gal/year by 2030 and the Dahlby empirical US-federal MCPF range $\mu_W \in [1.10, 1.30]$, the actual welfare gain from switching the NPV-binding linear $\sigma^{\text{lin}} = \sigma_{\text{prod}}$ to a CfD-with-floor analog in Regime A is $(\mu_W - 1) \cdot \$0.292/\text{gal} \in [\$0.029, \$0.088]/\text{gal}$ of producer output (Equation (14)). At 3 B gal/year scale this translates to roughly \$87 M–\$264 M/year of efficiency gain, equivalent to roughly 2%–6% of the NPV-binding linear-credit outlay the model compares against (\$4.08 B/year at $\sigma_{\text{prod}} = \$1.36/\text{gal} \times 3$ B gal; the post-2025 statutory outlay at \$0.74–\$0.83/gal is smaller still).[1] At the GREET-implied CO_2 abatement intensity of ≈ 10.4 kg CO_2e/gal SAF — the ≈ 89 g CO_2e/MJ petroleum-jet baseline at 131.5 MJ/gal (≈ 11.7 kg/gal) less the forestry-residue FT pathway's 8.6–12.9 kg CO_2e/mmBtu (≈ 1.1–1.6 kg/gal) — the welfare gain converts to roughly \$2.8–\$8.5/ton CO_2 abated across $\mu_W \in [1.10, 1.30]$ ($\approx \$5.6$/ton at the $\mu_W = 1.20$ headline), i.e., about 1.5%–4.5% of the EPA central social cost of carbon (\$190/ton, 2020 dollars, Bistline et al. 2024) — modest in magnitude but sign-determinate within the empirical MCPF range. (A separate *first-best fiscal-slack diagnostic* — \$7.00 per gallon of annual capacity over the asset life, $\approx \$1.04$/gal flow-equivalent — quantifies the market-completion ceiling on direct discount-rate substitution in the appendix derivation but does not constitute an achievable welfare gain within the legally feasible policy-instrument class $\mathcal{S}$.)

Alternatively, a DOE Title XVII loan guarantee that compresses the pioneer-plant hurdle from $r_h = 12.5\%$

[1] The 2%–6% denominator anchor relies on the absorption convention for the depreciation tax shield, giving $\sigma_{\text{prod}} = \$1.36/\text{gal}$ and fiscal-outlay denominator \$4.08 B/year. Under the rigorous depreciation derivation (Section 7.8), $\sigma_{\text{prod}}^{\text{rigorous}} = \$1.02/\text{gal}$ and the denominator drops to \$3.06 B/year; the welfare-gain ratio rises proportionally to roughly 3%–9% of the rigorous-convention fiscal outlay. The qualitative conclusion (modest, sign-determinate MCPF-channel welfare gain) is robust to the convention; only the denominator anchor and the ratio scale shift.

to $r_h \approx 7\%$ (typical for federally guaranteed projects) lowers σ_{prod} from \$1.36/gal to \$0.73/gal—a \$0.63/gal fiscal substitution at the same producer-participation NPV. At 3 B gal/year scale this substitutes \$1.88 B/year of 45Z linear-credit outlay onto Title XVII risk absorption (whose carrying cost is the guarantee-fee residual after federal-borrowing-rate financing). The fiscal substitution rate—fraction of 45Z outlay redirectable to Title XVII at equivalent producer participation—is therefore ≈ 46% at the empirical baseline, providing a concrete dollar handle for the Q3 2026 CFR rulemaking cycle.

Rulemaking recommendation. Translating the 46% Title XVII fiscal-substitution rate into a concrete Q3 2026 CFR rulemaking comment-letter proposal requires specifying four parameters. We recommend the following Treasury+DOE joint proposal as the practical instantiation of the framework developed in this paper.

(a) Title XVII tenor. A 12-year guarantee tenor, matching the 10-year project life of Proposition A.3 plus a 2-year construction-period bridge, aligns with DOE Loan Programs Office norms for FOAK biomass-to-liquid facilities (U.S. Department of Energy, U.S. Department of Transportation, and U.S. Department of Agriculture 2022) and with the IRS Asset Class 49.5 depreciation envelope (Rev. Proc. 87-56). 12-year tenor is the minimum that delivers the full $r_h = 12.5\%$ to $r_h \approx 7\%$ compression underlying the 46% substitution rate.

(b) Fee structure. A 1.5% origination fee plus a 0.5%/year annual guarantee fee, totaling 7.5% lifecycle cost on a 12-year tenor. This fee structure is a modeling convention of ≈ 0.5% per year of guaranteed principal. LPO's actual Title XVII structure differs — a one-time facility fee (0.6% of guaranteed principal up to \$2 B, 0.1% above), flat annual maintenance fees, and a one-time risk-based credit subsidy charge — so the recurring-fee convention here is an annualized approximation, not an LPO fee schedule. At the SAF Grand Challenge 3 B gal/year scale and FT-FR baseline calibration, this delivers a net fiscal substitution of \$1.70 B/year (i.e., \$1.88 B/year gross substitution minus \$0.14 B/year in expected guarantee-fee revenue and \$0.04 B/year in expected default-loss provision at the 3 B gal scale) onto Title XVII contingent liability, while the federal balance sheet absorbs the residual default risk at an actuarial Title XVII loss rate of ≈ 2–3.7% of disbursements, *inclusive* of the Solyndra cohort: GAO reports cumulative default costs of \$807 M, 3.7% of disbursements as of 2014 (U.S. Government Accountability Office 2015), and the DOE Loan Programs Office's Annual Portfolio Status Report shows \$1.03 B of actual and estimated losses — 3.1% of disbursements as of FY 2023 (U.S. Department of Energy, Loan Programs Office 2024); on the larger post-2023 disbursement base the loss share is commensurately lower. We treat 2%–3.7% as the plausible range and 2% as the central point estimate; the \$0.04 B/year default-loss provision below scales linearly in this rate. A stressed rate of 5% (above any published historical figure) would raise the provision to \$0.10 B/year, shrinking the net Title XVII substitution from \$1.70 B/year to \$1.64 B/year—a 3.5% reduction in the headline number.

(c) 45Z statutory amendment proposal. Add a new subsection Internal Revenue Code (IRC) §45Z(g) "Contract-for-Difference Election" authorizing the IRS, in coordination with DOE Title XVII, to elect CI-tier-conditional CfD-with-floor administration of the 45Z credit in lieu of linear per-gallon payment. The election would be irrevocable at Stage 1 facility commissioning (consistent with Assumption 9's form-commitment treatment) and would trigger Title XVII loan-guarantee eligibility at the 12-year tenor + fee structure of (a)–(b). This statutory amendment preserves 45Z's per-output payment architecture (consistent with the IRC §45 family defended in Assumption 11) while enabling the Borch–Wilson risk-sharing channel of Theorem B.3.

(d) Predicted fiscal substitution at the proposed combination. At the (a)–(c) parameters, 3 B gal/year SAF

Grand Challenge scale, and the FT-FR baseline calibration, the predicted net fiscal substitution is approximately \$1.70 B/year onto Title XVII contingent liability, net of \$0.14 B/year in expected guarantee-fee revenue and \$0.04 B/year in expected Title XVII default-loss provisions. Net 45Z linear-credit outlay declines from \$4.08 B/year (at $\sigma_{\text{prod}} = \$1.36/\text{gal} \times 3$ B gal) to \$2.19 B/year (at $\sigma_{\text{prod}}^{\text{compressed}} = \$0.73/\text{gal} \times 3$ B gal), a 46% reduction. The pathway-specific calibrations of Section 7.9 show that the substitution rate is larger for high-CAPEX FT and PtL pathways and smaller for low-CAPEX ATJ-corn; an extension of (c) to a pathway-conditional Title XVII fee schedule would further increase aggregate fiscal substitution under the SAF Grand Challenge pathway-portfolio target.

Engagement with the Q3 2026 CFR rulemaking. The proposal above is calibrated to the Q3 2026 CFR rulemaking cycle opened by H.R. 1. As of submission, Treasury and the IRS have indicated through IRS Notices 2025-10/2025-11 (Internal Revenue Service 2025) and the CRS *In Focus* brief on 45Z (Congressional Research Service 2025) that the active rulemaking question concerns CI-score methodology (45ZCF-GREET; CORSIA for SAF), not the linear-versus-state-contingent form question this paper addresses. Industry comment letters in the post-H.R. 1 comment dockets focus predominantly on CI-score qualification, feedstock LCA methodology, and the pathway-specific eligibility envelope. The instrument-form question (linear vs. CfD-with-floor) is therefore the natural extension of the current rulemaking and lies within Treasury's IRC §45Z(b) rate-setting authority. The Title XVII fiscal-substitution proposal is addressable to the DOE Loan Programs Office on its current authority and does not require congressional action.

9. Conclusion

The as-enacted IRA Section 45Z producer credit — \$1.30–\$1.45/gal for forestry-residue pathways under the statute's carbon-intensity scaling — is roughly 2.6 times the \$0.52/gal capacity-restoring benchmark derived from the same SE-US supply-chain data (Li and Yu 2026). This paper shows that the gap is mostly financial, not behavioral. At the Paper-1 horizon of $T = 10$ years, financial frictions (the project-finance hurdle-IRR premium at $r_h = 12.5\%$) account for 87.8% of the \$0.84/gal wedge; producer risk aversion (the mean-CVaR downside-premium at $\lambda_{\text{prod}} = 0.60$) accounts for 12.2%. This decomposition is the central empirical contribution. The attribution is a point estimate at $T = 10$ and shifts strongly with the project horizon: at $T = 5$ financial frictions account for 95.8% of the wedge, while at $T = 20$ risk aversion accounts for 71.0%. The right policy question is therefore not whether producers are too risk-averse but whether pioneer-plant investors face a project-finance hurdle that public policy should address.

Policy implications. H.R. 1 (One Big Beautiful Bill Act, enacted July 2025) eliminated the SAF premium for fuel produced after December 31, 2025 — cutting the CI-scaled credit for this pathway from \$1.30–\$1.45/gal to \$0.74–\$0.83/gal — and the accompanying CFR Q3 2026 rulemaking cycle puts the financial-frictions component at the center of the policy debate. The post-2025 credit falls \$0.53–\$0.62/gal below σ_{prod} at the baseline calibration, leaving pioneer-plant projects financially infeasible at the median SAF price projection. The paper's second result addresses the instrument question: under the assumptions of this model (complete information on producer financials, restricted policy-instrument class $\mathcal{S}$, Regime A NPV-binding), a contract-for-difference with a break-even floor weakly Pareto dominates the linear credit, strictly when public funds are costly ($\mu_W > 1$), which holds at the empirical calibration $\mu_W = 1.20$. The CfD-with-floor concentrates transfer payments in low-price scenarios, equalizing the marginal-utility ratio between the two agents (Borch-Wilson risk-sharing) while reducing expected fiscal outlay relative to a constant per-gallon payment. The ranking is reinforced by Paper 1's benefit-cost accounting on the same calibration: the deadweight loss the risk correction removes is an order of magnitude smaller than the marginal-cost-of-public-funds cost of delivering it through a flat credit, so the flat credit fails a benefit-cost test at any $\mu_W > 1.02$, while the floor and CfD forms — which concentrate outlay in the states where the correction binds — pass (Li and Yu 2026). The concrete four-part Treasury+DOE proposal that translates these results into the Q3 2026 CFR rulemaking cycle—tenor, fee structure, the proposed IRC §45Z(g) "CfD Election" amendment, and the predicted \$1.70 B/year fiscal substitution—is developed in Section 8.

If Treasury cannot mandate a state-contingent transfer form under current 45Z statutory language, the policy analog is a federal loan guarantee or partial credit enhancement that effectively lowers the pioneer-plant hurdle rate from $r_h = 12.5\%$ toward the social cost of capital. The financial-frictions wedge is the target, and instruments that reduce r_h are welfare-equivalent to a CfD-with-floor at the same fiscal cost.

Limitations. Four restrictions of the current model deserve acknowledgment.

(i) *Representative-producer abstraction.* The model uses a single representative producer; heterogeneity in λ_{prod} and r_h is not modeled. The screening implications of heterogeneous hurdle rates—which type the regulator should target, and whether a menu of contracts is superior—are deferred to Paper 3.

(ii) *Public observability of producer financials.* λ_{prod} and r_h are treated as publicly observed; asymmetric information on either parameter changes the optimal contract form. Section 7.7 quantifies the dominance-gap shrinkage under private r_h: at the calibration-envelope uncertainty ($\sigma_r = 0.014$, the $[0.10, 0.15]$ envelope read as uniform), the Component 2 attribution share remains at 81%, well above the 75% threshold, and the variance of private r_h would have to inflate by 2.25× before the headline 12.2%/87.8% attribution qualitatively weakens. Heterogeneous-investor screening under strictly private r_h via direct-mechanism design is deferred to Paper 3.

(iii) *Closed-form ΔW residual.* The full numerical welfare wedge ΔW at the SE-US 248/34 SP-MILP instance is reserved for a follow-up companion; the $O\big((\mu_W - 1)^2\big)$ residual in Equation (14) and the binary facility-siting dynamics are not fully resolved by the closed-form analysis.

(iv) *Attribution-ratio conditioning.* The 12.2%/87.8% attribution ratio depends on the Farooq et al. (2025) meta-analysis of pre-45Z SAF TEA studies and on the two-state KDE collapse of Li and Yu (2026); the *level* of Component 2 (\$0.74/gal) is structurally invariant to both the λ_{prod} identification (Table 7) and the discretization choice (Table 8).

As post-45Z project-finance data accumulate, the λ_{prod} and r_h posteriors will update; the policy-robust headline is the financial-frictions *level*, not the attribution *ratio*.

Future work. Three directions extend this analysis. Paper 3 introduces asymmetric information on λ_{prod} and r_h, with heterogeneous-investor screening and optimal mechanism design under private hurdle rates. A numerical companion resolves the full SP-MILP at 248/34 and quantifies ΔW at the discrete facility-siting optimum, including the second-order welfare terms. A learning-by-doing extension would endogenize $(r_h, \kappa_{\text{CAPEX}})$ along the FOAK-to-nth-of-a-kind (NOAK) cost trajectory (Lazard 2024), asking how the required subsidy evolves as pioneer-plant learning reduces both the capital cost and the equity risk premium.

A production credit calibrated to the project-finance hurdle rate is not a subsidy to producers in the ordinary sense; it is the public cost of operating within the missing-market friction (Remark 3.2) that prevents pioneer plants from clearing the private investment hurdle at the social cost of capital.

References

Aldy, Joseph E. and Robert N. Stavins. 2012. "The Promise and Problems of Pricing Carbon: Theory and Experience." *The Journal of Environment & Development* 21 (2): 152–180. 10.1177/1070496512442508. Carbon pricing vs PTC comparison; cited for clean-energy welfare-economics context.

Arrow, Kenneth J. and Robert C. Lind. 1970. "Uncertainty and the Evaluation of Public Investment Decisions." *American Economic Review* 60 (3): 364–378.

Bistline, John E. T., Kimberly A. Clausing, Neil Mehrotra, James H. Stock, and Catherine Wolfram. 2024. "Climate Policy Reform Options in 2025." 32168, National Bureau of Economic Research. NBER Working Paper w32168.

Bistline, John E. T., Neil R. Mehrotra, and Catherine Wolfram. 2023. "Economic Implications of the Climate Provisions of the Inflation Reduction Act." , Brookings Institution. Spring 2023 issue, pp. 77–157; also NBER Working Paper w31267.

Boomsma, Trine Krogh, Nigel Meade, and Stein-Erik Fleten. 2012. "Renewable Energy Investments under Different Support Schemes: A Real Options Approach." *European Journal of Operational Research* 220 (1): 225–237. 10.1016/j.ejor.2012.01.017.

Borch, Karl. 1962. "Equilibrium in a Reinsurance Market." *Econometrica* 30 (3): 424–444. 10.2307/1909887.

Borenstein, Severin. 2008. "The Market Value and Cost of Solar Photovoltaic Electricity Production." *Center for the Study of Energy Markets Working Paper Series* (CSEM WP 176). Foundational reference for "subsidy-cost vs. social-value" framework for renewables credits.

Congressional Budget Office. 2025. "Estimated Budgetary Effects of Public Law 119-21, the One Big Beautiful Bill Act." , U.S. Congress, Congressional Budget Office. https://www.cbo.gov/publication/61570. July 21, 2025. The enacted law aligns the SAF 45Z rate ($1.75/gal) with the non-SAF rate ($1.00/gal) for fuel produced after December 31, 2025; the 45Z package is scored by JCT as incorporated in the CBO estimate.

Congressional Research Service. 2025. "The Section 45Z Clean Fuel Production Credit." IF12502, Congressional Research Service. https://www.congress.gov/crs-product/IF12502. Policy primer on 45Z post-IRA enactment; CRS *In Focus* brief.

Crux Climate. 2025. "The State of Clean Energy Finance: 2025 Mid-Year Transferable Tax Credit Market Intelligence Report." , Crux Climate, Inc. https://www.cruxclimate.com/reports/2025-mid-year-market-intelligence-report.

Cullen, Joseph A. 2013. "Measuring the Environmental Benefits of Wind-Generated Electricity." *American Economic Journal: Economic Policy* 5 (4): 107–133. 10.1257/pol.5.4.107.

Farooq, Zeenat, Elisabeth Wetterlund, Sennai Mesfun, and Erik Furusjö. 2025. "Uncovering the Economic Potential of Sustainable Aviation Fuel Production Pathways: A Meta-Analysis of Techno-Economic Studies." *Energy Conversion and Management* 341: 120076. 10.1016/j.enconman.2025.120076.

Federal Aviation Administration. 2021. "Aviation Climate Action Plan." , U.S. Department of Transportation, Federal Aviation Administration. https://www.faa.gov/sites/faa.gov/files/2021-11/Aviation_Climate_Action_Plan.pdf.

Froot, Kenneth A., David S. Scharfstein, and Jeremy C. Stein. 1993. "Risk Management: Coordinating Corporate Investment and Financing Policies." *Journal of Finance* 48 (5): 1629–1658.

Gollier, Christian. 2001. *The Economics of Risk and Time*. Cambridge, MA: MIT Press. Public-finance treatment of social discount rate and risk pricing; baseline for the planner mean-CVaR weight calibration.

Graham, John R. and Campbell R. Harvey. 2001. "The Theory and Practice of Corporate Finance: Evidence from the Field." *Journal of Financial Economics* 60 (2–3): 187–243. 10.1016/S0304-405X(01)00044-7.

Holland, Stephen P., Jonathan E. Hughes, and Christopher R. Knittel. 2009. "Greenhouse Gas Reductions under Low Carbon Fuel Standards?" *American Economic Journal: Economic Policy* 1 (1): 106–146. 10.1257/pol.1.1.106.

Holmström, Bengt and Paul Milgrom. 1987. "Aggregation and Linearity in the Provision of Intertemporal Incentives." *Econometrica* 55 (2): 303–328. 10.2307/1913238.

Holmström, Bengt and Jean Tirole. 1998. "Private and Public Supply of Liquidity." *Journal of Political Economy* 106 (1): 1–40.

Internal Revenue Service. 2025. "Notice 2025-11: Section 45Z Clean Fuel Production Credit; Emissions Rates and Related Guidance." IRS Notice 2025-11. January 2025; initial emissions-rate table and CI methodologies (45ZCF-GREET for non-SAF; CORSIA for SAF); accompanied by Notice 2025-10.

International Air Transport Association. 2023. "Net Zero Roadmaps." , International Air Transport Association. https://www.iata.org/en/programs/sustainability/flynetzero/roadmaps/. Suite of five roadmaps (June 2023; Policy and Finance updated September 2024); the Energy and New Fuels Infrastructure roadmap covers SAF supply.

International Energy Agency. 2023. "The Role of E-fuels in Decarbonising Transport." , International Energy Agency. https://www.iea.org/reports/the-role-of-e-fuels-in-decarbonising-transport. December 2023; power-to-liquid (PtL) SAF cost projections used for the PtL entry in tab:pathway-sigma-prod.

Joint Committee on Taxation. 2022. "Estimated Budget Effects of the Revenue Provisions of Title I – Committee on Finance, of an Amendment in the Nature of a Substitute to H.R. 5376 (Inflation Reduction Act of 2022)." JCX-18-22, U.S. Congress, Joint Committee on Taxation. https://www.jct.gov/publications/2022/jcx-18-22/. August 9, 2022; original IRA revenue score, including §45Z.

Lazard. 2024. "Lazard Levelized Cost of Energy+ Analysis — Version 17." , Lazard. https://www.lazard.com/media/gjyffoqd/lazards-lcoeplus-june-2024.pdf.

Li, Pengzhen and T. Edward Yu. 2026. "Price Risk, Irreversible Investment, and the Welfare Benchmark for the Section 45Z Clean Fuel Production Credit." Working paper, University of Tennessee, Knoxville.

Maccheroni, Fabio, Massimo Marinacci, and Aldo Rustichini. 2006. "Ambiguity Aversion, Robustness, and the Variational Representation of Preferences." *Econometrica* 74 (6): 1447–1498. 10.1111/j.1468-0262.2006.00716.x.

Metcalf, Gilbert E. 2019. "The Distributional Impacts of U.S. Energy Policy." *Energy Policy* 129: 926–929. 10.1016/j.enpol.2019.01.076. Cited for clean-energy welfare-economics context.

Milgrom, Paul and Ilya Segal. 2002. "Envelope Theorems for Arbitrary Choice Sets." *Econometrica* 70 (2): 583–601. 10.1111/1468-0262.00296.

National Renewable Energy Laboratory. 2024. "2024 Transportation Annual Technology Baseline." , National Renewable Energy Laboratory, Golden, CO. https://atb.nrel.gov/transportation/2024. Aviation-fuels section; capital-charge-factor methodology for biomass-to-liquid SAF pioneer plants, an anchor on hurdle IRR independent of the DOE 2022 SAF Grand Challenge Roadmap.

Newbery, David M. 2023. "Efficient Renewable Electricity Support: Designing an Incentive-compatible Support Scheme." *The Energy Journal* 44 (3): 1–22. 10.5547/01956574.44.3.dnew.

Rockafellar, R. Tyrrell and Stanislav Uryasev. 2000. "Optimization of Conditional Value-at-Risk." *Journal of Risk* 2 (3): 21–41. 10.21314/JOR.2000.038.

Sallee, James M. 2011. "The Surprising Incidence of Tax Credits for the Toyota Prius." *American Economic Journal: Economic Policy* 3 (2): 189–219. 10.1257/pol.3.2.189.

Shapiro, Alexander, Darinka Dentcheva, and Andrzej Ruszczyński. 2014. *Lectures on Stochastic Programming: Modeling and Theory*. MOS-SIAM Series on Optimization. 2nd ed. Philadelphia: SIAM. 10.1137/1.9781611973433.

Tan, Eric C.D̃., Michael Talmadge, Abhijit Dutta, Jesse Hensley, Josh Schaidle, Mary Biddy, David Humbird, Lesley J. Snowden-Swan, Jeff Ross, Danielle Sexton, et al. 2015. "Process Design and Economics for the Conversion of Lignocellulosic Biomass to Hydrocarbons via Indirect Liquefaction: Thermochemical Research Pathway to High-Octane Gasoline Blendstock Through Methanol/Dimethyl Ether Intermediates." , National Renewable Energy Laboratory, NREL/TP-5100-62402. https://www.osti.gov/servlets/purl/1215006.

Tanzil, Abid H., Kristin Brandt, Michael Wolcott, Xiao Zhang, and Manuel Garcia-Perez. 2021a. "Strategic assessment of sustainable aviation fuel production technologies: Yield improvement and cost reduction opportunities." *Biomass and Bioenergy* 145: 105942. 10.1016/j.biombioe.2020.105942. Pathway-specific CAPEX/OPEX calibration for the FT entry in tab:pathway-sigma-prod.

Tanzil, Abid H., Xiao Zhang, Michael Wolcott, Kristin Brandt, Claudio Stöckle, Ganti Murthy, and Manuel Garcia-Perez. 2021b. "Evaluation of dry corn ethanol bio-refinery concepts for the production of sustainable aviation fuel." *Biomass and Bioenergy* 146: 105937. 10.1016/j.biombioe.2020.105937. ATJ-corn calibration in tab:pathway-sigma-prod.

United States Code. 2025. "26 U.S.C. § 6418 — Transfer of Certain Credits." Legal Information Institute, Cornell Law School. https://www.law.cornell.edu/uscode/text/26/6418.

U.S. Department of Energy, U.S. Department of Transportation, and U.S. Department of Agriculture. 2022. "SAF Grand Challenge Roadmap: Flight Plan for Sustainable Aviation Fuel." , U.S. Departments of Energy, Transportation, and Agriculture. https://www.energy.gov/sites/default/files/2022-09/beto-saf-gc-roadmap-report-sept-2022.pdf.

U.S. Department of Energy, Loan Programs Office. 2024. "Annual Portfolio Status Report, Fiscal Year 2023." , U.S. Department of Energy. https://www.energy.gov/edf/annual-portfolio-status-report.

U.S. Government Accountability Office. 2015. "DOE Loan Programs: Current Estimated Net Costs Include $2.2 Billion in Credit Subsidy, Plus Other Risks." GAO-15-438, U.S. Government Accountability Office. https://www.gao.gov/products/gao-15-438. Historical portfolio-cost and default documentation for Title XVII loan guarantees; actuarial anchor for the fiscal-substitution computation. Current portfolio status: GAO-25-106631 (May 2025).

Weyl, E. Glen and Michal Fabinger. 2013. "Pass-Through as an Economic Tool: Principles of Incidence under Imperfect Competition." *Journal of Political Economy* 121 (3): 528–583. 10.1086/670401.

Wilson, Robert. 1968. "The Theory of Syndicates." *Econometrica* 36 (1): 119–132. 10.2307/1909607.

Wu, Mengying, Kristen McCormack, William A. Scott, Aaron Smith, Jingran Zhang, and James H. Stock. 2025. "Policy Options to Achieve US Sustainable Aviation Fuel Targets." 34326, National Bureau of Economic Research. NBER Working Paper w34326.

A. Proofs of the Closed-Form sigma_prod Result

This appendix collects the proofs of the Paper 2 closed-form results on the producer's financial-equivalence subsidy σ_{prod} (Propositions A.1, A.3, A.5 and A.7), the optimality of CfD-with-floor in the restricted policy-instrument class $\mathcal{S}$ (Theorem B.3), and the marginal-cost-of-public-funds (MCPF) corollary (Corollary A.10). Throughout we work in the collapsed CVaR regime $\alpha \geq \pi_H$ that contains the $\alpha = 0.95$ baseline (cf. Paper 1 Theorem 1, regime A); the split-regime $\alpha < \pi_H$ extensions are obtained by replacing π_H with the Paper 1 piecewise factor $\min\{\alpha\pi_L/(1-\alpha),\, \pi_H\}$, exactly as in Proposition A.1 below.

Notation conventions. We retain the Paper 1 conventions (Paper 1 §3.5) and write $A_H \equiv a_H - c$, $A_L \equiv a_L - c$, $\mathbb{E}[A] \equiv \pi_H A_H + \pi_L A_L$, $K \equiv b + k$. For the project-finance machinery we write $\text{AF}(r, T) \equiv \sum_{t=1}^{T}(1+r)^{-t}$ for the annuity factor at discount rate r over horizon T; u for the capacity utilization factor; κ_{CAPEX} for the CAPEX intensity (\$/gallon-annual-capacity); τ for the effective tax rate; $\bar{p_{\text{SAF}}} \equiv \mathbb{E}[p_{\text{SAF}}]$ and $\bar{c_{\text{op}}} \equiv \mathbb{E}[c_{\text{op}}]$ for unconditional kernel-density-estimated (KDE) means; and the Rockafellar–Uryasev (2000) mean–CVaR functional

$$\mathbb{E}_{\text{prod}}[X] \;\equiv\; (1-\lambda_{\text{prod}})\,\mathbb{E}[X] \;+\; \lambda_{\text{prod}}\;\text{CVaR}_\alpha[X], \tag{A.1}$$

under the producer's risk-aversion weight $\lambda_{\text{prod}} \in [0, 1]$ (by Assumption 7). The planner's analogue uses λ_{planner} and appears in the welfare functional defined in Section 3.4.

A.1. Inheritance of Paper 1: $\sigma^*_{\text{planner}}$

PROPOSITION A.1 (Welfare-equivalence subsidy, Paper 1 inheritance). *Under Assumptions 1 to 8, the welfare-equivalence subsidy $\sigma^*_{\text{planner}}(\alpha)$ that restores the risk-neutral planner output level Q^*_{RN} in the collapsed CVaR regime $\alpha \geq \pi_H$ admits the closed form*

$$\sigma^*_{\text{planner}}(\alpha) \;=\; \lambda_{\text{planner}} \cdot (a_H - a_L) \cdot \min\left\{ \tfrac{\alpha\,\pi_L}{1-\alpha},\; \pi_H \right\}. \tag{A.2}$$

In the baseline $\alpha = 0.95 \geq \pi_H = 0.472$, this collapses to $\sigma^*_{\text{planner}} = \lambda_{\text{planner}} \cdot \pi_H \cdot (a_H - a_L)$, which equals \$0.517*/gal at $\lambda_{\text{planner}} = 0.5$, $\pi_H = 0.472$, $a_H - a_L =$* \$2.19*/gal.*

PROOF OF PROPOSITION A.1. The result is Paper 1 Theorem 1 (Li and Yu 2026, eq. (3.5)) verbatim. Both the strict-concavity ingredient (Paper 1 Lemma 1) and the Rockafellar–Uryasev CVaR representation under the two-state collapse (Assumptions 1, 4 and 6) carry over without modification because Paper 2 retains A1–A8 and the welfare functional of the planner is unchanged. We therefore omit the line-by-line derivation and refer to Paper 1 Appendix A.2 (proof of Theorem 1). The piecewise structure on $\{\alpha \geq \pi_H\} \cup \{\alpha < \pi_H\}$ and the continuity at the regime boundary $\alpha = \pi_H$ are also inherited (Paper 1 Corollary 1′). □

remark A.2. *Under Paper 2, $\sigma^*_{\text{planner}}$ plays the role of a* reference point *in the welfare functional space W, not the role of σ_{prod} (which lives in the project-finance NPV space, cf. Proposition A.3). The wedge $\sigma_{\text{prod}} - \sigma^*_{\text{planner}}$ is the headline empirical object decomposed in Proposition A.5.*

A.2. The financial-equivalence subsidy: sigma_prod

PROPOSITION A.3 (Financial-equivalence subsidy via project-finance NPV = 0). *Under Assumptions 1 to 8 and Assumptions 9 to 13, in the collapsed CVaR regime $\alpha \geq \pi_H$, the producer's project-finance NPV at the hurdle IRR r_h equals zero at the unique per-unit constant subsidy*

$$\boxed{\sigma_{\text{prod}} \;=\; \underbrace{\frac{\kappa_{\text{CAPEX}}}{u \cdot \text{AF}(r_h, T) \cdot (1-\tau)}}_{\textit{hurdle-IRR-premium term}} + \underbrace{\lambda_{\text{prod}} \cdot \pi_H \cdot (a_H - a_L)}_{\textit{risk-aversion term}} - \underbrace{\left[\bar{p_{\text{SAF}}} - \bar{c_{\text{op}}}\right]}_{\textit{operating-margin offset}} \;.} \tag{A.3}$$

Equivalently, σ_{prod} is the unique solution in $[0, \infty)$ of $\text{NPV}(\sigma_{\text{prod}}; r_h, \kappa_{\text{CAPEX}}, u, \tau, T) = 0$ *at $Q(\omega) = Q^*(\omega)$, where $Q^*(\omega)$ is the producer's Stage 2 best response (Lemma A.4 below).*

PROOF OF PROPOSITION A.3. The proof proceeds in six steps.

Step 1: Per-scenario after-tax operating cash flow. The producer's per-scenario gross operating profit at a constant per-unit subsidy σ is $[p_{\text{SAF}}(\omega) + \sigma - c_{\text{op}}(\omega)] \cdot Q(\omega)$ (payoff function in Section 3.4). Under Assumption 13 the producer pays effective corporate tax at rate τ on operating margin and benefits from a depreciation tax shield, so the per-scenario after-tax cash flow is

$$\text{CF}(\sigma, Q; \omega) \;=\; \left[p_{\text{SAF}}(\omega) + \sigma - c_{\text{op}}(\omega)\right] \cdot Q(\omega) \cdot (1-\tau) \;+\; \tau \cdot D_t, \tag{A.4}$$

where D_t is the per-period depreciation deduction. Under straight-line depreciation over the project horizon T at zero salvage, $D_t = \kappa_{\text{CAPEX}} \cdot \bar{Q}/(u \cdot T)$, where $\bar{Q} = u \cdot \kappa_{\text{CAPEX}}^{-1} \cdot \text{CAPEX}$ is the annual production capacity (Section 3.5). The depreciation tax shield is deterministic and is folded into the recovery requirement in Step 3.

Step 2: Risk-adjusted expected per-period cash flow. The producer evaluates the random per-period cash flow under mean–CVaR weighting (A.1). The Rockafellar–Uryasev (2000) auxiliary-function representation (by Assumption 6, the denominator $1-\alpha$ is finite) is

$$\text{CVaR}_\alpha\left[\text{CF}\right] \;=\; \sup_{\zeta \in \mathbb{R}} \left\{ \zeta - \tfrac{1}{1-\alpha}\, \mathbb{E}\left[(\zeta - \text{CF})_+\right] \right\}, \tag{A.5}$$

and by Assumption 4 the bad-tail event is unambiguously the low state L at any $Q > 0$ (because $p_{\text{SAF}}(L) < p_{\text{SAF}}(H) \Rightarrow \text{CF}(\sigma, Q; L) < \text{CF}(\sigma, Q; H)$ under Assumption 5). In the collapsed regime $\alpha \geq \pi_H$, exactly as in Paper 1 (eq. (A.3) of Paper 1's Appendix A), the discrete CVaR collapses to the state-L cash flow:

$$\text{CVaR}_\alpha\left[\text{CF}(\sigma, Q; \omega)\right] \;=\; \text{CF}(\sigma, Q; L) \qquad (\alpha \geq \pi_H). \tag{A.6}$$

Combining (A.1) with (A.6), the risk-adjusted expected per-period cash flow per gallon of annual capacity is, after grouping the deterministic depreciation tax shield τD_t separately,

$$\begin{aligned} \mathbb{E}_{\text{prod}}\left[\tfrac{\text{CF}}{\bar{Q}}\right] &= (1-\lambda_{\text{prod}})\, \mathbb{E}\left[(p_{\text{SAF}} + \sigma - c_{\text{op}})\, Q/\bar{Q}\right](1-\tau) \\ &\quad + \lambda_{\text{prod}}\, (p_{\text{SAF}}(L) + \sigma - c_{\text{op}}(L))\, Q(L)/\bar{Q}\, (1-\tau) \;+\; \tau D_t/\bar{Q}. \end{aligned} \tag{A.7}$$

Step 3: NPV per gallon of annual capacity. The producer evaluates the present value at the hurdle IRR r_h

(Assumption 13). Under Assumption 13's finite horizon T, the NPV per gallon of annual capacity is

$$\frac{\mathrm{NPV}(\sigma; r_h, \kappa_{\mathrm{CAPEX}}, u, \tau, T)}{\bar{Q}} = \mathrm{AF}(r_h, T) \cdot \mathbb{E}_{\mathrm{prod}}\left[\mathrm{CF}(\sigma, Q^*; \omega)/\bar{Q}\right] - \frac{\kappa_{\mathrm{CAPEX}}}{u}, \tag{A.8}$$

where the upfront CAPEX outlay $\kappa_{\mathrm{CAPEX}} \cdot \bar{Q}/u$ enters divided by $\bar{Q}$ and where $Q^*(\omega)$ is the producer's Stage 2 best response taken as the representative-investor quadratic-capacity solution (cf. Lemma A.4 below).

Step 4: Set NPV = 0 and solve for σ. The financial-equivalence subsidy is the constant σ_{prod} that sets the left-hand side of (A.8) to zero. We adopt the standard TEA convention used by Farooq et al. (2025) and Lazard (2024) of evaluating CAPEX recovery on a pre-depreciation-shield basis (equivalently: treating D_t as recovered fully in the first period, so that τD_t enters as a one-time CAPEX reduction absorbed into κ_{CAPEX} at the calibration step). Under this convention, the depreciation tax-shield term $\tau D_t/\bar{Q}$ drops out of the periodic cash flow in (A.7), and substituting the resulting $\mathbb{E}_{\mathrm{prod}}[\mathrm{CF}]$ into the NPV = 0 condition yields, after separating the σ-dependent and σ-independent terms,

$$\mathrm{AF}(r_h, T)(1-\tau) \cdot \left[\mathbb{E}_{\mathrm{prod}}[(p_{\mathrm{SAF}} + \sigma_{\mathrm{prod}} - c_{\mathrm{op}})\, Q^*/\bar{Q}]\right] = \frac{\kappa_{\mathrm{CAPEX}}}{u}. \tag{A.9}$$

Dividing through by $\mathrm{AF}(r_h, T)(1-\tau)$ and isolating the σ-dependent expectation,

$$\mathbb{E}_{\mathrm{prod}}\left[(p_{\mathrm{SAF}} + \sigma_{\mathrm{prod}} - c_{\mathrm{op}})\, Q^*/\bar{Q}\right] = \frac{\kappa_{\mathrm{CAPEX}}}{u \cdot \mathrm{AF}(r_h, T) \cdot (1-\tau)}. \tag{A.10}$$

For the representative-investor evaluation at the stationary capacity $Q^*(\omega)/\bar{Q} \to 1$ (capacity-binding in the SAF-positive representative scenario; the deviations are second-order in the representative-investor sense, cf. Paper 1 §3.5 representative-investor reduction), σ_{prod} enters linearly and we solve directly:

$$\sigma_{\mathrm{prod}} = \frac{\kappa_{\mathrm{CAPEX}}}{u \cdot \mathrm{AF}(r_h, T) \cdot (1-\tau)} - \mathbb{E}_{\mathrm{prod}}[p_{\mathrm{SAF}} - c_{\mathrm{op}}]. \tag{A.11}$$

Step 5: Substitute the mean–CVaR identity using Paper 1's 2-state collapse. The mean–CVaR expectation on the operating margin decomposes as

$$\begin{aligned}
\mathbb{E}_{\mathrm{prod}}[p_{\mathrm{SAF}} - c_{\mathrm{op}}] &= (1-\lambda_{\mathrm{prod}})\, \mathbb{E}[p_{\mathrm{SAF}} - c_{\mathrm{op}}] + \lambda_{\mathrm{prod}}\, \mathrm{CVaR}_\alpha[p_{\mathrm{SAF}} - c_{\mathrm{op}}] \\
&\overset{(A.6)}{=} (1-\lambda_{\mathrm{prod}})\, (\bar{p_{\mathrm{SAF}}} - \bar{c_{\mathrm{op}}}) + \lambda_{\mathrm{prod}}\, (p_{\mathrm{SAF}}(L) - c_{\mathrm{op}}(L)) \\
&= (\bar{p_{\mathrm{SAF}}} - \bar{c_{\mathrm{op}}}) - \lambda_{\mathrm{prod}} \left[(\bar{p_{\mathrm{SAF}}} - \bar{c_{\mathrm{op}}}) - (p_{\mathrm{SAF}}(L) - c_{\mathrm{op}}(L))\right].
\end{aligned} \tag{A.12}$$

The bracketed term is precisely the upside-probability times spread: $(\bar{p_{\mathrm{SAF}}} - \bar{c_{\mathrm{op}}}) - (p_{\mathrm{SAF}}(L) - c_{\mathrm{op}}(L)) = \pi_H \cdot [(p_{\mathrm{SAF}}(H) - c_{\mathrm{op}}(H)) - (p_{\mathrm{SAF}}(L) - c_{\mathrm{op}}(L))]$ by $\bar{X} = \pi_H X(H) + \pi_L X(L)$ with $\pi_L = 1 - \pi_H$. Under Assumption 2 and the Paper 1 two-state-collapse calibration of the KDE ($p_{\mathrm{SAF}}(H) = a_H$, $p_{\mathrm{SAF}}(L) = a_L$ at the representative-investor output) and the operating-expenditure (OPEX) equalization convention ($c_{\mathrm{op}}(H) = c_{\mathrm{op}}(L) = \bar{c_{\mathrm{op}}}$, the Paper 1 baseline OPEX-state-invariance assumption used in §3.5), the bracket simplifies to $a_H - a_L$ and we obtain

$$\mathbb{E}_{\mathrm{prod}}[p_{\mathrm{SAF}} - c_{\mathrm{op}}] = (\bar{p_{\mathrm{SAF}}} - \bar{c_{\mathrm{op}}}) - \lambda_{\mathrm{prod}} \cdot \pi_H \cdot (a_H - a_L). \tag{A.13}$$

Step 6: Substitute back into (A.11). Combining (A.11) and (A.13),

$$\sigma_{\text{prod}} \;=\; \frac{\kappa_{\text{CAPEX}}}{u \cdot \mathrm{AF}(r_h, T) \cdot (1-\tau)} \;+\; \lambda_{\text{prod}} \cdot \pi_H \cdot (a_H - a_L) \;-\; \left[\bar{p_{\text{SAF}}} - \bar{c_{\text{op}}}\right], \tag{A.14}$$

which is the boxed expression (A.3). Uniqueness in $[0, \infty)$ is immediate from the linearity of the LHS of (A.10) in σ and the strict positivity of $\mathrm{AF}(r_h, T) \cdot (1-\tau)$ under Assumption 13 ($r_h > -1, \tau < 1$). □

LEMMA A.4 (Stage 2 producer best response). *Under Assumptions 2 to 3, given (σ, y, ω), the Stage 2 producer's per-scenario profit $\pi^{\text{op}}(\sigma, y, Q; \omega) = [p_{\text{SAF}}(\omega) + \sigma]\, Q - (b/2)Q^2 - (k/2)Q^2$ (under the representative-investor reduction of Paper 1 §3.5) is strictly concave in Q with $\partial^2 \pi^{\text{op}} / \partial Q^2 = -(b+k) = -K < 0$. The interior best response*

$$Q^*(\sigma, y; \omega) \;=\; \frac{a(\omega) + \sigma - c}{K} \tag{A.15}$$

exists, is unique, and is positive by Assumption 5.

PROOF. Strict concavity of π^{op} follows by direct differentiation. The interior FOC $a(\omega) + \sigma - c - K \cdot Q = 0$ yields (A.15); positivity at the optimum follows since $a_L > c$ by Assumption 5 and $\sigma \geq 0$. □

A.3. Wedge decomposition

PROPOSITION A.5 (Two-component wedge decomposition). *Under the conditions of Proposition A.3, the wedge between the producer's financial-equivalence subsidy σ_{prod} and the planner's welfare-equivalence subsidy $\sigma^*_{\text{planner}}$ (Proposition A.1) admits the two-component closed-form decomposition*

$$\sigma_{\text{prod}} - \sigma^*_{\text{planner}} \;=\; \underbrace{(\lambda_{\text{prod}} - \lambda_{\text{planner}}) \cdot \pi_H \cdot (a_H - a_L)}_{\textit{Component 1: risk-aversion wedge}} \;+\; \underbrace{\Delta\mathrm{IRR}_{\text{prem}}}_{\textit{Component 2: financial-frictions wedge}}, \tag{A.16}$$

where the Component 2 closed form is

$$\Delta\mathrm{IRR}_{\text{prem}} \;\equiv\; \frac{\kappa_{\text{CAPEX}}}{u \cdot \mathrm{AF}(r_h, T) \cdot (1-\tau)} \;-\; \left[\bar{p_{\text{SAF}}} - \bar{c_{\text{op}}}\right]. \tag{A.17}$$

PROOF OF PROPOSITION A.5. Subtract (A.2) (collapsed regime, $\sigma^*_{\text{planner}} = \lambda_{\text{planner}} \pi_H (a_H - a_L)$) from the closed form (A.14) and group the λ-times-π_H-times-spread terms:

$$\begin{aligned}
\sigma_{\text{prod}} - \sigma^*_{\text{planner}} &= \frac{\kappa_{\text{CAPEX}}}{u\,\mathrm{AF}(r_h, T)(1-\tau)} + \lambda_{\text{prod}} \pi_H (a_H - a_L) \\
&\quad - (\bar{p_{\text{SAF}}} - \bar{c_{\text{op}}}) - \lambda_{\text{planner}} \pi_H (a_H - a_L) \\
&= (\lambda_{\text{prod}} - \lambda_{\text{planner}})\, \pi_H\, (a_H - a_L) \;+\; \underbrace{\left[\frac{\kappa_{\text{CAPEX}}}{u\,\mathrm{AF}(r_h, T)(1-\tau)} - (\bar{p_{\text{SAF}}} - \bar{c_{\text{op}}})\right]}_{\equiv\, \Delta\mathrm{IRR}_{\text{prem}}}.
\end{aligned}$$

This is (A.16). The Component 1 term is in the welfare-functional space (it has the Paper 1 closed-form structure $\lambda \cdot \pi_H \cdot (a_H - a_L)$ with λ replaced by $\lambda_{\text{prod}} - \lambda_{\text{planner}}$); the Component 2 term is in the project-finance space and vanishes only when the CAPEX recovery requirement equals the operating-margin earnings. □

remark A.6 (Empirical decomposition at the baseline calibration)**.** *At the empirical baseline (*$\lambda_{\text{prod}} = 0.60$*,* $\lambda_{\text{planner}} = 0.50$*,* $\pi_H = 0.472$*,* $a_H - a_L = \$2.19$*/gal,* $\kappa_{\text{CAPEX}} = \$11$*/gal-annual-capacity,* $u = 0.85$*,* $r_h = 0.125$*,* $T = 10$*,* $\tau = 0.21$*,* $\bar{p_{\text{SAF}}} = \$6.00$*/gal,* $\bar{c_{\text{op}}} = \$3.78$*/gal):* $\text{AF}(0.125, 10) = 5.5364$*, the CAPEX recovery term equals* $\$11/(0.85 \cdot 5.5364 \cdot 0.79) = \$2.959$*/gal, the operating margin is* $\$2.220$*/gal, the financial-frictions wedge is* $\Delta\text{IRR}_{\text{prem}} = \$2.959 - \$2.220 = \$0.739$*/gal, the risk-aversion wedge is* $(0.60 - 0.50) \cdot 0.472 \cdot \$2.19 = \$0.103$*/gal, the total wedge is* $\$0.842$*/gal, and the headline* $\sigma_{\text{prod}} = \$0.517 + \$0.842 = \$1.359$*/gal* $\in [\$1.30, \$1.45]$*, matching the empirical 45Z band. The attribution is* $\$0.103/\$0.842 \approx 12.2\%$ *(risk-aversion) and* $\$0.739/\$0.842 \approx 87.8\%$ *(financial-frictions).*

A.4. Comparative statics

PROPOSITION A.7 (Comparative statics for σ_{prod} in the collapsed regime). *Under the conditions of Proposition A.3, in the collapsed regime* $\alpha \geq \pi_H$*, the partial derivatives of* σ_{prod} *in* (A.14) *with respect to the producer's preference and project-finance primitives, and of the wedge* $\sigma_{\text{prod}} - \sigma^*_{\text{planner}}$ *with respect to the discount-rate spread, are*

$$\frac{\partial \sigma_{\text{prod}}}{\partial \lambda_{\text{prod}}} = \pi_H \cdot (a_H - a_L) \; > \; 0, \tag{A.18}$$

$$\frac{\partial \sigma_{\text{prod}}}{\partial r_h} = \frac{\kappa_{\text{CAPEX}}}{u(1-\tau)} \cdot \frac{-\partial \, \text{AF}(r_h, T)/\partial r_h}{[\text{AF}(r_h, T)]^2} \; > \; 0, \tag{A.19}$$

$$\frac{\partial \sigma_{\text{prod}}}{\partial \kappa_{\text{CAPEX}}} = \frac{1}{u \cdot \text{AF}(r_h, T) \cdot (1-\tau)} \; > \; 0, \tag{A.20}$$

$$\frac{\partial \sigma_{\text{prod}}}{\partial \tau} = \frac{\kappa_{\text{CAPEX}}}{u \cdot \text{AF}(r_h, T) \cdot (1-\tau)^2} \; > \; 0, \tag{A.21}$$

$$\frac{\partial \sigma_{\text{prod}}}{\partial u} = -\frac{\kappa_{\text{CAPEX}}}{u^2 \cdot \text{AF}(r_h, T) \cdot (1-\tau)} \; < \; 0, \tag{A.22}$$

$$\frac{\partial \sigma_{\text{prod}}}{\partial \pi_H} = \lambda_{\text{prod}} \cdot (a_H - a_L) \; > \; 0, \tag{A.23}$$

$$\frac{\partial \sigma_{\text{prod}}}{\partial (a_H - a_L)} = \lambda_{\text{prod}} \cdot \pi_H \; > \; 0, \tag{A.24}$$

and

$$\begin{aligned} \frac{\partial(\sigma_{\text{prod}} - \sigma^*_{\text{planner}})}{\partial r_h} &= -\frac{\kappa_{\text{CAPEX}}}{u\,(1-\tau)\;\text{AF}(r_h, T)^2} \cdot \frac{\partial \, \text{AF}(r_h, T)}{\partial r_h} \\ &= -\frac{\kappa_{\text{CAPEX}}}{u\,(1-\tau)\;\text{AF}(r_h, T)^2} \cdot \left[\frac{T\,(1 + r_h)^{-T-1}}{r_h} - \frac{\text{AF}(r_h, T)}{r_h}\right] \; > \; 0, \end{aligned} \tag{A.25}$$

where the exact closed form $\partial \, \text{AF}/\partial r_h = T(1+r_h)^{-T-1}/r_h - \text{AF}(r_h, T)/r_h < 0$ *replaces the single-term first-order Taylor approximation displayed in earlier versions of this proof, which Theorem Step 4c (of Theorem B.3) showed to overstate the wedge magnitude by roughly six-fold at the empirical baseline. The partial derivative of* σ_{prod} *with respect to the project horizon* T *is* ***strictly monotone decreasing with attenuating slope****. For fixed* $r_h > 0$*,* $\text{AF}(r_h, T)$ *is monotone increasing concave saturating at* $1/r_h$*, so* $1/\text{AF}(r_h, T)$ *is monotone decreasing on all* $T \geq 1$*, asymptotically approaching* r_h *from above. Hence* $\partial \sigma_{\text{prod}}/\partial T = -\kappa_{\text{CAPEX}}/[u(1-\tau)\;\text{AF}(r_h, T)^2] \cdot \partial \, \text{AF}/\partial T < 0$

everywhere on $T \geq 1$, with $|\partial\sigma_{\text{prod}}/\partial T|$ approaching zero as $T \to \infty$. There is no sign change. Table 2 confirms this numerically across $T \in \{5, 10, 15, 20\}$: σ_{prod} equals \$3.00, \$1.36, \$0.87, \$0.66, respectively (strict decrease).

Proof of Proposition A.7. *Step 1: Closed-form partials.* Equations (A.18), (A.23), (A.24) follow directly from the $\lambda_{\text{prod}}\pi_H(a_H - a_L)$ summand of (A.14) by linearity. Equations (A.20), (A.21), (A.22), (A.19) follow by differentiation of the hurdle-IRR-premium summand $\kappa_{\text{CAPEX}}/[u\ \text{AF}(r_h, T)(1-\tau)]$ with respect to the indicated primitive (the operating-margin offset $\bar{p_{\text{SAF}}} - \bar{c_{\text{op}}}$ is independent of $(r_h, \kappa_{\text{CAPEX}}, \tau, u)$ by the Stage 0a observability convention, Assumptions 10 and 13).

Step 2: Signs of AF *partials.* $\partial\ \text{AF}(r_h, T)/\partial r_h = \sum_{t=1}^{T}(-t)(1+r_h)^{-t-1} < 0$ since each summand is negative. Substituting into (A.19) gives the displayed sign. The $(1-\tau)^{-2}$ factor in (A.21) is positive under Assumption 13 ($\tau < 1$).

Step 3: Wedge partial in $(r_h - r_{\text{RN}})$. Fixing r_{RN} (planner-side) and differentiating $\sigma_{\text{prod}} - \sigma^*_{\text{planner}}$ with respect to r_h (only the Component 2 $\Delta\text{IRR}_{\text{prem}}$ depends on r_h once r_{RN} is fixed), expand $\text{AF}(r_h, T)$ around r_h by the closed form $\text{AF}(r_h, T) = (1 - (1+r_h)^{-T})/r_h$ and differentiate:

$$\frac{\partial\ \text{AF}(r_h, T)}{\partial r_h} = -\frac{1-(1+r_h)^{-T}}{r_h^2} + \frac{T\,(1+r_h)^{-T-1}}{r_h}. \tag{A.26}$$

Substituting into (A.19) and gathering at the empirical baseline $r_h = 0.125$, $T = 10$, $r_{\text{RN}} = 0.03$, the local linear approximation around the wedge's value at $r_h = r_{\text{RN}}$ gives the displayed first-order approximation (A.25): the leading term in the expansion is $\kappa_{\text{CAPEX}} \cdot T/[u(1+r_h)^{T+1}(1-\tau)]$, which is manifestly positive.

Step 4: T-monotonicity (strictly decreasing with attenuating slope). Differentiating the closed form $\text{AF}(r_h, T) = [1 - (1+r_h)^{-T}]/r_h$ in T yields

$$\frac{\partial\ \text{AF}(r_h, T)}{\partial T} = \frac{(1+r_h)^{-T} \cdot \ln(1+r_h)}{r_h} > 0, \tag{A.27}$$

which is itself *strictly decreasing* in T, so AF rises monotonically and saturates at $1/r_h$ as $T \to \infty$. Consequently $1/\text{AF}(r_h, T)$ is strictly monotone decreasing on $T \geq 1$, and therefore $\sigma_{\text{prod}} = \kappa_{\text{CAPEX}}/[u\ \text{AF}(r_h, T)\ (1-\tau)] + \cdots$ is strictly monotone decreasing in T with attenuating slope, asymptotically approaching the T-invariant terms as $T \to \infty$. There is no sign change. The numerical Robustness Table 2 confirms strict monotone decrease across $T \in \{5, 10, 15, 20\}$: σ_{prod} equals \$3.00, \$1.36, \$0.87, \$0.66, respectively. □

remark A.8 (Representative-investor reduction error bound)**.** *The closed-form σ_{prod} derivation in Proposition A.3 invokes the representative-investor reduction $Q^*(\omega)/\bar{Q} \to 1$ at the SAF-positive scenario (Paper 1 §3.5). Under Assumption 5 and the capacity-binding convention of Assumption 3, the producer's Stage-2 best response satisfies $Q^*(\omega) = \bar{Q}$ exactly whenever the per-gallon margin $p_{\text{SAF}}(\omega) + \sigma - c_{\text{op}}(\omega) > 0$. At the empirical baseline ($\sigma = \sigma_{\text{prod}} = \$1.36$/gal, $c_{\text{op}} = \$3.78$/gal, $p_{\text{SAF}}(L) = a_L = \$4.97$/gal), the low-state margin equals $4.97 + 1.36 - 3.78 = \$2.55$/gal, strictly positive; the high-state margin is strictly larger by $p_{\text{SAF}}(H) - p_{\text{SAF}}(L) = \$2.19$/gal. Both states are therefore SAF-positive at baseline, and $Q^*(\omega) = \bar{Q}$* exactly *in both states—not "second-order". The reduction is exact, not asymptotic, in the empirically relevant calibration envelope.*

remark A.9 (Option E: σ_{prod} and $\sigma^*_{\text{planner}}$ as objects in different functional spaces)**.** *The closed-form* $\sigma_{\text{prod}} = \kappa_{\text{CAPEX}}/[u\ \text{AF}(r_h, T)(1-\tau)] + \lambda_{\text{prod}}\,\pi_H\,(a_H - a_L) - (\bar{p_{\text{SAF}}} - \bar{c_{\text{op}}})$ *of Proposition A.3 and the Paper 1 closed-form* $\sigma^*_{\text{planner}} = \lambda_{\text{planner}}\,\pi_H\,(a_H - a_L)$ *of Proposition A.1 are derived from* different functionals on different domains*:*

- $\sigma^*_{\text{planner}} = \arg\max_\sigma \mathbb{E}_{\text{planner}}[W(\sigma)]$ *solves the planner's mean–CVaR welfare-equivalence FOC (welfare functional space W);*
- $\sigma_{\text{prod}} = \inf\{\sigma : \text{NPV}(\sigma; r_h, \kappa_{\text{CAPEX}}, u, \tau, T) \geq 0\}$ *solves the producer's project-finance financial-equivalence NPV-binding condition (financial functional space* NPV*).*

These two equilibrium concepts are not nested under a single parameter restriction; their wedge decomposes into Component 1 (risk-aversion: different λ in the two functionals) and Component 2 (financial-frictions: different discount rates r_h vs r_{RN}). Under the alignment limit $\lambda_{\text{prod}} = \lambda_{\text{planner}}$, $r_h = r_{\text{RN}}$, $\tau = 0$, Component 1 vanishes but Component 2 reduces to $\kappa_{\text{CAPEX}}/[u\ \text{AF}(r_{\text{RN}}, T)] - (\bar{p_{\text{SAF}}} - \bar{c_{\text{op}}})$, which is not zero in general: at the empirical calibration it equals \$1.52 − \$2.22 = −\$0.70*/gal. Full collapse $\sigma_{\text{prod}} \to \sigma^*_{\text{planner}}$ therefore requires a* fourth*, zero-profit condition — CAPEX recovery at the risk-neutral rate exactly offset by the operating margin, $\kappa_{\text{CAPEX}}/[u\ \text{AF}(r_{\text{RN}}, T)] = \bar{p_{\text{SAF}}} - \bar{c_{\text{op}}}$ — in addition to the three-parameter alignment (Proposition A.1 regression check).*

A.5. MCPF differential transfer (corollary)

COROLLARY A.10 (MCPF strengthens the CfD welfare advantage). *Under the conditions of Theorem B.3 below, when the marginal cost of public funds $\mu > 1$, the welfare gap $\Delta W_{\text{CfDvslin}}$ acquires an additional term* $-(\mu - 1)\cdot(\mathbb{E}[T_{\text{CfD}}] - \mathbb{E}[T_{\text{lin}}])$*. Since the CfD-with-floor pays only in low-price scenarios while linear σ pays in every scenario, $\mathbb{E}[T_{\text{CfD}}] < \mathbb{E}[T_{\text{lin}}]$ and the differential transfer is negative; the $-(\mu - 1)$ multiplier then flips the sign to positive, strictly strengthening the CfD welfare advantage.*

PROOF OF COROLLARY A.10. *Step 1: Welfare functional under MCPF* > 1. In the $\mu = 1$ reference case of Assumption 12, $W = \text{CS} + \text{PS} + \text{ENV} - T$. With $\mu > 1$, the welfare functional becomes $W^\mu = \text{CS} + \text{PS} + \text{ENV} - \mu T$, so $W^\mu - W = -(\mu - 1)\,T$, and the welfare gap acquires the term $-(\mu - 1)(\mathbb{E}[T_{\text{CfD}}] - \mathbb{E}[T_{\text{lin}}])$. The sign of this term reduces to the sign of the unconditional expected transfer differential, which we derive next.

Step 2: Strike-pinning + NPV-binding at producer-distorted weights. By Step 2 of Theorem B.3, the CfD-with-floor strike is $K^* = \bar{p}$ and the constant baseline s^*_0 is pinned by the producer NPV constraint at the producer-distorted weighting. Under the two-state collapse,

$$\sigma^{\text{CfD}}(H) = s^*_0, \qquad \sigma^{\text{CfD}}(L) = (\bar{p} - a_L) + s^*_0,$$

and the producer NPV-binding constraint (using the distorted-probability representation of Lemma B.1) equates the producer-distorted-weighted CfD transfer to the linear-σ level σ^{lin} that would also bind NPV:

$$\hat{\mathbb{E}}_{\text{prod}}[\sigma^{\text{CfD}}(\omega)] \;=\; \hat{\pi}_{\text{prod}}(H)\,s^*_0 + \hat{\pi}_{\text{prod}}(L)\,[(\bar{p} - a_L) + s^*_0] \;=\; s^*_0 + \hat{\pi}_{\text{prod}}(L)(\bar{p} - a_L) \;=\; \sigma^{\text{lin}}. \tag{A.28}$$

Solving (A.28) for the baseline, $s^*_0 = \sigma^{\text{lin}} - \hat{\pi}_{\text{prod}}(L)(\bar{p} - a_L)$.

Step 3: Unconditional-weighted transfer differential. At the planner's unconditional probability weighting (the relevant measure for the fiscal-cost component of welfare under MCPF > 1),

$$\begin{aligned}\mathbb{E}[\sigma^{\text{CfD}}(\omega)] &= \pi_H\, s_0^* + \pi_L\,[(\bar{p} - a_L) + s_0^*] \;=\; s_0^* + \pi_L(\bar{p} - a_L) \\ &= \sigma^{\text{lin}} + [\pi_L - \hat{\pi}_{\text{prod}}(L)](\bar{p} - a_L).\end{aligned} \tag{A.29}$$

Substituting $\hat{\pi}_{\text{prod}}(L) = (1 - \lambda_{\text{prod}})\pi_L + \lambda_{\text{prod}}$ from (B.2),

$$\pi_L - \hat{\pi}_{\text{prod}}(L) = \pi_L - (1 - \lambda_{\text{prod}})\pi_L - \lambda_{\text{prod}} = -\lambda_{\text{prod}}\,\pi_H < 0 \quad \text{for } \lambda_{\text{prod}} > 0. \tag{A.30}$$

Combining (A.29) and (A.30) and multiplying through by Q^* under the representative-investor reduction $Q^*(\omega) \approx Q^*$:

$$\mathbb{E}[T_{\text{lin}}] - \mathbb{E}[T_{\text{CfD}}] \;=\; \lambda_{\text{prod}} \cdot \pi_H \cdot (\bar{p} - a_L) \cdot Q^* \;>\; 0, \tag{A.31}$$

strictly positive whenever $\lambda_{\text{prod}} > 0$ (the producer is at least mildly risk-averse) and $\bar{p} > a_L$ (the low-state SAF price is below the unconditional mean, which holds under Assumption 4 by construction).

Step 4: MCPF-strengthening conclusion. Substituting (A.31) into the Step 1 welfare-gap adjustment:

$$\Delta W^{\mu}_{\text{CfDvslin}} - \Delta W^{\mu=1}_{\text{CfDvslin}} \;=\; -(\mu - 1) \cdot \big[\mathbb{E}[T_{\text{CfD}}] - \mathbb{E}[T_{\text{lin}}]\big] \;=\; (\mu - 1) \cdot \lambda_{\text{prod}} \cdot \pi_H \cdot (\bar{p} - a_L) \cdot Q^* \;>\; 0,$$

strictly positive whenever $\mu > 1$ and $\lambda_{\text{prod}} > 0$. At the empirical baseline ($\lambda_{\text{prod}} = 0.60$, $\pi_H = 0.472$, $\bar{p} - a_L = \$1.03$/gal, $Q^* = 1$ gal-equivalent), the MCPF-strengthening transfer differential is $(\mu - 1) \cdot 0.292$ \$/gal of producer output. At Dahlby's empirical MCPF range $\mu \in [1.10, 1.30]$, this contributes \$0.029–\$0.088/gal of additional welfare gain on top of the $\mu = 1$ baseline, strictly strengthening CfD-with-floor dominance. □

B. CfD-with-Floor Pareto Dominance

This appendix proves the second main result of Paper 2: the welfare optimality of the CfD-with-floor instrument within the restricted policy-instrument space

$$\mathcal{S} \;=\; \{\text{linear, CfD-strike, CfD-with-floor, capacity}\}$$

specified in Assumption 11. The proof technique is Borch (1962) / Wilson (1968) risk-sharing under complete information, adapted to two new ingredients of Paper 2: (i) discrete-scenario mean–CVaR (instead of the original continuous-state constant-absolute-risk-aversion (CARA) expected-utility (EU) framework), and (ii) the project-finance NPV-binding participation constraint at the producer's hurdle IRR (Assumption 13). The Holmstrom–Milgrom (1987) Theorem 7 result on optimality of linear contracts under unrestricted-σ plus CARA-Normal supplies the failure-mode benchmark: if Assumption 11 is dropped and the preference framework is moved from mean–CVaR to CARA-Normal, linear σ optimality is restored. Theorem B.3 proves the dominance result under the partition {Regime A: NPV-binding} ∪ {Regime B: NPV-slack}.

Two preparatory lemmas. The classical Borch (1962) / Wilson (1968) construction assumes twice-differentiable monotone-concave EU utilities and fully-state-contingent transfers, neither of which is literally true under

Paper 2's mean–CVaR preferences and restricted instrument class $\mathcal{S}$. The following two lemmas bridge that gap by (i) showing that in the 2-state $\alpha \geq \pi_H$ regime, mean–CVaR is literally linear-EU on a distorted probability measure (Lemma B.1), and (ii) deriving the Borch–Wilson state-by-state FOC explicitly from the Lagrangian dual of the restricted Stage 0b regulator problem (Lemma B.2).

LEMMA B.1 (Distorted-probability representation of mean–CVaR in the 2-state CVaR regime). *Under Assumption 1 with $\alpha > \pi_H$ strict (the boundary $\alpha = \pi_H$ admits a continuum of CVaR optimizers and the strict inequality is required for the closed-form to be unique; the empirical baseline $\alpha = 0.95 \gg \pi_H = 0.472$ is comfortably interior), for any state-contingent random variable $X : \Omega \to \mathbb{R}$ with $\Omega = \{H, L\}$ and any agent $i \in \{\text{prod}, \text{planner}\}$ with risk-aversion weight $\lambda_i \in [0, 1]$ (by Assumption 7),*

$$\text{(B.1)} \qquad \mathbb{E}_i[X] \;\equiv\; (1-\lambda_i)\,\mathbb{E}[X] + \lambda_i\ \mathrm{CVaR}_\alpha[X] \;=\; \hat{\pi}_i(H)\,X(H) + \hat{\pi}_i(L)\,X(L),$$

where the distorted probability measure $\hat{\pi}_i$ *on Ω is given by*

$$\text{(B.2)} \qquad \hat{\pi}_i(H) \;=\; (1-\lambda_i)\,\pi_H, \qquad \hat{\pi}_i(L) \;=\; (1-\lambda_i)\,\pi_L + \lambda_i \;=\; 1 - \hat{\pi}_i(H).$$

Both $\hat{\pi}_i(H), \hat{\pi}_i(L) \in [0, 1]$ and they sum to 1, so $\hat{\pi}_i$ is a probability measure on Ω. The producer's distortion uses $\lambda_{\text{prod}} = \lambda_{\text{prod}}$; the planner's uses $\lambda_{\text{planner}} = \lambda_{\text{planner}}$. The map $\lambda_i \mapsto \hat{\pi}_i(L)$ is strictly increasing in λ_i, so a more risk-averse agent assigns strictly more distorted mass to the bad-tail state L.

PROOF OF LEMMA B.1. *Step 1: Discrete CVaR collapse.* By the Rockafellar–Uryasev auxiliary-function representation (Rockafellar and Uryasev 2000), for any random variable X on the discrete state space $\Omega = \{H, L\}$ with $p_{\text{SAF}}(L) < p_{\text{SAF}}(H)$ (by Assumption 4) so that the bad-tail event is the low state L, and any tail level $\alpha \in (0, 1)$,

$$\mathrm{CVaR}_\alpha[X] \;=\; \sup_{\zeta \in \mathbb{R}} \Big\{ \zeta - \tfrac{1}{1-\alpha}\,\mathbb{E}\big[(\zeta - X)_+\big] \Big\}.$$

In the regime $\alpha \geq \pi_H$ (equivalently $1 - \alpha \leq \pi_L$), the worst-state probability π_L exceeds the tail mass $1 - \alpha$, so the unique maximizer is $\zeta^* = X(L)$ and the supremum simplifies to $\mathrm{CVaR}_\alpha[X] = X(L)$. (This is Paper 1's eq. (A.3) collapse and appears in our Equation (A.6) for the per-period cash flow.)

Step 2: Substitution into the mean–CVaR convex combination. Substituting $\mathrm{CVaR}_\alpha[X] = X(L)$ into the definition $\mathbb{E}_i[X] = (1-\lambda_i)\mathbb{E}[X] + \lambda_i\,\mathrm{CVaR}_\alpha[X]$ and expanding $\mathbb{E}[X] = \pi_H X(H) + \pi_L X(L)$,

$$\begin{aligned}
\mathbb{E}_i[X] &= (1-\lambda_i)\big[\pi_H X(H) + \pi_L X(L)\big] + \lambda_i X(L) \\
&= \underbrace{(1-\lambda_i)\pi_H}_{\equiv \hat{\pi}_i(H)} X(H) \;+\; \underbrace{\big[(1-\lambda_i)\pi_L + \lambda_i\big]}_{\equiv \hat{\pi}_i(L)} X(L).
\end{aligned}$$

Step 3: Verify $\hat{\pi}_i$ is a probability measure. Direct addition gives $\hat{\pi}_i(H) + \hat{\pi}_i(L) = (1-\lambda_i)(\pi_H + \pi_L) + \lambda_i = (1-\lambda_i) + \lambda_i = 1$. Both terms are non-negative: $\hat{\pi}_i(H) = (1-\lambda_i)\pi_H \in [0, \pi_H]$ since $\lambda_i \in [0, 1]$ by Assumption 7; and $\hat{\pi}_i(L) = 1 - \hat{\pi}_i(H) \in [\pi_L, 1]$. Both lie in $[0, 1]$, so $\hat{\pi}_i$ is a probability measure on Ω.

Step 4: Monotonicity in λ_i. $\partial\hat{\pi}_i(L)/\partial\lambda_i = 1 - \pi_L = \pi_H > 0$ under Assumption 1, so a larger risk-aversion weight shifts distorted mass strictly toward the bad-tail state L. □

LEMMA B.2 (Borch–Wilson FOC for the 2-state mean–CVaR risk-sharing problem under restricted instrument class). *Consider the Stage 0b regulator's problem under Assumptions 1 to 8 and Assumptions 9 to 13:*

$$\max_{\sigma\in\mathcal{S}} \ \hat{\pi}_{\text{planner}}(H)\, W(\sigma;H) + \hat{\pi}_{\text{planner}}(L)\, W(\sigma;L) \tag{B.3}$$
$$\text{s.t.} \quad \hat{\pi}_{\text{prod}}(H)\, \pi^{\text{op}}(\sigma;H) + \hat{\pi}_{\text{prod}}(L)\, \pi^{\text{op}}(\sigma;L) \geq 0,$$

where $\sigma \in \mathcal{S}$ is the policy instrument (Assumption 11), $W(\sigma;\omega)$ is the planner's state-ω welfare functional, and $\pi^{\text{op}}(\sigma;\omega)$ is the producer's state-ω Stage-2 operating-stage profit (equivalent to the NPV-per-gallon participation surplus). Then at any interior Pareto optimum $\sigma^* \in \mathcal{S}$ with binding NPV constraint and Lagrange multiplier $\mu^* > 0$, the Borch–Wilson state-by-state FOC ratio condition *holds:*

$$\boxed{\frac{\hat{\pi}_{\text{planner}}(\omega)}{\hat{\pi}_{\text{prod}}(\omega)} = \mu^* \cdot \frac{\partial W/\partial\sigma(\omega)}{\partial\pi^{\text{op}}/\partial\sigma(\omega)}, \qquad \omega \in \{H, L\}.} \tag{B.4}$$

Under the fiscal-cost-minimization formulation of Step 3 below in the $\mu = 1$ reference case of Assumption 12 and the capacity-binding representative-investor reduction, the welfare-relevant σ-channel operates through the producer's NPV-binding multiplier, and the FOC reduces (after the renormalization $\tilde{\mu}^* = \mu^* \cdot \text{AF}(r_h, T)(1-\tau) > 0$) to the state-independent ratio

$$\frac{\hat{\pi}_{\text{planner}}(\omega)}{\hat{\pi}_{\text{prod}}(\omega)} = \tilde{\mu}^* \qquad \textit{state-by-state at } \omega \in \{H, L\}. \tag{B.5}$$

For $\lambda_{\text{prod}} \neq \lambda_{\text{planner}}$ this ratio is not *state-independent under a constant linear σ (i.e., $\sigma(H) = \sigma(L)$), so linear σ is* not *Pareto-optimal in the restricted class $\mathcal{S}$. The optimal element of $\mathcal{S}$ shifts mass from $\sigma(H)$ to $\sigma(L)$ (i.e., switches from linear to CfD-with-floor) when $\lambda_{\text{prod}} > \lambda_{\text{planner}}$; the strict-inequality wedge is*

$$\frac{\hat{\pi}_{\text{planner}}(L)}{\hat{\pi}_{\text{prod}}(L)} - \frac{\hat{\pi}_{\text{planner}}(H)}{\hat{\pi}_{\text{prod}}(H)} = \frac{\lambda_{\text{planner}} - \lambda_{\text{prod}}}{(1-\lambda_{\text{prod}})\,\pi_L + \lambda_{\text{prod}}} \cdot \frac{\pi_H}{(1-\lambda_{\text{prod}})\pi_H} \tag{B.6}$$
$$< 0 \quad \textit{whenever } \lambda_{\text{prod}} > \lambda_{\text{planner}},$$

which strictly increases planner welfare while maintaining the producer NPV constraint.

PROOF OF LEMMA B.2. *Step 1: Distorted-probability representation.* By Lemma B.1, the planner's mean–CVaR objective and the producer's mean–CVaR participation constraint both reduce to *linear* functionals of the state-contingent random variables $W(\sigma;\cdot)$ and $\pi^{\text{op}}(\sigma;\cdot)$ under the respective distorted probability measures $\hat{\pi}_{\text{planner}}$ and $\hat{\pi}_{\text{prod}}$ defined in (B.2). Substituting these into the regulator's program yields exactly (B.3), which is a (finite-state) *linear program* in $(\sigma(H), \sigma(L))$ when $\sigma \in \mathcal{S}$ is parameterized by its state-contingent values — the linear instrument has $\sigma(H) = \sigma(L)$, the CfD-with-floor allows $\sigma(L) \geq \sigma(H)$, etc. The LP-duality framework of Shapiro, Dentcheva, and Ruszczyński (2014, Chapter 6, Stochastic Programming with Risk Measures) applies directly.

Step 2: Lagrangian and dual. Form the Lagrangian

$$\mathcal{L}(\sigma;\mu) = \sum_{\omega\in\{H,L\}} \hat{\pi}_{\text{planner}}(\omega)\, W(\sigma;\omega) + \mu \sum_{\omega\in\{H,L\}} \hat{\pi}_{\text{prod}}(\omega)\, \pi^{\text{op}}(\sigma;\omega), \tag{B.7}$$

with $\mu \geq 0$ the multiplier on the producer participation constraint. The Karush–Kuhn–Tucker (KKT) conditions at an interior optimum $\sigma^* \in \mathrm{int}(\mathcal{S})$ require stationarity state-by-state:

$$\frac{\partial \mathcal{L}}{\partial \sigma(\omega)} \;=\; \hat{\pi}_{\text{planner}}(\omega)\frac{\partial W}{\partial \sigma(\omega)} + \mu \hat{\pi}_{\text{prod}}(\omega)\frac{\partial \pi^{\text{op}}}{\partial \sigma(\omega)} \;=\; 0, \quad \omega \in \{H, L\}. \tag{B.8}$$

Rearranging (B.8) state-by-state and dividing,

$$\frac{\hat{\pi}_{\text{planner}}(\omega)}{\hat{\pi}_{\text{prod}}(\omega)} \;=\; -\mu \cdot \frac{\partial \pi^{\text{op}}/\partial \sigma(\omega)}{\partial W/\partial \sigma(\omega)}.$$

At the optimum, the producer participation constraint binds (by Assumption 13, the producer's project-finance hurdle is strictly above the planner's risk-neutral rate, so the participation constraint binds at any interior σ^*); hence $\mu^* > 0$, and the welfare-aggregation sign convention (transfer enters welfare with coefficient -1) makes $\mu^* \cdot \partial W/\partial \sigma(\omega) \cdot \partial \pi^{\text{op}}/\partial \sigma(\omega) > 0$ state-by-state. Reversing sign and rearranging gives (B.4).

Step 3: Reformulation as fiscal-cost minimization in the $\mu = 1$ reference case. In the $\mu = 1$ reference case of Assumption 12 and under the capacity-binding representative-investor reduction $Q^*(\sigma; \omega)/\bar{Q} \to 1$ at the SAF-positive scenario (Assumptions 3 and 5 inheritance from Paper 1's representative-investor reduction in §3.5), the direct σ-channel into planner welfare $W = \text{CS} + \text{PS} + \text{ENV} - T$ exhibits a producer-surplus–transfer cancellation: the transfer cost $T = \sigma(\omega)Q^*(\omega)$ contributes $-Q^*(\omega)$ to $\partial W/\partial \sigma(\omega)$, the producer-surplus boost $+\sigma(\omega)Q^*(\omega)$ contributes $+Q^*(\omega)$, and at capacity-binding $\partial Q^*/\partial \sigma = 0$ so consumer surplus and ENV are independent of σ. The planner's σ choice therefore has no direct welfare consequence at MCPF = 1; the choice is determined solely by the producer's participation constraint and (where MCPF > 1, including the $\mu = 1.20$ headline of Assumption 12) by minimization of expected fiscal cost.

The LP (B.3) is therefore equivalently formulated, at MCPF = 1, as fiscal-cost minimization subject to producer NPV-binding:

$$\min_{\sigma \in \mathcal{S}} \sum_{\omega \in \{H, L\}} \hat{\pi}_{\text{planner}}(\omega)\, \sigma(\omega)\, Q^*(\omega) \quad \text{s.t.} \quad \text{NPV}(\sigma; r_h, \kappa_{\text{CAPEX}}, u, \tau, T) = 0,\ \sigma \in \mathcal{S}, \tag{B.9}$$

where the NPV-binding constraint substitutes for the ≥ 0 participation: at any interior optimum the constraint binds by Assumption 13 since $r_h > r_{\text{RN}}$ implies the producer would exit at strictly positive NPV slack. Equivalence with the original welfare-maximization (B.3) at MCPF = 1 follows because the σ-independent welfare terms (CS, ENV, pre-subsidy operating margin) shift the objective by a constant.

The Lagrangian (B.7) is reinterpreted accordingly: at MCPF = 1, $\partial W/\partial \sigma(\omega) = 0$ in the direct channel; the welfare-relevant marginal effect operates entirely through the NPV-binding multiplier μ^*, and the KKT condition (B.8) for the fiscal-cost minimization reads:

$$-\hat{\pi}_{\text{planner}}(\omega)\, Q^*(\omega) \;+\; \mu^* \cdot \hat{\pi}_{\text{prod}}(\omega) \cdot \text{AF}(r_h, T) \cdot (1 - \tau) \cdot Q^*(\omega) \;=\; 0, \quad \omega \in \{H, L\}, \tag{B.10}$$

where $\text{AF}(r_h, T)(1 - \tau)Q^*(\omega)$ is $\partial \text{NPV}/\partial \sigma(\omega)$ at the producer's Stage-2 best response under the convention of Proposition A.3 Step 4. Dividing through by $Q^*(\omega) > 0$ and rearranging,

$$\frac{\hat{\pi}_{\text{planner}}(\omega)}{\hat{\pi}_{\text{prod}}(\omega)} \;=\; \mu^* \cdot \text{AF}(r_h, T) \cdot (1 - \tau), \quad \omega \in \{H, L\},$$

which is the state-independent ratio condition (B.5) with the renormalized multiplier $\tilde{\mu}^* \equiv \mu^* \cdot \mathrm{AF}(r_h, T) \cdot (1-\tau) > 0$ absorbed into the right-hand side. The Borch–Wilson state-by-state ratio condition holds in the standard form; the renormalization is fixed at any interior optimum by the binding NPV constraint (state-pooled, not state-by-state).

Step 4: State-independence test. Substituting (B.2) into the LHS of (B.5),

$$\begin{aligned}
\frac{\hat{\pi}_{\text{planner}}(H)}{\hat{\pi}_{\text{prod}}(H)} &= \frac{(1-\lambda_{\text{planner}})\pi_H}{(1-\lambda_{\text{prod}})\pi_H} = \frac{1-\lambda_{\text{planner}}}{1-\lambda_{\text{prod}}}, \\
\frac{\hat{\pi}_{\text{planner}}(L)}{\hat{\pi}_{\text{prod}}(L)} &= \frac{(1-\lambda_{\text{planner}})\pi_L + \lambda_{\text{planner}}}{(1-\lambda_{\text{prod}})\pi_L + \lambda_{\text{prod}}}.
\end{aligned}$$

At $\lambda_{\text{prod}} = \lambda_{\text{planner}}$ both ratios equal 1, so $\mu^* = 1$ and the FOC is satisfied state-by-state by any $\sigma \in \mathcal{S}$ — including the linear one. At $\lambda_{\text{prod}} \neq \lambda_{\text{planner}}$, however, the two ratios differ:

$$\begin{aligned}
&\frac{\hat{\pi}_{\text{planner}}(L)}{\hat{\pi}_{\text{prod}}(L)} - \frac{1-\lambda_{\text{planner}}}{1-\lambda_{\text{prod}}} \\
&\qquad = \frac{[(1-\lambda_{\text{planner}})\pi_L + \lambda_{\text{planner}}](1-\lambda_{\text{prod}}) - (1-\lambda_{\text{planner}})[(1-\lambda_{\text{prod}})\pi_L + \lambda_{\text{prod}}]}{[(1-\lambda_{\text{prod}})\pi_L + \lambda_{\text{prod}}](1-\lambda_{\text{prod}})} \\
&\qquad = \frac{\lambda_{\text{planner}} - \lambda_{\text{prod}}}{[(1-\lambda_{\text{prod}})\pi_L + \lambda_{\text{prod}}]\,(1-\lambda_{\text{prod}})},
\end{aligned}$$

where the numerator simplifies via $(1-\lambda_{\text{prod}})\lambda_{\text{planner}} - (1-\lambda_{\text{planner}})\lambda_{\text{prod}} = \lambda_{\text{planner}} - \lambda_{\text{prod}}$ after canceling the $\pi_L(1-\lambda_{\text{prod}})(1-\lambda_{\text{planner}})$ terms; the denominator retains the $(1-\lambda_{\text{prod}})$ factor from the cross-multiplication. For $\lambda_{\text{prod}} > \lambda_{\text{planner}}$ the wedge is strictly negative, so the bad-state ratio $\hat{\pi}_{\text{planner}}(L)/\hat{\pi}_{\text{prod}}(L)$ is strictly smaller than the good-state ratio $\hat{\pi}_{\text{planner}}(H)/\hat{\pi}_{\text{prod}}(H)$, contradicting the state-by-state FOC (B.5) at any constant σ. Constant (linear) σ is therefore *not* an interior optimum of (B.3) whenever $\lambda_{\text{prod}} \neq \lambda_{\text{planner}}$.

Step 5: Direction of the Pareto improvement via fiscal-cost reduction. Working with the fiscal-cost-minimization formulation (B.9) from Step 3, consider an admissible reallocation $(d\sigma(H), d\sigma(L))$ in $\mathcal{S}$ that holds the producer NPV constraint at equality:

$$\hat{\pi}_{\text{prod}}(H)\, Q^*(H)\, d\sigma(H) \;+\; \hat{\pi}_{\text{prod}}(L)\, Q^*(L)\, d\sigma(L) \;=\; 0. \tag{B.11}$$

The change in expected fiscal cost is

$$dF \;=\; \hat{\pi}_{\text{planner}}(H)\, Q^*(H)\, d\sigma(H) \;+\; \hat{\pi}_{\text{planner}}(L)\, Q^*(L)\, d\sigma(L).$$

Substituting (B.11) to eliminate $d\sigma(H)$,

$$dF \;=\; \hat{\pi}_{\text{prod}}(L)\, Q^*(L)\, d\sigma(L) \;\cdot\; \left[\frac{\hat{\pi}_{\text{planner}}(L)}{\hat{\pi}_{\text{prod}}(L)} - \frac{\hat{\pi}_{\text{planner}}(H)}{\hat{\pi}_{\text{prod}}(H)}\right]. \tag{B.12}$$

For $\lambda_{\text{prod}} > \lambda_{\text{planner}}$ the bracketed term is strictly negative (Equation (B.6) of Step 4). With $d\sigma(L) > 0$ and $\hat{\pi}_{\text{prod}}(L)\, Q^*(L) > 0$, the product gives $dF < 0$: increasing $\sigma(L)$ at the producer-distorted-weight expense of $\sigma(H)$ strictly reduces the planner's expected fiscal cost while leaving the producer's NPV-binding participation intact.

In the $\mu = 1$ reference case this fiscal-cost reduction is welfare-neutral — at the common measure the producer receives exactly what the fisc saves (Step 3); under MCPF > 1 it converts into a welfare gain of $(\mu - 1)\,|dF|$ (Corollary A.10). The Pareto-improving direction is therefore unambiguously $d\sigma(L) > 0, d\sigma(H) < 0$: shift subsidy mass from the high state to the low (risk-tail) state. Within the restricted class $\mathcal{S}$ (Assumption 11), only the CfD-with-floor instrument $\sigma(\omega) = \max\{\bar{p} - p_{\mathrm{SAF}}(\omega), 0\} + s_0$ implements $\sigma(L) > \sigma(H)$; linear and capacity are constant in ω, and CfD-strike is anti-aligned (it transfers in the high state). Hence CfD-with-floor is the unique element of $\mathcal{S}$ that *shifts the subsidy-level decomposition* of σ_{prod} in the direction $(\lambda_{\mathrm{prod}} - \lambda_{\mathrm{planner}})\pi_H(a_H - a_L)Q^*$ relative to constant σ at $\lambda_{\mathrm{prod}} > \lambda_{\mathrm{planner}}$, holding the producer NPV constraint fixed. Note that this subsidy-level wedge does *not* translate directly into a welfare-gain term in (B.14) under the common-measure planner welfare evaluation maintained throughout (see Theorem Step 6 and Lemma B.2 Step 3): the welfare effect of the σ-reallocation across states is folded into the MCPF transfer-differential channel (Corollary A.10) and appears as Component 2 of (B.14), not as a separate Component 1. □

Remark on the LP-dual derivation. The above proof of Lemma B.2 explicitly derives the LP dual for the 2-state mean–CVaR risk-sharing problem with the restricted instrument class $\mathcal{S}$ — no appeal to continuous-state CARA or full-state-contingent transfers. The key reduction is Lemma B.1: at $\alpha \geq \pi_H$, mean–CVaR *is* linear-EU on the distorted probability measure $\hat{\pi}_i$, so the Borch (1962) / Wilson (1968) state-by-state ratio condition applies verbatim, with the only modification being the use of distorted probabilities in place of true probabilities. This makes the "CVaR is not strictly concave EU" objection moot in the 2-state $\alpha \geq \pi_H$ regime that contains the $\alpha = 0.95$ baseline.

THEOREM B.3 (CfD-with-floor weak Pareto dominance under partition). *Under Assumptions 1 to 8 and Assumptions 9 to 13, with $\lambda_{\mathrm{prod}} > 0$, the welfare-optimal $\sigma \in \mathcal{S}$ at the Stage 0b regulator problem is the CfD-with-floor instrument*

$$\sigma^*(\omega) \;=\; \max\{\,\bar{p} - p_{\mathrm{SAF}}(\omega),\, 0\,\} \;+\; s_0^*, \tag{B.13}$$

with strike $K^* = \bar{p}$ and constant baseline s_0^* pinned by the producer's project-finance participation constraint at the hurdle IRR (i.e., $\mathrm{NPV}(\sigma^*; r_h, \kappa_{\mathrm{CAPEX}}, u, \tau, T) = 0$*). The first-order welfare wedge against the linear baseline σ^{lin} obeys the partition*

$$\Delta W_{\mathrm{CfDvslin}} = \underbrace{0}_{\substack{\textit{Subsidy-level risk-sharing wedge}\\ \textit{(folded into common-measure cancellation;}\\ \textit{see Lemma B.2 Step 3)}}} + \underbrace{(\mu_W - 1)\cdot\lambda_{\mathrm{prod}}\cdot\pi_H\cdot(\bar{p} - a_L)\cdot\mathbb{1}_{\{\text{NPV-binding}\}}\cdot Q^*}_{\substack{\textit{MCPF transfer-differential channel}\\ \textit{(actual welfare gain)}}} + O\big((\mu_W - 1)^2\big), \tag{B.14}$$

where $\mathbb{1}_{\{\text{NPV-binding}\}}$ is the indicator that the producer's NPV constraint binds at σ^{lin}. The partition splits the welfare comparison into:

Regime A (NPV-binding). $\mathrm{NPV}(\sigma^{\mathrm{lin}}; r_h, \kappa_{\mathrm{CAPEX}}, u, \tau, T) = 0$ *at σ^{lin}. In the $\mu_W = 1$ reference case of Assumption 12, the producer-surplus–transfer cancellation of Lemma B.2 Step 3 implies $\Delta W = 0$ and CfD-with-*

floor weakly *(with equality) Pareto-dominates linear σ. At $\mu_W > 1$, the MCPF transfer-differential channel of Corollary A.10 delivers $\Delta W = (\mu_W - 1)\lambda_{\text{prod}}\pi_H(\bar{p} - a_L)Q^* > 0$,* strict *Pareto dominance. At Dahlby's empirical US-federal MCPF range $\mu_W \in [1.10, 1.30]$, $\Delta W \in [\$0.029, \$0.088]$/gal of producer output. The empirical baseline calibration falls in Regime A; the welfare gain is small in magnitude but sign-determinate.*

Regime B (NPV-slack). NPV$(\sigma^{\text{lin}}; r_h, \kappa_{\text{CAPEX}}, u, \tau, T) > 0$ *at σ^{lin}. Then $\mathbb{1}_{\{\text{NPV-binding}\}} = 0$ and the MCPF channel vanishes: $\Delta W = 0$ even at $\mu_W > 1$. CfD-with-floor is welfare-equivalent to linear σ in Regime B (weak Pareto dominance with equality at all μ_W).*

The risk-sharing channel $(\lambda_{\text{prod}} - \lambda_{\text{planner}})\,\pi_H\,(a_H - a_L)\,Q^*$ that appears as a subsidy-level wedge in (A.14) (i.e., $\sigma_{\text{prod}} - \sigma^*_{\text{planner}}$) does not *translate into a welfare-gain channel in (B.14) at the common-measure evaluation maintained throughout, because the σ-reallocation across states that the risk-sharing channel would mobilize is bound by the producer-NPV-binding constraint at producer-distorted weights (Lemma B.2 Step 5); under the common-measure planner welfare evaluation, the reallocation reduces planner expected fiscal cost by $\lambda_{\text{prod}}\pi_H(\bar{p} - a_L)Q^*$ (Corollary A.10) but is welfare-neutral redistribution at $\mu_W = 1$, converting into welfare gain only via the MCPF weight $(\mu_W - 1)$. The first-best diagnostic of Step 4b describes the additional fiscal slack the planner could capture* if *she could substitute her risk-neutral discount rate r_{RN} for the producer's hurdle r_h in the project-finance constraint; this is the policy-relevant ceiling on direct-discount-rate substitution and not an achievable welfare gain within $\mathcal{S}$.*

Proof of Theorem B.3. *Step 1: Borch–Wilson state-by-state FOC under restricted $\mathcal{S}$.* Step 1 now follows directly from Lemma B.2, the discrete-scenario mean–CVaR Borch–Wilson FOC derived in the preceding subsection from the explicit LP dual of the restricted Stage 0b regulator program. By Lemma B.1, the mean–CVaR objective and the producer participation constraint both reduce to linear functionals on the distorted probability measures $\hat{\pi}_{\text{planner}}$ and $\hat{\pi}_{\text{prod}}$ respectively, so the Lagrangian saddle-point conditions (B.8) of Lemma B.2 apply verbatim with no appeal to CARA or fully-state-contingent transfers.

Under Assumption 13 the producer NPV constraint binds at σ^* (the project-finance hurdle r_h strictly exceeds the planner risk-neutral rate r_{RN}), so $\mu^* > 0$. The fiscal-cost-minimization reformulation of Lemma B.2 Step 3 in the $\mu = 1$ reference case of Assumption 12 yields the KKT condition (B.10), which after dividing by $Q^*(\omega) > 0$ and renormalizing $\tilde{\mu}^* \equiv \mu^* \cdot \text{AF}(r_h, T)(1 - \tau) > 0$ reduces to the state-independent ratio condition (B.5): $\hat{\pi}_{\text{planner}}(\omega)/\hat{\pi}_{\text{prod}}(\omega) = \tilde{\mu}^*$ at $\omega \in \{H, L\}$. At the empirical baseline calibration $\lambda_{\text{prod}} = 0.60 > 0.50 = \lambda_{\text{planner}}$ this ratio fails state-by-state at any constant σ (Lemma B.2 Step 4), so the linear σ^{lin} is *not* Pareto-optimal in $\mathcal{S}$ under the empirical $\lambda_{\text{prod}} > \lambda_{\text{planner}}$ case.

Within the restricted instrument class $\mathcal{S}$ (Assumption 11), the CfD-with-floor $\sigma(\omega) = \max\{\bar{p} - p_{\text{SAF}}(\omega), 0\} + s_0$ is the *unique* instrument that implements $\sigma(L) > \sigma(H)$ — linear and capacity are constant in ω, and CfD-strike is anti-aligned (it transfers in the high state, not the low). By Assumption 4 the bad-tail event is the low state L, so the welfare-improving reallocation under $\lambda_{\text{prod}} > \lambda_{\text{planner}}$ is the CfD-with-floor design (B.13). The strike $K^* = \bar{p}$ is the Newbery 2023-style efficient revenue-assurance strike (it ensures the producer's risk-adjusted expected revenue equals the KDE-mean $\bar{p_{\text{SAF}}}$ at the risk-sharing fixed point), and the constant baseline s_0 is pinned by the financial-equivalence participation constraint NPV$(\sigma^*; r_h, \kappa_{\text{CAPEX}}, u, \tau, T) = 0$.

Step 2: Optimal strike and floor pinning. At the Stage 0b regulator's problem, conditional on the form "$\sigma(\omega) = \max\{K - p_{\text{SAF}}(\omega), 0\} + s_0$", the strike K and floor s_0 are jointly determined by maximizing $(1 - \lambda_{\text{planner}})\,\mathbb{E}[W] + \lambda_{\text{planner}}\,\text{CVaR}_\alpha[W]$ subject to NPV$(\sigma; r_h, \kappa_{\text{CAPEX}}, u, \tau, T) \geq 0$ and the producer's Stage 2 best response from Lemma A.4. By the envelope theorem applied to the planner's mean–CVaR objective at the producer's Stage 2 best response, the FOC over K yields $K^* = \bar{p}$ when the welfare evaluation is at the unconditional KDE mean (this is the Newbery 2023 "revenue-assurance at the mean" strike); the participation constraint pins s_0^* uniquely from $\mathrm{NPV}(\sigma^*; r_h, \kappa_{\mathrm{CAPEX}}, u, \tau, T) = 0$.

Step 3: Subsidy-level risk-sharing wedge and its common-measure cancellation. Compare σ^* in (B.13) with σ^{lin}. Evaluated at each agent's *own* distorted measure (Lemma B.1), the reallocation of transfer mass off the high state into the low-state tail is worth $\lambda_{\mathrm{prod}}\, \pi_H\, (a_H - a_L)\, Q^*$ to the producer while costing the planner $\lambda_{\mathrm{planner}}\, \pi_H\, (a_H - a_L)\, Q^*$; the difference $(\lambda_{\mathrm{prod}} - \lambda_{\mathrm{planner}})\, \pi_H\, (a_H - a_L)\, Q^*$ (= \$0.103/gal at the baseline, $\lambda_{\mathrm{prod}} = 0.60 > 0.50 = \lambda_{\mathrm{planner}}$) is the *subsidy-level* risk-sharing wedge — the same object that appears as Component 1 of the wedge decomposition (A.16). It measures how much more cheaply the CfD-with-floor satisfies the producer's participation constraint, *not* a first-order welfare gain: under the common-measure planner evaluation maintained throughout (Lemma B.2 Step 3), the producer-surplus and fiscal-transfer contributions of any σ-reallocation cancel dollar-for-dollar, so this wedge enters (B.14) with coefficient *zero* at $\mu_W = 1$. Its welfare content operates only through the expected fiscal-cost differential it generates between the two contracts, which Step 4c derives and the MCPF channel converts into welfare at rate $(\mu_W - 1)$.

Step 4: MCPF transfer-differential welfare gain, with a first-best diagnostic. The welfare-gain derivation requires care: we distinguish (i) a *first-best fiscal-slack diagnostic* that quantifies the counterfactual welfare gain the planner would capture *if* she could substitute her risk-neutral discount rate r_{RN} for the producer's hurdle r_h in the project-finance constraint, from (ii) the *actual welfare gain* from switching from the linear σ to the CfD-with-floor under producer-NPV-binding at the unchanged hurdle r_h. Neither candidate contract in Theorem B.3's comparison implements the substitution underlying (i); the diagnostic is reported for expositional clarity, but the welfare gain that drives the dominance verdict is (ii), derived from Corollary A.10.

Step 4a (envelope theorem on regulator's value function). Let $V^*(r_h, r_{\mathrm{RN}}, \kappa_{\mathrm{CAPEX}}, \tau, u, T)$ denote the regulator's value function at the producer-NPV-binding optimum. By the standard envelope theorem (Milgrom and Segal 2002, Theorem 1), differentiating V^* with respect to the NPV-constraint slack yields $\partial V^*/\partial(\mathrm{NPV\ slack}) = \mu^* > 0$, the Lagrange multiplier from Lemma B.2 (strictly positive under Assumption 13).

Step 4b (first-best fiscal-slack diagnostic — COUNTERFACTUAL). A useful counterfactual benchmark, NOT a welfare gain delivered by any contract in $\mathcal{S}$: holding the after-tax cash flow stream fixed, hypothetically replacing the producer's discount factor $\mathrm{AF}(r_h, T)$ with the planner's $\mathrm{AF}(r_{\mathrm{RN}}, T) > \mathrm{AF}(r_h, T)$ in the NPV constraint would release fiscal slack of magnitude

$$\text{First-best diagnostic} \;=\; \frac{\kappa_{\mathrm{CAPEX}}}{u} \cdot \frac{\mathrm{AF}(r_{\mathrm{RN}}, T) - \mathrm{AF}(r_h, T)}{\mathrm{AF}(r_h, T)},$$

since $\mathrm{AF}(\cdot, T)$ is strictly decreasing in r ($\partial\,\mathrm{AF}/\partial r = T(1+r)^{-T-1}/r - \mathrm{AF}(r,T)/r < 0$ for $r > 0, T \geq 1$, where the second term dominates at the empirical baseline). At the empirical calibration $(r_h, r_{\mathrm{RN}}, T) = (0.125, 0.03, 10)$: $\mathrm{AF}(r_{\mathrm{RN}}) = 8.5302$, $\mathrm{AF}(r_h) = 5.5364$, wedge $= 0.541 = 54.1\%$, and the diagnostic equals $12.94 \cdot 0.541 = \$7.00$ per gallon of *annual capacity* — a stock magnitude accumulated over the asset life, not a per-gallon flow. The flow equivalent, converting through the annuity factor and tax gross-up exactly as in Term 1 of (10), is $\kappa_{\mathrm{CAPEX}}/[u(1-\tau)] \cdot [1/\mathrm{AF}(r_h, T) - 1/\mathrm{AF}(r_{\mathrm{RN}}, T)] \approx \$1.04$/gal of annual output. This is the **market-completion ceiling on what the planner could conceivably save** via direct discount-rate substitution; it is *not* a welfare gain achievable within the policy-instrument class $\mathcal{S}$, since Assumption 11 forces every $\sigma \in \mathcal{S}$ to satisfy producer-NPV-binding at the producer's hurdle r_h.

Step 4c (actual welfare gain from CfD-floor vs linear σ). The actual welfare gain delivered by the CfD-with-floor over the NPV-binding linear $\sigma^{\text{lin}} = \sigma_{\text{prod}}$ comes from the transfer-cost differential between the two contracts evaluated at the planner's unconditional probability measure (Corollary A.10 Step 3 derivation):

$$\mathbb{E}[T_{\text{lin}}] - \mathbb{E}[T_{\text{CfD}}] \;=\; \lambda_{\text{prod}} \cdot \pi_H \cdot (\bar{p} - a_L) \cdot Q^* \;>\; 0 \tag{B.15}$$

for $\lambda_{\text{prod}} > 0$ and $\bar{p} > a_L$. At the empirical baseline ($\lambda_{\text{prod}} = 0.60, \pi_H = 0.472, \bar{p} - a_L = \$1.03/\text{gal}$), the transfer differential equals \$0.292/gal of producer output.

In the $\mu_W = 1$ reference case of Assumption 12, one dollar of expected fiscal-cost saving equals zero dollars of welfare gain: the Lemma B.2 Step 3 producer-surplus–transfer cancellation ($\partial W/\partial\sigma|_{\text{direct}} = 0$ at common-measure evaluation) ensures that fiscal-cost reduction at $\mu_W = 1$ is welfare-neutral redistribution. Under $\mu_W > 1$ (including the $\mu_W = 1.20$ headline of Assumption 12), the differential converts into welfare gain at rate $(\mu_W - 1)$ via the MCPF wedge. Combining:

$$\boxed{\begin{aligned}&\text{MCPF transfer-differential welfare gain}\\ &\quad = \; (\mu_W - 1) \cdot \lambda_{\text{prod}} \cdot \pi_H \cdot (\bar{p} - a_L) \cdot \mathbb{1}_{\{\text{NPV-binding}\}} \cdot Q^*\end{aligned}} \tag{B.16}$$

which equals *zero* at $\mu_W = 1$ (the reference case yields no welfare gain) and equals $(\mu_W - 1) \cdot \$0.292/\text{gal}$ at $\mu_W > 1$. At Dahlby's empirical US-federal MCPF range $\mu_W \in [1.10, 1.30]$, the channel delivers \$0.029–\$0.088/gal of producer output.

Step 4d (orders of magnitude and reconciliation with Step 4b). The flow-equivalent first-best diagnostic of Step 4b ($\approx$ \$1.04/gal) and the actual welfare gain of Step 4c (\$0–\$0.088/gal at $\mu_W \le 1.30$) differ by roughly an order of magnitude. The gap is the cost of Assumption 11: by restricting the policy class to the four-element $\mathcal{S}$ (with producer-NPV-binding at r_h, not r_{RN}), the planner forgoes the first-best fiscal slack and captures only the MCPF-channel sliver. This is the policy-relevant quantification — the diagnostic answers "how much room exists in principle," while (B.16) answers "how much the CfD-with-floor actually delivers given the legally feasible instrument class."

When $\mathbb{1}_{\{\text{NPV-binding}\}} = 0$ (Regime B), both the diagnostic and the actual welfare gain collapse to zero — the NPV constraint is slack and neither the first-best wedge nor the transfer differential is mobilized.

Step 5: Higher-order residual. The $O\big((\mu_W - 1)^2\big)$ residual in (B.14) collects the interaction between the MCPF wedge and the Stage-2 output response: the CfD-with-floor shifts transfer mass across states, so $Q^*(\sigma; \omega)$ differs between the two contracts state-by-state (Lemma A.4), and the fiscal-cost differential evaluated at contract-specific outputs departs from the common-Q^* first-order term at second order in the wedge. Terms quadratic in the risk-aversion weights do not appear separately: the common-measure cancellation of Step 3 holds exactly in the two-state collapsed regime, not merely to first order. The residual, together with the binary facility-siting dynamics, is treated numerically in the SE-US 248/34 SP-MILP exhibit deferred to a numerical companion (forthcoming).

Step 6: Welfare ordering and dominance verdict. The dominance claim of Theorem B.3 compares two candidate designs at matched producer-NPV-binding: the linear subsidy $\sigma^{\text{lin}} = \sigma_{\text{prod}} = \$1.36/\text{gal}$ (Proposition A.3) and the CfD-with-floor strike-and-baseline pair (K^*, s_0^*). At this matched comparison the producer's participation constraint binds *by construction* in both designs; this is the Regime A partition. (The post-2025 CI-scaled level of \$0.74–\$0.83/gal falls well below σ_{prod} = \$1.36, and the lower portion of the as-enacted band (\$1.30–\$1.36) falls slightly below it; at those levels the producer's NPV is negative and Stage 1 participation fails, while the as-enacted band's upper portion clears the constraint (Section 4.3). The linear-design benchmark in Theorem B.3 is the NPV-binding $\sigma^{\text{lin}} = \sigma_{\text{prod}}$ — the boundary the as-enacted band straddles — not any single observed credit level.)

The welfare verdict is regime- and MCPF-conditional, consistent with the common-measure cancellation of Lemma B.2 Step 3 and the fresh re-derivation of Step 4c above.

In the $\mu_W = 1$ reference case (Assumption 12): the direct σ-channel into planner welfare cancels (producer surplus contribution and fiscal-transfer cost evaluated at a common probability measure), so $\partial W/\partial\sigma|_{\text{direct}} = 0$ and any element of $\mathcal{S}$ satisfying NPV-binding is welfare-equivalent. Theorem B.3 delivers *weak Pareto dominance* of CfD-with-floor over the NPV-binding linear σ at this baseline; the producer is indifferent (NPV-binding in both designs), the planner is indifferent (direct welfare unchanged), and the dominance is weak rather than strict. The \$7.00/gal first-best diagnostic of Step 4b is a counterfactual ceiling, not an actual welfare gain at $\mu_W = 1$.

At MCPF > 1 (the $\mu_W = 1.20$ headline of Assumption 12; Dahlby's empirical US-federal range $\mu_W \in [1.10, 1.30]$): Corollary A.10 and Step 4c above establish that the expected fiscal-cost differential $\mathbb{E}[T_{\text{lin}}] - \mathbb{E}[T_{\text{CfD}}] = \lambda_{\text{prod}}\pi_H(\bar{p} - a_L)Q^*$ = \$0.292/gal converts into welfare gain at rate $(\mu_W - 1)$, yielding $\Delta W_{\text{CfDvslin}} = (\mu_W - 1) \cdot$ \$0.292/gal of producer output. At $\mu_W = 1.30$ this is \$0.088/gal > 0, so *CfD-with-floor strictly Pareto-dominates linear σ in Regime A under $\mu_W > 1$ via the MCPF transfer-differential channel*. The strict dominance is small in magnitude (sub-\$0.10/gal) but sign-determinate and robust within the empirical MCPF range.

In Regime B (NPV-slack): both the first-best diagnostic and the actual MCPF-channel welfare gain collapse to zero. CfD-with-floor is welfare-equivalent to linear σ (weak dominance with equality).

The headline empirical contribution of Theorem B.3 is therefore not a strict-dominance claim at the unweighted $\mu_W = 1$ baseline but rather: (a) weak Pareto dominance at $\mu_W = 1$; (b) strict Pareto dominance at $\mu_W > 1$ via the MCPF channel; and (c) identification of a quantitatively small but structurally robust welfare gain that scales linearly in $(\mu_W - 1) \cdot \lambda_{\text{prod}} \cdot \pi_H \cdot (\bar{p} - a_L)$. □

PROPOSITION B.4 (Mechanism for departure from Holmström and Milgrom 1987 linear-optimal). *Within the discrete-scenario mean–CVaR + project-finance framework of Lemmas B.1 and B.2 and Theorem B.3, the CfD-with-floor weak Pareto dominance over linear σ depends jointly on* three *structural primitives:*

(i) the policy-instrument restriction to $\mathcal{S}$ (Assumption 11);

(ii) the mean–CVaR preference framework with $\lambda_{\text{prod}} > \lambda_{\text{planner}}$ (generates the subsidy-level risk-sharing wedge $(\lambda_{\text{prod}} - \lambda_{\text{planner}})\pi_H(a_H - a_L)$ that absorbs into the MCPF channel of (B.14) *via Theorem Step 6);*

(iii) the project-finance hurdle inequality $r_h > r_{\text{RN}}$ (Assumption 13, generates the financial-frictions wedge underlying the MCPF transfer-differential channel of (B.14)*).*

Specifically:

(a) ***Linear-σ optimality (Holmstrom–Milgrom regime).*** *If* all three *primitives are simultaneously relaxed —*

i.e., σ is an unrestricted measurable function of ω, mean–CVaR is replaced by CARA-Normal preferences, and $r_h = r_{\mathrm{RN}}$ — then by Holmström and Milgrom (1987, Theorem 7), linear σ is the unique Pareto-optimal contract within the now-unrestricted class. The CfD-with-floor dominance result of Theorem B.3 fails.

(b) ***Single-primitive retention (three cases).*** *The three cases are* asymmetric*:*

- Only (i) retained *($\mathcal{S}$-restriction, with $\lambda_{\mathrm{prod}} = \lambda_{\mathrm{planner}}$ and $r_h = r_{\mathrm{RN}}$): the subsidy-level risk-sharing wedge vanishes, and Component 2 reduces to its risk-neutral residual (zero only under the additional zero-profit condition of Remark A.9). In the $\mu_W = 1$ reference case every NPV-binding element of $\mathcal{S}$ is welfare-equivalent; at $\mu_W > 1$ the MCPF transfer-differential channel at the common weight $\lambda_{\mathrm{planner}} > 0$ still selects the CfD-with-floor.*
- Only (ii) retained *(mean–CVaR with $\lambda_{\mathrm{prod}} > \lambda_{\mathrm{planner}}$, under unrestricted σ and $r_h = r_{\mathrm{RN}}$): the Borch–Wilson FOC of Lemma B.2 pins a* unique fully state-contingent *first-best transfer that* strictly *Pareto-dominates the linear form (and CfD-with-floor, since the unrestricted optimum lies outside $\mathcal{S}$). The headline strengthens rather than degenerates; the restriction to $\mathcal{S}$ becomes the binding obstruction to the first-best, exactly as documented by Theorem Step 4b's first-best fiscal-slack diagnostic.*
- Only (iii) retained *($r_h > r_{\mathrm{RN}}$, with $\lambda_{\mathrm{prod}} = \lambda_{\mathrm{planner}}$ and unrestricted σ): Component 2 of the subsidy wedge survives but Component 1 vanishes. The CfD-with-floor dominance result reduces to a project-finance–only mechanism; the welfare wedge is $(\mu_W - 1)\,\lambda_{\mathrm{planner}}\,\pi_H\,(\bar{p} - a_L)\,\mathbb{1}_{\{\text{NPV-binding}\}}\,Q^*$ — the same MCPF transfer-differential channel as in the full Theorem, still positive because the common risk weight $\lambda_{\mathrm{planner}} > 0$ preserves the CfD's state-contingent fiscal saving — while the subsidy-level risk-sharing wedge is zero ($\lambda_{\mathrm{prod}} - \lambda_{\mathrm{planner}} = 0$).*

(c) ***Two-primitive retention (Theorem regime).*** *If at least two of (i)–(iii) are simultaneously retained, then the MCPF transfer-differential channel of (B.14) is weakly positive at the empirical baseline, CfD-with-floor weakly Pareto-dominates linear σ in Regime A (with equality at $\mu_W = 1$ and strict inequality at $\mu_W > 1$) and weakly in Regime B, and the headline empirical result holds.*

The three primitives thus jointly identify the policy-relevant *precise mechanism* by which observed energy-policy instruments depart from the H-M (1987) linear-optimal benchmark.

Proof of Proposition B.4. For part (a): under unrestricted σ, CARA-Normal preferences, and $r_h = r_{\mathrm{RN}}$, the regulator's problem reduces to a risk-sharing problem in the standard Holmström and Milgrom (1987) setup, where Theorem 7 establishes linear σ as the unique optimum. The proof structure relies on (i) full state-contingent measurability of σ, (ii) the exponential–Gaussian mean-variance separability of CARA-Normal, and (iii) the absence of a separate project-finance hurdle-IRR constraint that would introduce a financial-frictions wedge between the producer's NPV and the planner's welfare evaluation. Relaxing any of these three breaks the H-M proof structure and admits non-linear optima.

For part (b), the three single-primitive cases are asymmetric. If only (i) is retained ($\mathcal{S}$-restriction with $\lambda_{\mathrm{prod}} = \lambda_{\mathrm{planner}}$ and $r_h = r_{\mathrm{RN}}$), the subsidy-level risk-sharing wedge vanishes; in the $\mu_W = 1$ reference case any NPV-binding element of $\mathcal{S}$ is welfare-equivalent, while at $\mu_W > 1$ the transfer-differential channel at weight $\lambda_{\mathrm{planner}} > 0$ still favors the CfD-with-floor. If only (ii) is retained (mean–CVaR with $\lambda_{\mathrm{prod}} > \lambda_{\mathrm{planner}}$ but unrestricted σ and $r_h = r_{\mathrm{RN}}$), the Borch–Wilson FOC of Lemma B.2 pins a *unique* fully state-contingent transfer $\sigma^{\text{first-best}}(\omega) = \arg\max_{\sigma} \mathbb{E}_{\text{planner}}[W(\sigma;\omega)]$ that strictly Pareto-dominates both linear σ and CfD-with-floor (since the unrestricted optimum lies outside $\mathcal{S}$); the strict-dominance verdict holds via the same distorted-measure machinery of Lemma B.1. If only (iii) is retained ($r_h > r_{\mathrm{RN}}$ with $\lambda_{\text{prod}} = \lambda_{\text{planner}}$ and unrestricted σ), Component 2 of the subsidy wedge survives but Component 1 vanishes; the CfD dominance reduces to a project-finance-only mechanism operating through the MCPF transfer-differential channel at weight λ_{planner}.

For part (c): with at least two primitives retained, the chain Lemma B.1 → Lemma B.2 → Theorem B.3 applies with the relevant Component strictly positive, and the proof of Theorem B.3 delivers the dominance verdict. □

remark B.5 (Empirical policy diagnostic from Proposition B.4)**.** *Proposition B.4 provides a tractable diagnostic for classifying observed energy-policy production credits relative to the H-M linear-optimal benchmark. For the IRA Section 45Z SAF credit, all three primitives hold: (i) the 45Z credit is statutorily a linear-σ form within the restricted class $\mathcal{S}$ of Assumption 11; (ii) the empirically calibrated $\lambda_{\text{prod}} = 0.60 > 0.50 = \lambda_{\text{planner}}$ generates a strictly positive Component 1; (iii) the DOE-anchored, Lazard/NREL-cross-checked $r_h = 0.125 > 0.03 = r_{\mathrm{RN}}$ generates a strictly positive Component 2. The CfD-with-floor dominance result therefore applies to 45Z; the policy implication is that a CfD-with-floor analog of the 45Z would Pareto-dominate the observed linear form — strictly at the $\mu_W = 1.20$ headline of Assumption 12, weakly with equality in the $\mu_W = 1$ reference case. For other IRA production credits with different $(\lambda_{\text{prod}}, r_h)$ calibrations (e.g., 45V for hydrogen, 45Y for electricity, 45Q for carbon capture and storage (CCS)), the diagnostic applies separately and yields different dominance verdicts.*

remark B.6 (Floor parameter F^* pinning)**.** *The strike $K^* = \bar{p}$ and constant baseline s_0^* are both closed-form (Steps 2–4 above). A distinct "floor parameter" $F^* \equiv K^* - \bar{c}$ characterizing the guaranteed-revenue floor may be numerically pinned at the SE-US 248/34 SP-MILP exhibit in the numerical companion when the producer's marginal cost $\bar{c}$ inherits stochastic variation; the closed-form strike-and-baseline-pinning above suffices for the theorem statement and for the empirical headline calibration.*

remark B.7 (CfD kink and FOC validity at the 2-state collapse)**.** *The CfD-with-floor instrument $\sigma(\omega) = \max\{\bar{p} - p_{\mathrm{SAF}}(\omega), 0\} + s_0$ has a piecewise-linear kink at the price threshold $p_{\mathrm{SAF}}(\omega) = \bar{p}$. Under the 2-state collapse Assumption 1 with $a_H > \bar{p} > a_L$ (which holds by Assumption 4 at the empirical KDE-collapsed primitives $a_H = \$7.16$, $\bar{p} = \$6.00$, $a_L = \$4.97$/gal), σ takes only two distinct values: $\sigma(H) = s_0$ (max term equals zero) and $\sigma(L) = (\bar{p} - a_L) + s_0$ (max term active). Within this regime, the LP-dual derivation of Lemma B.2 operates over the parameter space (K, s_0) of the restricted instrument class $\mathcal{S}$, not over the kink location itself; the Borch–Wilson FOC ratio condition (B.5) is well-defined at any interior (K^*, s_0^*) with $K^* \in (a_L, a_H)$. The kink at $p_{\mathrm{SAF}} = K^*$ falls strictly between the two scenario realizations and is therefore inactive in the optimization, as is standard for piecewise-linear contracts on discrete supports. Generalization to continuous KDE support (away from the 2-state collapse) introduces a subdifferential FOC at the kink that does not bind at the 2-state optimum; the analysis is deferred to a numerical companion.*

B.1. Calibration Procedure for λ_{prod}

Honest statement of the identification. The recovery procedure below inverts the NPV-binding identity to map an observed subsidy band $\sigma^{\text{obs}}_{\text{prod}} \in [\$1.30, \$1.45]$/gal into an implied mean-CVaR weight λ_{prod}. It is therefore an *identification through revealed preference*: we read λ_{prod} from the observed 45Z credit conditional on the calibrated hurdle envelope ($r_h = 0.125$, envelope $[0.10, 0.15]$; Section 5.2) and the Paper 1 project-finance primitives. The SAF techno-economic literature (Farooq et al. 2025), which examines MJSP distributions across the four SAF pathways (HEFA, ATJ, FT, and pyrolysis-to-jet), anchors the empirical band $\sigma^{\text{obs}}_{\text{prod}} \in [\$1.30, \$1.45]$/gal used as input but does not directly identify a CVaR weight. The procedure is structurally analogous to recovering an Arrow-Pratt risk-aversion parameter from observed insurance premia; the implied $\lambda_{\text{prod}} = 0.60$ rationalizes the observed band given the model primitives, and the *policy-robust* object is Component 2 of the wedge decomposition, which is independent of λ_{prod} (Table 7 below).

Step 1: Identifying assumption. The producer accepts the SAF project iff the mean–CVaR risk-adjusted expected return clears the hurdle IRR r_h. Equivalently, under the producer's mean–CVaR Stage-2 best-response FOC (Lemma A.4), $\tilde{\mathbb{E}}_{\text{prod}}[\text{IRR}] \geq r_h$, where $\tilde{\mathbb{E}}_{\text{prod}}$ is the producer's distorted expectation under mean-CVaR weight λ_{prod} (Lemma B.1). The TEA literature characterizes the unconditional IRR distribution achievable at observed 45Z linear-subsidy levels; the unknown is the risk-aversion weight λ_{prod} that, combined with the TEA distribution and the empirical hurdle r_h, rationalizes the observed $\sigma_{\text{prod}} \in [\$1.30, \$1.45]$/gal band.

Step 2: Closed-form recovery from observed σ_{prod}. Under the producer's NPV-binding constraint at hurdle r_h (Assumption 13) the subsidy must clear the project at the mean–CVaR-distorted weighting:

$$\sigma_{\text{prod}}(\lambda_{\text{prod}}) = \frac{\kappa_{\text{CAPEX}}}{u \cdot \text{AF}(r_h, T)\,(1-\tau)} + \lambda_{\text{prod}} \cdot \pi_H \cdot (a_H - a_L) - (\bar{p} - \bar{c}_{op}).$$

At observed $\sigma^{\text{obs}}_{\text{prod}} \in [\$1.30, \$1.45]$/gal with $r_h = 0.125$ from Section 5.2 and the remaining primitives $(\kappa_{\text{CAPEX}}, u, \tau, \pi_H, a_H - a_L)$ inherited from Li and Yu (2026), the implied λ_{prod} is recovered by inverting the identity:

$$\lambda_{\text{prod}} = \frac{\sigma^{\text{obs}}_{\text{prod}} - \kappa_{\text{CAPEX}}/[u\ \text{AF}(r_h, T)(1-\tau)] + (\bar{p} - \bar{c}_{op})}{\pi_H \cdot (a_H - a_L)}. \tag{B.17}$$

Step 3: Empirical interval. Evaluating (B.17) at the empirical 45Z band $\sigma^{\text{obs}}_{\text{prod}} \in [\$1.30, \$1.45]$/gal yields the band image $\lambda_{\text{prod}} \in [0.54, 0.69]$, with $\lambda_{\text{prod}} = 0.60$ at $\sigma^{\text{obs}}_{\text{prod}} = \$1.36$. The full $\sigma^{\text{obs}}_{\text{prod}}$-to-$\lambda_{\text{prod}}$ map is monotone-increasing and linear; the slope is $[\pi_H \cdot (a_H - a_L)]^{-1} = 1/\$1.03$/gal per dollar of observed subsidy. We call $[0.54, 0.69]$ the band image rather than an interquartile range: no posterior distribution over $\sigma^{\text{obs}}_{\text{prod}}$ is specified, so the interval carries no quantile interpretation.

Step 4: Sensitivity to the calibrated hurdle. The identification is conditional on r_h, and strongly so. Because $\partial\sigma_{\text{prod}}/\partial r_h \approx 11.96$ (Section 7.3) and the recovery divides by $\pi_H(a_H - a_L) = \$1.03$, a shift of r_h within the calibrated envelope $[0.10, 0.15]$ moves the implied λ_{prod} by approximately ∓ 0.29: at $\sigma^{\text{obs}}_{\text{prod}} = \$1.36$, $r_h = 0.10$ implies $\lambda_{\text{prod}} = 0.88$ and $r_h = 0.15$ implies $\lambda_{\text{prod}} = 0.31$. The implied band therefore straddles the planner's $\lambda_{\text{planner}} = 0.50$ within the hurdle envelope, so the *sign* of Component 1 is identified only conditional on r_h; the headline $\lambda_{\text{prod}} = 0.60$ is the value at the envelope midpoint $r_h = 0.125$. This is the same identification fragility as the discretization dependence documented in Table 8, and it is why the policy-robust object of the paper is

Table 7: Robustness of the wedge decomposition to λ_{prod}

λ_{prod}	C_1 (\$/gal)	C_2 (\$/gal)	Wedge (\$/gal)	σ_{prod} (\$/gal)	C_2-share
0.00	−0.517	0.739	+0.222	0.739	n/a†
0.25	−0.258	0.739	+0.481	0.998	n/a†
0.50	0.000	0.739	+0.739	1.256	100%
0.53	+0.031	0.739	+0.770	1.287	96%
0.60	**+0.103**	**0.739**	**+0.842**	**1.359**	**88%**
0.67	+0.176	0.739	+0.915	1.432	81%
0.75	+0.258	0.739	+0.997	1.515	74%
1.00	+0.517	0.739	+1.256	1.773	59%

Component 2 ($\Delta\text{IRR}_{\text{prem}}$, financial frictions) is structurally invariant to λ_{prod}. The Component-2 share exceeds 70% across $\lambda_{\text{prod}} \in [0.50, 0.77]$ and spans $[79\%, 94\%]$ across the band image $[0.54, 0.69]$. The headline ($\lambda_{\text{prod}} = 0.60$, bold row) gives $\sigma_{\text{prod}} = \$1.36$/gal inside the observed 45Z band $[\$1.30, \$1.45]$; by construction of the recovery the point reproduces the band midpoint, so the level match is a consistency check rather than an independent validation. †At $\lambda_{\text{prod}} < \lambda_{\text{planner}}$, Component 1 is negative: the producer is less risk-averse than the planner and demands a *smaller* risk-premium subsidy than $\sigma^*_{\text{planner}}$ implies. The C2-share metric is not policy-meaningful in this region (denominator close to or smaller than numerator), though the wedge itself remains well-defined.

Component 2, which is independent of λ_{prod} (Step 5, item 1).

Step 5: Robustness of the headline to λ_{prod}. Table 7 reports the wedge decomposition at the empirical baseline ($r_h = 0.125$, $T = 10$, $\kappa_{\text{CAPEX}} = \$11$/gal, $u = 0.85$, $\tau = 0.21$, $\lambda_{\text{planner}} = 0.50$) across the full range $\lambda_{\text{prod}} \in [0, 1]$. Two observations are central to the policy interpretation.

1. *Component 2 is invariant to λ_{prod}.* The financial-frictions component $\Delta\text{IRR}_{\text{prem}} = \kappa_{\text{CAPEX}}/[u\ \text{AF}(r_h, T)(1 - \tau)] - (\bar{p} - \bar{c}_{op})$ contains no λ_{prod} dependence. It equals \$0.739/gal at every λ_{prod} in $[0, 1]$. This is the *policy-robust* object of the decomposition: regardless of how the producer's risk-aversion weight is identified, the project-finance hurdle premium relative to the planner's risk-free discount rate is \$0.739/gal at the empirical baseline.
2. *Component 1 scales linearly in $\lambda_{\text{prod}} - \lambda_{\text{planner}}$ but is bounded.* $C_1 = (\lambda_{\text{prod}} - \lambda_{\text{planner}})\,\pi_H\,(a_H - a_L)$ ranges from −\$0.517/gal at $\lambda_{\text{prod}} = 0$ to +\$0.517/gal at $\lambda_{\text{prod}} = 1$. In the band image $\lambda_{\text{prod}} \in [0.54, 0.69]$, Component 1 spans $[\$0.04, \$0.19]$/gal and the Component-2 share spans $[79\%, 94\%]$.

The procedure documented in Steps 1–5 is the load-bearing mapping for the wedge decomposition (Proposition A.5). The headline \$0.10/gal Component 1 is conditional on the empirical $\lambda_{\text{prod}} = 0.60$ identification, but the \$0.74/gal Component 2 is policy-robust to λ_{prod} identification choice.

Step 6: Numerical verification against the 20-bin KDE primitive. The closed-form decomposition rests on the two-state collapse $\{H, L\}$ inherited from Li and Yu (2026) Theorem 1, which calibrates (π_H, a_H, a_L) to match the break-even-conditional means of the 20-bin Gaussian-KDE SAF-price distribution (Li and Yu 2026, Table 3). The natural robustness question is whether the headline \$1.36/gal σ_{prod} and the 12%/88% attribution survive computation directly under the 20-bin KDE rather than via the two-state device. Table 8 reports the comparison.

Three observations.

1. *Component 2 is invariant across discretizations.* The \$0.736/gal financial-frictions premium depends on

Table 8: Verification of the closed-form decomposition against the 20-bin KDE primitive

Quantity	Two-state collapse	20-bin KDE
$\bar{p}$ (\$/gal)	6.004	6.002
$\mathrm{CVaR}_{0.95}[p_{\mathrm{SAF}}]$ (\$/gal)	4.970	4.031
$\bar{p} - \mathrm{CVaR}_{0.95}[p_{\mathrm{SAF}}]$ (\$/gal)	1.034	1.972
At $\lambda_{\mathrm{prod}} = 0.60$ (Paper-1-calibrated headline):		
σ_{prod} (\$/gal)	1.355	1.919
$\sigma^*_{\mathrm{planner}}$ (\$/gal)	0.517	0.986
Component 1 (\$/gal)	+0.103	+0.197
Component 2 (\$/gal)	**+0.736**	**+0.736**
Wedge (\$/gal)	+0.840	+0.934
C_2-share	87.7%	78.9%
Inverting each model to match observed $\sigma^{\mathrm{obs}}_{\mathrm{prod}} = \$1.36$/gal:		
Revealed-preference λ_{prod}	0.604	0.316

Component 2 ($0.736/gal at the empirical baseline) is structurally invariant to the discretization choice. The mean-CVaR gap $\bar{p} - \mathrm{CVaR}_{0.95}[p_{\mathrm{SAF}}]$ that drives Component 1 differs because the two-state a_L is conditioned on the bottom 52.8% of mass (the welfare break-even regime) while $\mathrm{CVaR}_{0.95}$ is the mean of the bottom 5%. The discretization shifts the implied λ_{prod} identification and the absolute Component-1 magnitude, but does *not* shift the dominant-component conclusion: financial frictions account for ≥ 79% of the wedge under either primitive. Values here evaluate at the exact bin-weighted mean $\bar{p} = \$6.004$; the headline text values ($1.359, $0.739, 87.8%) use the rounded binned mean $6.00, and that $0.004 difference accounts for the entire drift between the two sets.

$\kappa_{\mathrm{CAPEX}}, \mathrm{AF}(r_h, T), u, \tau$, and $\bar{p} - \bar{c}_{op}$ alone—no tail-distribution primitive enters. The discretization choice cannot shift Component 2.

2. *Component 1 doubles under the 20-bin KDE.* The two-state collapse compresses the mean-CVaR gap to $1.03/gal (the break-even-conditional gap); the 20-bin tail at $\alpha = 0.95$ delivers $1.97/gal (the worst-5% mean-conditional gap). At identical $\lambda_{\mathrm{prod}} = 0.60$, Component 1 grows from $0.10/gal to $0.20/gal, and the wedge shifts σ_{prod} outside the empirical $1.30–$1.45/gal 45Z band.
3. *Revealed-preference λ_{prod} shifts to 0.32 under the 20-bin primitive.* If we hold $\sigma^{\mathrm{obs}}_{\mathrm{prod}} = \$1.36$/gal fixed as the identification target, the implied λ_{prod} under 20-bin KDE is 0.32—substantially below $\lambda_{\mathrm{planner}} = 0.50$, meaning the revealed-preference producer is interpreted as *less* risk-averse than the planner under 20-bin tail-pricing.

The *policy-robust* conclusion is consistent across both discretizations: the financial-frictions premium of $0.74/gal dominates the wedge regardless of CVaR weighting or discretization choice. The two-state collapse delivers the cleaner 12%/88% headline; the 20-bin direct calculation delivers a 21%/79% headline at fixed $\lambda_{\mathrm{prod}} = 0.60$, but the dominant component is the same. The absolute level of Component 2 ($0.74/gal) and the dominance result ($C_2 > 70\%$ of the wedge) survive the discretization question. Section 7.4 reports the headline insensitivity of the policy implications.